FirstPass STUDIO

USER GUIDE

Version 1 · October 2026 · by PIVT

ABOUT THIS GUIDE

Graphics for the stream, from the first whistle to the last.

FirstPass Studio draws broadcast graphics for a netball livestream, fed straight from PlayHQ and sent to your switcher. This guide starts with the way most nights run, one operator and Studio doing the rest, then covers running it by hand, setting up a competition, sponsors, athletes and the Stream Deck.

Screenshots use the fictional Elite Netball League.



CHAPTER 1 OF 11

What Studio is

FirstPass Studio is a Mac app that draws broadcast graphics for a netball livestream and sends them to your vision switcher. The scores, clocks, ladders, fixtures and line-ups come from PlayHQ's public data, so there is nothing to type in during the game.

IN THIS CHAPTER

1 The words used in this guide
2 The pages you work in
3 The sample competition in this guide

Studio does one job on stream night: graphics. Your cameras, switcher and streaming encoder stay as they are. Studio hands the switcher a picture with transparency (fill and key, or a chroma colour) so the graphics sit over the vision.

The words used in this guide

Studio
The whole thing: the Mac app, the pages you work in, and the graphics it puts on screen.

On air, off air
Studio is on air while a Studio window is open. On air it talks to PlayHQ, serves the graphics and answers the Stream Deck. Closing the last window takes it off air.

A graphic, or a page
One thing that can go on screen: the scoreboard, a ladder, a lower third, an animation. Each is a web page that Studio draws.

A source
A graphic placed in the scene, with its own position, options and fade. The scoreboard is one source; a ladder is another.

The scene
Studio's own 1920 × 1080 frame, built from sources switched on and off. It is what the switcher receives.

A scene (on the Scenes list)
A saved look: which sources are on and which are off. "Live game" and "Clear" are scenes.

A sequence
A list of steps played in order from one press: a run-up, a quarter-time break, a full-time wrap.

The competition
Everything that belongs to one league: its PlayHQ link, names, rules, colours, logos, sponsors and athletes. Studio keeps a setup per competition and uses one at a time.

The data folder
Where Studio saves everything, outside the app, so updates never touch it.

The pages you work in

Every Studio window has the same bar at the top, with the FirstPass Studio mark and the tabs in three groups by what you are doing. **Run** is the night itself: **Output**, where the show is built and sent out, and **Scoreboard control**, which runs the board. **Plan** is before the night: the **Production Planner** plans the court and the commentators, and **Socials** makes the night's pictures for social media. **Set up** is done once: **Recordings**, **Competitions setup**, **Sponsors**, **Athletes** and **Devices**. **Docs** sits on its own at the right: it is this guide, with the bar on top. In the app, ⌘1 to ⌘9 and ⌘0 go to Output, Production Planner, Scoreboard control, Recordings, Socials, Competitions setup, Sponsors, Athletes, Devices and Docs.

The right of the bar carries the status chips, on every page, so you can see the night from wherever you are: **ON AIR** in orange like a tally light, with a dot per output that is solid in the output's colour while it sends and hollow while it doesn't (or **Off air**); the game on the board, its dot orange before the clock has started, green from the first clock start or score until an hour after the game ends, then grey; PlayHQ, whose dot is the link alone (green when PlayHQ answers, amber when a reading is overdue, red when it keeps failing, grey when scores are by hand) and whose words give the quarter and clock, or **game completed** once PlayHQ closes the game (**full time** when the last quarter is run down by hand); the sequence playing with its step and the time left on it (and what the run order plays after it), the triggered sequence counting down to start or the run order's **Next**, **ATEM** while a switcher is set up (green when it answers, red when it doesn't), and **REC** while a game records (grey while Record games waits for one). The chips are status, not links; the tabs take you to a page.



The bar on every page: the tabs grouped as Run, Plan and Set up, and the chips for the night: off air, the game on the board, PlayHQ with the quarter and clock, and what the run order plays next.

The sample competition in this guide

The screenshots use a fictional league, the **Elite Netball League**, with eight clubs named after animals (Falcons, Panthers, Stingrays, Cobras, Grizzlies, Hawks, Rhinos and Wolves) across Premier, Men's and Under 21 divisions, playing at the Metro Arena. Its

sponsors are fictional too. Everything you see is what Studio draws for any competition once its setup is in.

Getting started

Install the app, import or make your competition, and learn the one rule that matters on stream night: keep a Studio window open.

IN THIS CHAPTER

1 Install
2 On air and off air
3 The 1P menu
4 Before you go on air

Install

1. Download `FirstPass Studio.dmg`, drag the app to Applications and open it. It runs on Apple silicon Macs (M1 and later).
2. The **1P** icon appears in the menu bar, off air. Choose **Open Studio**. A short splash shows, Studio goes on air and the Output page opens in the app's own window.
3. Go to **Competitions setup** and either **Import a template file** (your league's `.json`, with its logos and sponsors inside) or **New competition** and fill it in (Chapter 6).
4. Back on **Output**, add the graphics you need with **Add page**, and turn on the video output that feeds your switcher (Chapter 3).

On air and off air

The pages zoom like Safari's, for a small screen: **Zoom In** (⌘+), **Zoom Out** (⌘−) and **Actual Size** (⌘⌥0) in the View menu, which shows the level, or a pinch on the trackpad. Every Studio window zooms together and the level is kept on that Mac; while it isn't 100% a chip at the right of the bar shows it, and a click on the chip puts it back. The Docs tab, this guide, has its contents at the left on a wide window.

Studio only runs while a Studio window is open. Opening one (Open Studio, any page in the 1P menu, or ⌘N) puts it on air; closing the last one takes it off air, and the graphics and Stream Deck get no answer until a window opens again. Closing the last window while on air asks first, and so does ⌘Q, so a stray keypress can't stop a stream. Leave the Output page open all night.

The bar's chips show the game on the board, whether PlayHQ is answering (amber when overdue, red after a minute of errors) and which outputs are on air, on every page; the Output page's status board shows the score and the rest of the night. If anything is wrong, the page you keep open says so.

The 1P menu

The 1P icon in the menu bar is Studio's home: it opens Studio, its pages and this guide, backs the settings up and restores them, and has **Open at login**, **Appearance** (the Studio pages follow the Mac's light or dark look unless pinned here; the graphics never change) and **Check for updates**. Chapter 10 lists every item. Updates download in the background and install when the app quits, never mid-stream.

LINKS

Studio's links start with the Mac's Bonjour name, such as `http://matts-macbook-pro.local:8080/` , so the same link works from a phone, another computer or a Stream Deck on the same network. `http://localhost:8080` works on the Mac itself.

Before you go on air

1. **Open the app.** The Output page opens and Studio goes on air. Keep a window open all night.
2. **Right competition.** Competitions setup shows the one in use at the top. Check its PlayHQ link, the court to follow and the scoring window.
3. **Right game.** On Scoreboard control, the picked game is tonight's first on your court and Follow PlayHQ is on.
4. **Scores arrive.** Once the scorer has started on PlayHQ, the control page says the game is being live scored and the board moves. Nothing is read before the scoring window opens, so a quiet board before then is normal.
5. **Scene and output.** On Output, turn on the display, NDI or Blackmagic output your switcher takes, check fill and key or chroma, and show one graphic to see it land.
6. **Sound.** Play a few seconds of music and an animation with sound, and check the sound only comes out where you want it.
7. **Stream Deck.** Press a Source key and watch the Sources list follow it. A score key switches scoring to Manual, so put it back to Follow PlayHQ after testing.
8. **Sponsors.** Tonight's spots and ads are on; the sponsor report starts counting when the output turns on.

More in the Reference: **The Mac app**.

Game night, the automated way

This is how most nights run: one operator, PlayHQ scoring the game, a base scene that stays on, automatic layers that fire by themselves, and sequences that handle the run-up, the breaks and full time.

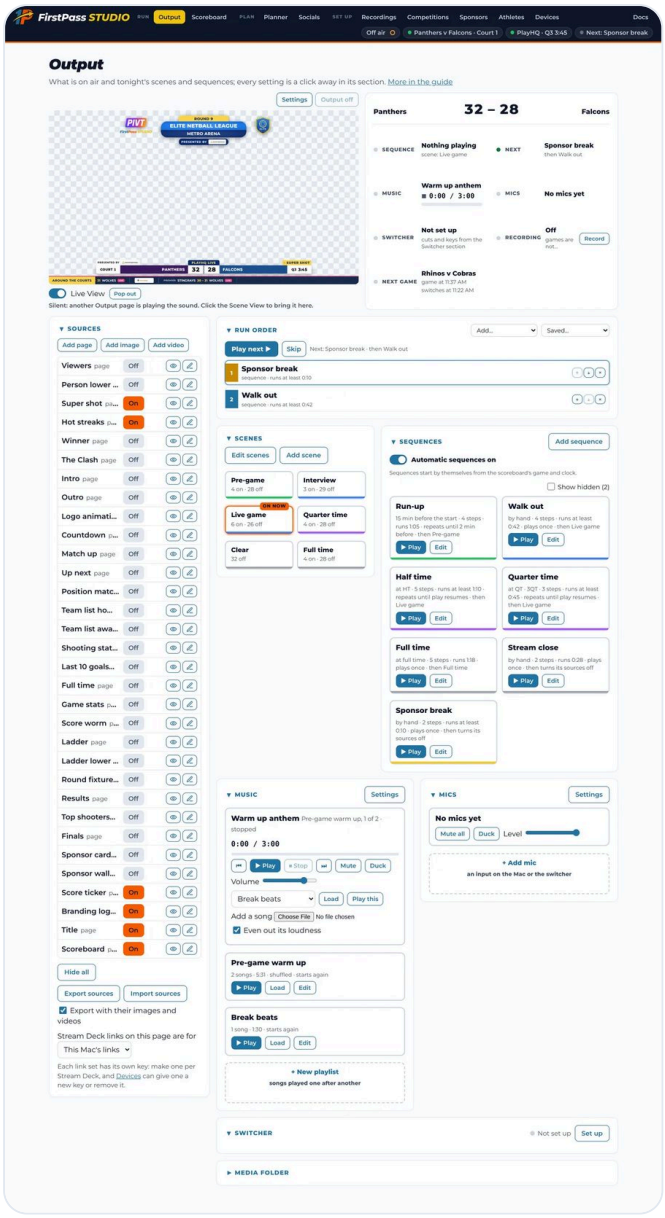
IN THIS CHAPTER

1 How it fits together
2 Scoreboard control
3 The Output page
4 The base scene
5 Automatic layers
6 Scenes
7 Sequences
8 Music, playlists and mics
9 Video output
10 A night, start to finish
11 The Production Planner
12 Socials
13 Recordings

How it fits together

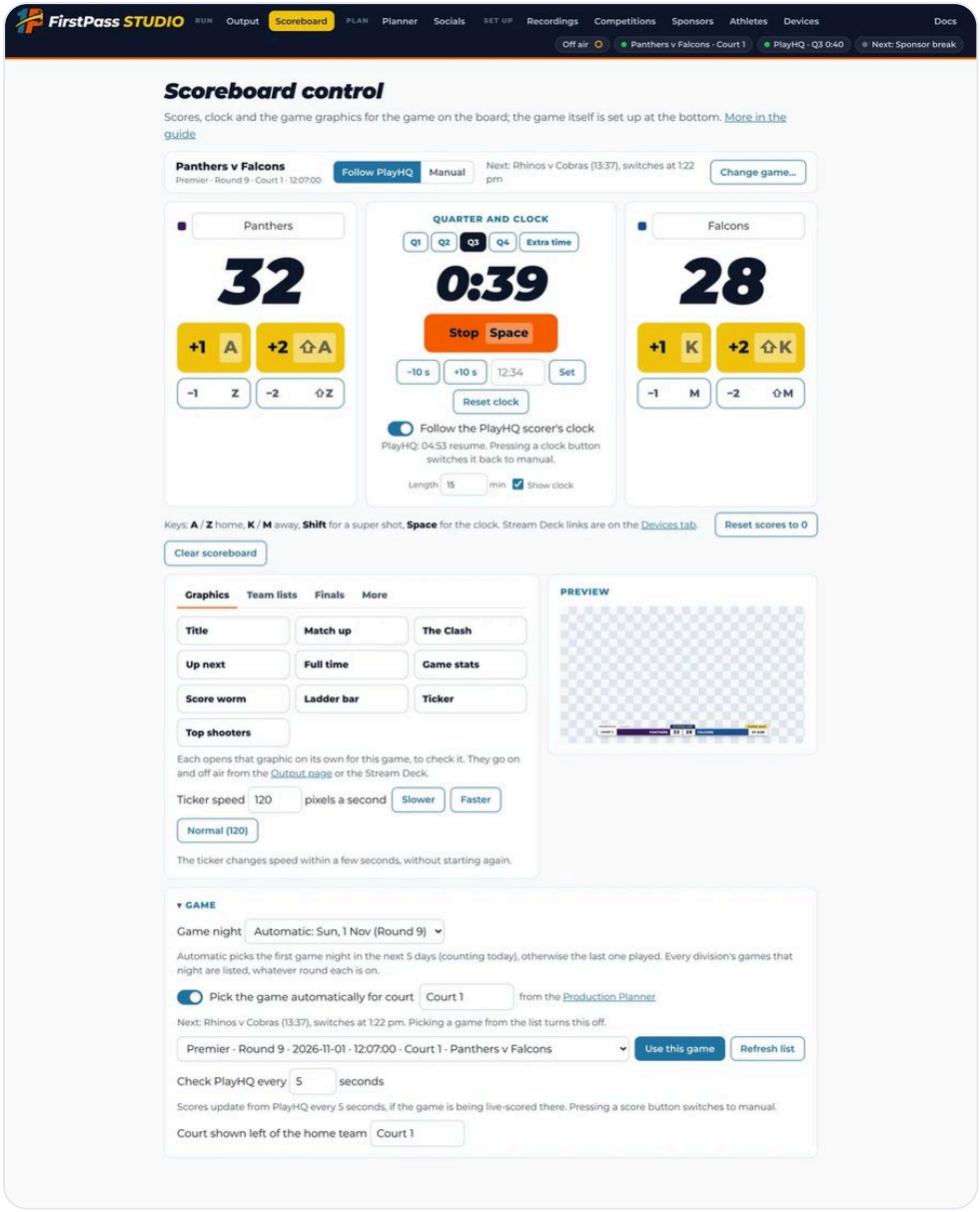
Four things do the work, and each one is set up once:

- **Follow PlayHQ.** The scoreboard reads the live score, quarter and clock from the same feed the PlayHQ game page uses, every 5 seconds by default (as often as every second), while someone live-scores the game on PlayHQ. Automatic game selection moves the board to each next game on your court as its electronic scoring starts.
- **The base scene.** A saved scene with the sources that stay on through the game: the scoreboard, the title banner, the branding logos and the score ticker.
- **Automatic layers.** Sources that stay on but only show when something happens: the super shot celebration, the hot streaks and game moments, the super shot tab on the scoreboard, the naming-rights sponsor's tab. They watch the board and play by themselves.
- **Sequences.** Steps played in order for the parts of the night that aren't live play: the run-up, the walk out, quarter and half time, full time and the close. A sequence can start by itself from the game's clock, carry music, and hand over to another sequence.



The Output page is the whole show on one page: video output, music, sequences, scenes, the Scene View, and the sources with their settings.

- **Scoreboard control**



Scoreboard control: the game in one bar, the two scores and the clock as the biggest things on the page with their keys printed on them, then the graphics, team lists and finals as tabs with a preview of the board.

The page is laid out for the person scoring. The bar at the top names the game on the board with its division, round, court and time, holds the **Follow PlayHQ** / **Manual** switch, and says which game the board will switch to next; **Change game…** opens the **Game** section at the bottom, where the game night, court, automatic selection and PlayHQ settings live (it folds to its title once the night is set). Under the bar are the two scores with their +1 and +2 pads, the quarter buttons and the clock with **Start** / **Stop**, and the keys printed on each button. Beneath them, tabs for the game graphics (each opens on its own, to check it), the team lists, the finals and the rest, with a preview of the board beside them.

ALWAYS A GAME ON

When Studio starts, it puts a game on the board with Follow PlayHQ on: the game on now on your court, otherwise the first game of tonight on that court. Games are picked by game night, not by round: the list holds every division's games that date, whatever round each is up to, so one division on Round 10 and another on Finals Round 1 sit side by side, each game showing its own round. Tonight is found by date (the first game night in the next 5 days, counting today), so a Tuesday night is found from the Friday before. You can pick another night from the **Game night** list, which names each night's rounds; its games on your court are then followed by their time (the one due now, otherwise the next to come). Set it back to Automatic afterwards.

AUTOMATIC GAME SELECTION

With **Pick the game automatically for court** ticked, the board shows the first game of the night on that court, then switches to each next game when its electronic scoring starts (**Electronic scoring starts N minutes before the game** on the setup page, 10 by default, 15 in the sample league), with the quarter back to Q1 and a full clock. In finals it keeps the final on until PlayHQ marks it finished, in case of extra time. Picking a game from the list by hand turns automatic selection off; tick it again to go back. When the Production Planner has a court for tonight, that court is set here when the night arrives, with **from the Production Planner** beside it. Changing it by hand still works, and the Planner flags it.

FOLLOW PLAYHQ

With **Follow PlayHQ** on, the score and quarter update every 5 seconds from PlayHQ's live scoring, and **Follow the PlayHQ scorer's clock** runs the clock from the scorer's clock. The clock matches the scores: switching to Manual, or pressing any score button, turns clock-following off as well, and the control page says so. Press a clock button to run the clock by hand while scores still follow PlayHQ. To line the clock up with the venue, type the venue's time in **Match venue clock**.

NOTHING BEFORE THE WINDOW

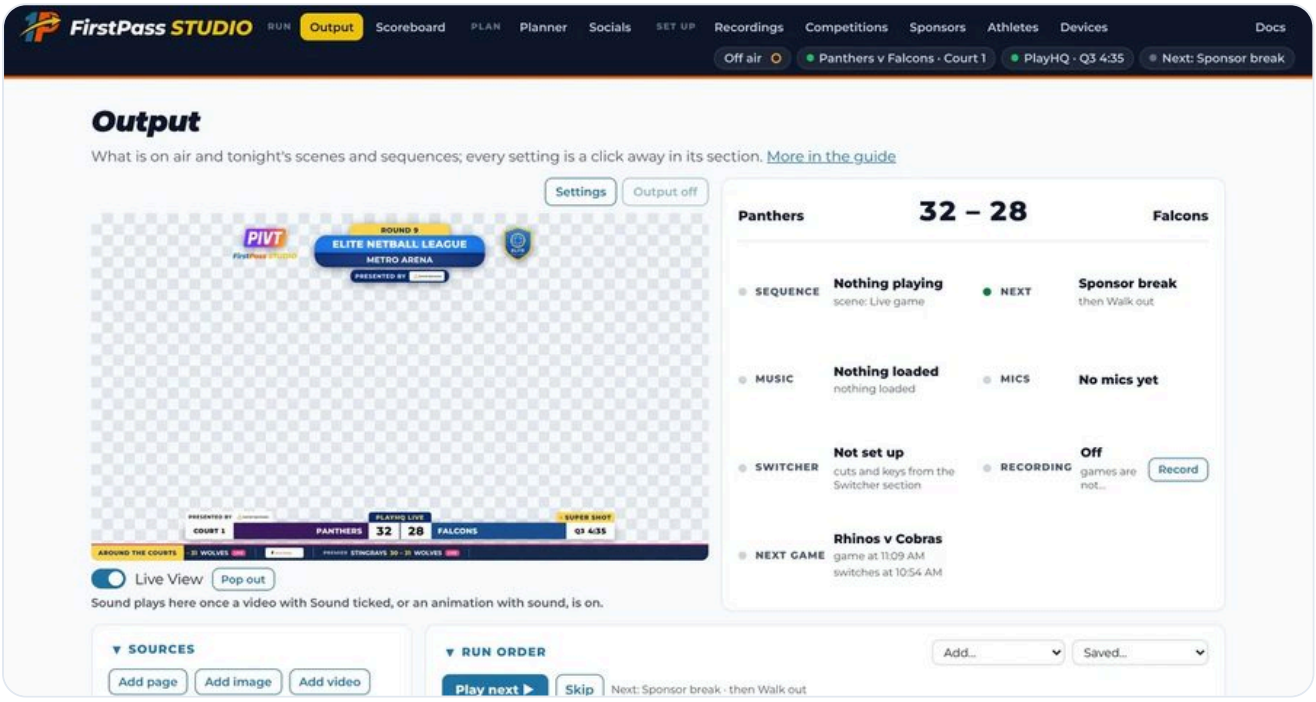
Studio doesn't ask PlayHQ's live feed for a game until its electronic scoring window opens, whether the game was picked automatically or by hand. The control page says when it will start checking.

More in the Reference: **Scoreboard control in detail**.

The Output page

The Output page builds Studio's own scene: sources drawn as layers on one 1920 × 1080 frame, top layer first, and sends it to a display, a Blackmagic device or NDI for your switcher.

It is one page, laid out for the night. The picked output's picture sits at the top with its tabs directly above it, **HD** or **4K**, **Settings** and **Output on** in a row over them, and a status board beside it with the score and a line each for the sequence playing, the next one armed (counting down, in days when it's days away), the music, the mics (with an amber line while they duck), the switcher, the recording (with **Record** or **Stop**) and, across the board, the next game: its teams, when it starts and when the scoreboard switches to it. Under them the Sources run down the left, with the scenes and sequences as tiles, then music and mics, the switcher and the media folder to their right. Every setting is a click away in its section: a source's settings open under its row, **Edit scenes** opens the scene grid, **Settings** on Music and Mics unfolds theirs, and **Set up** on the switcher its address and keys. A name that's too long for its space ends in …; hover over it to read it in full.



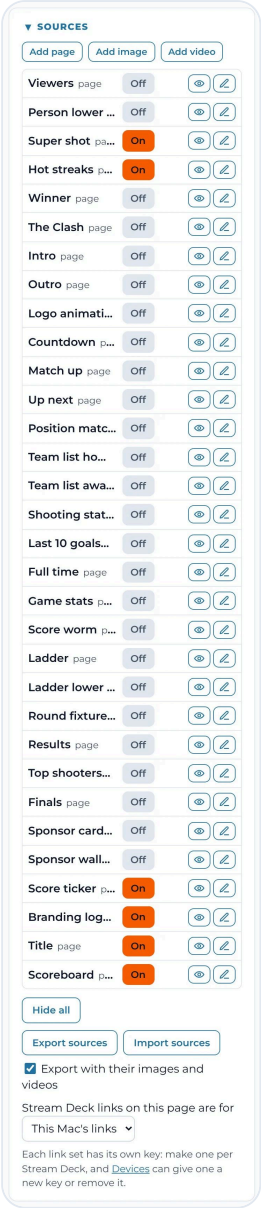
The top of the Output page in the third quarter: the picture with Settings and Output on over it, and the status board beside it with the score, the run order's next item, the music, the mics and the next game.

Every section folds away with the arrow beside its title (or a click on the title), so the ones you don't use, say Music or Switcher, take one line. Studio remembers which are folded on that Mac, even after it restarts, and buttons in a folded section's title row, like **Output on**, still work. The page carries no explanations; this guide has them, and **More in the guide** under each page's title opens its part.

SOURCES

Add page adds any Studio graphic (or any web address), **Add image** a PNG, JPG, GIF, WebP or SVG, and **Add video** an MP4, WebM or MOV. A new source goes in hidden with its settings open under its row, ready to change. Each row has its On / Off, an eye to preview it on its own and a pencil for its settings. The list is top layer first: drag a

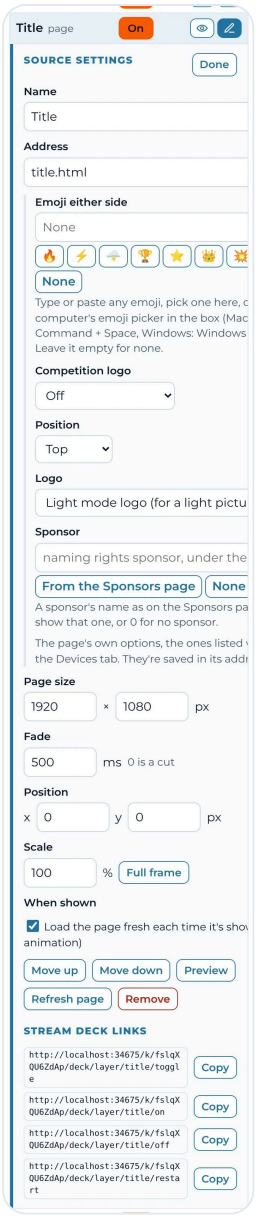
source up or down to move it between layers, or use **Move up** and **Move down** in its settings. **Add all unused pages** puts every Studio graphic in the scene at once, hidden.



The Sources list, top layer first. The orange On buttons are the base scene and the automatic layers; everything else waits for a scene, a sequence or a key.

SOURCE SETTINGS

Click a source, or its pencil, and its settings open under its row; **Done** closes them. A graphic's link options (position, heading, division, round, animate away and so on) show as fields under its address, so there is nothing to look up. **On / Off** fades it in or out over its own **Fade** time (0 is a cut). **Position and scale** place it on the frame. **Load the page fresh each time it's shown** (on by default) reloads a page as it fades in so its animation replays; untick it to keep a page running so it shows instantly. **Preview** (and the eye in its row) opens the source on its own, half size, on a checkerboard.



Source settings for the title banner: its options are fields, and the Stream Deck links for this source are copied from here.

Videos play from the beginning each time they're shown, can loop, and fade out when they finish unless looping; a video that fades out switches its source off, so its key goes dark and the next On plays it again. A ProRes 4444 `.mov` is converted to a transparent WebM when it is added, keeping its alpha. **Export sources** saves the whole scene with its media as one zip to keep or move to another Mac; **Import sources** puts one back.

More in the Reference: **The Output page in detail**.

The base scene

The base scene is what viewers see for most of the game. In the sample league it is the **Live game** scene: the scoreboard at the bottom, the title banner at the top with the PIVT and FirstPass Studio logos on the left and the competition logo on the right, and the score ticker along the bottom edge with the other courts' scores.



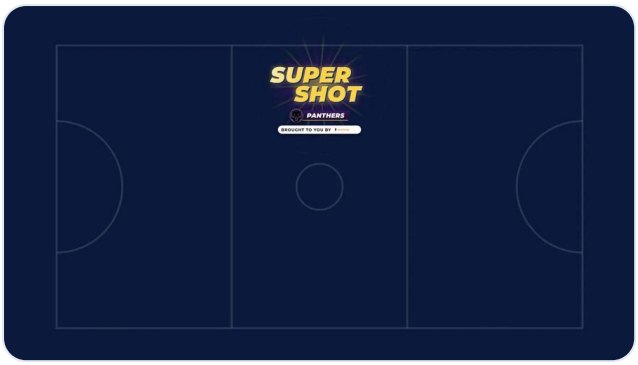
The Live game scene. The court behind it stands in for your vision; the switcher keys the graphics over the cameras.

- **Scoreboard.** Court in a pink box on the left, both teams in their club colours, the score at screen centre, quarter and clock on the right. "PlayHQ live" shows while scores follow PlayHQ; a "Presented by" tab carries the naming-rights sponsor.
- **Title.** Round, competition and venue, with the naming-rights sponsor's pill under it.
- **Branding logos.** PIVT and FirstPass Studio left of the title, the competition logo right of it, so other graphics can be turned on and off around them.
- **Score ticker.** Scrolling scores from tonight's other games, led by a Partner-and-up sponsor's tagline. Its speed is set on the control page or from the Stream Deck.

Build it once: turn those sources on, then press **Save what's on now** under Scenes and name it. From then on, one click (or one Stream Deck key) brings the show back to this look.

Automatic layers

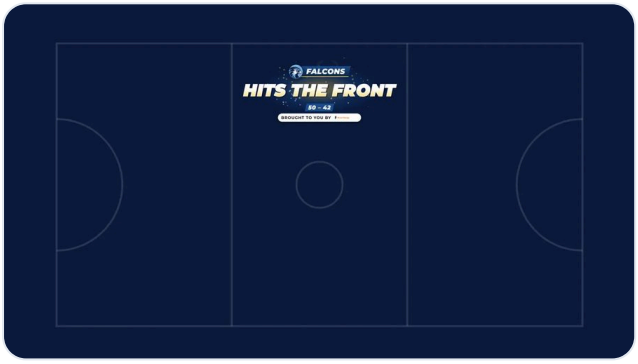
These sources are left on in the base scene and play by themselves from the scoreboard. They add no PlayHQ requests of their own; they ask Studio, which already has the score.



Super shot. A 3-second burst with the team's logo and name in their club colour, played for each successful super shot, from PlayHQ's super shot count or a +2 press. It plays small, centred just under the title.



Hot streaks. "Heating up" at 3 goals in a row, "On fire" at 5, "Goal for goal" when 10 in a row go one each way, from the goal order Studio works out from the score.



Game moments on the same page: hits the front, all square (level in the last 2 minutes), coming back, on the siren, back to back super shots. One goal plays one moment at most.

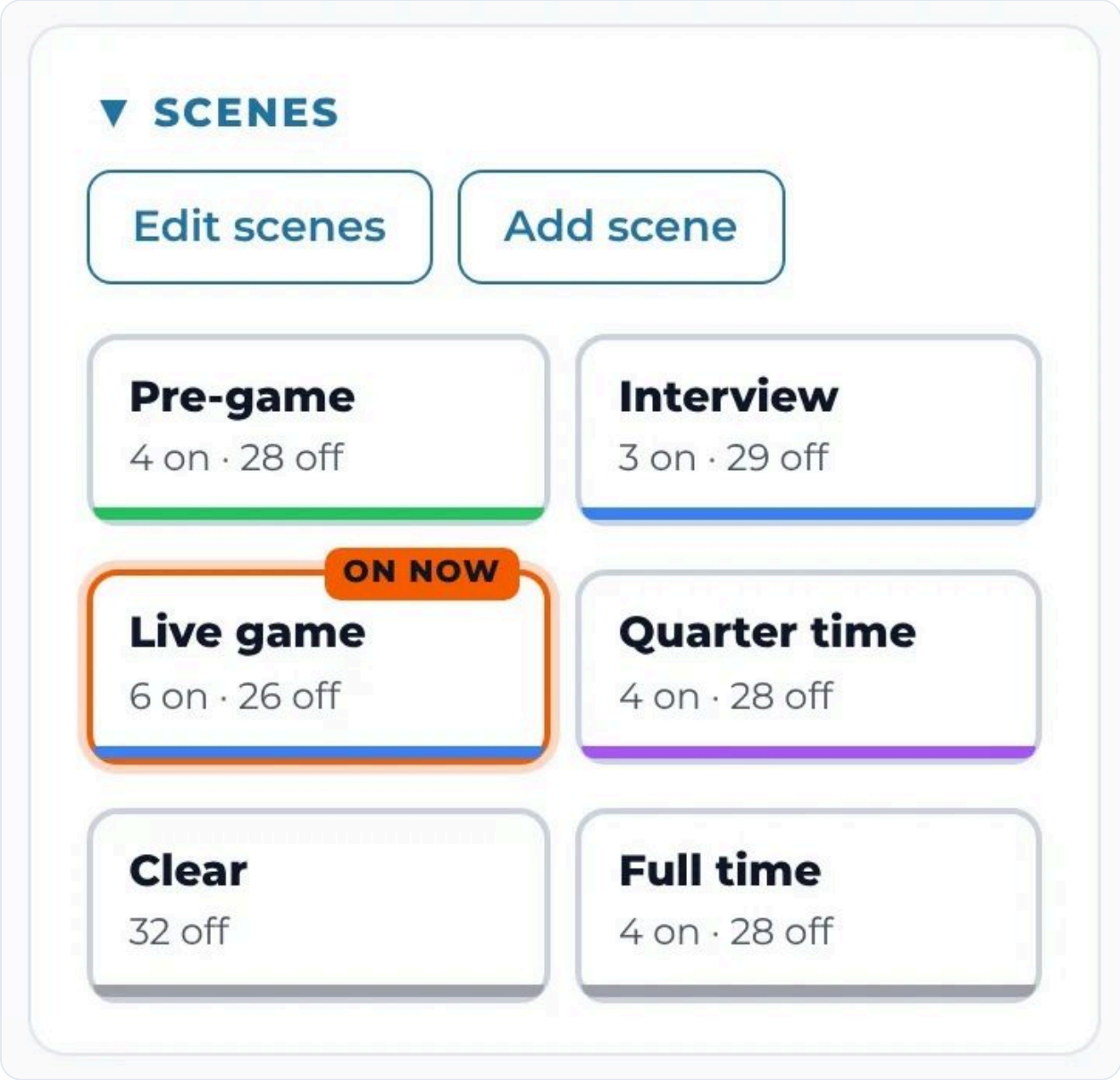


Super shot tab. In a super shot division, a yellow tab rises above the clock for the last minutes of each quarter (set per competition) and sinks away at the siren.

Each celebration can carry a sponsor's "Brought to you by" tag (Chapter 7). A division without super shots plays no super shot animation and counts no super goals; that is set on the competition setup page, per division if needed.

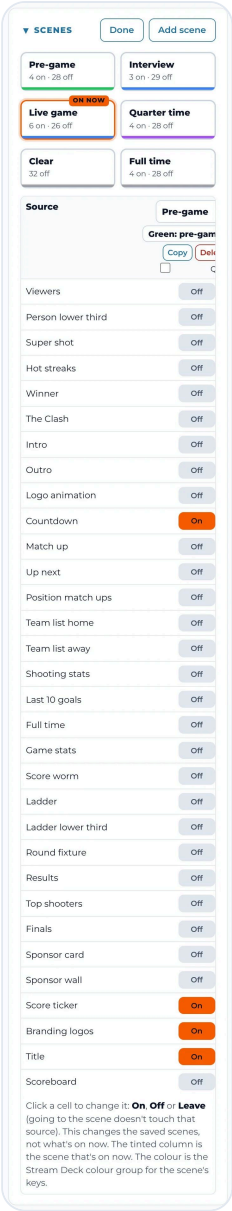
Scenes

A scene is a saved set of sources on and off. Clicking a scene goes to it, turning sources on and off to match while anything already on stays on, not reloaded. Scenes are listed by colour in the order of a stream: green pre-game, blue live, purple quarter time, grey post-game, yellow ads.



The sample league's scenes. Pre-game, Live game, Quarter time, Full time, a quiet Interview scene for an on-court chat, and Clear.

Turn a source on or off by hand after going to a scene and the scene is marked **changed**; the bar under the scenes offers **Back to saved**, **Save over** it, or **Save as new scene**. **Edit scenes** shows a grid, sources down and scenes across, to set each one On, Off or Leave, rename, colour, copy or delete scenes without changing what's on. A source added later is off in every scene.



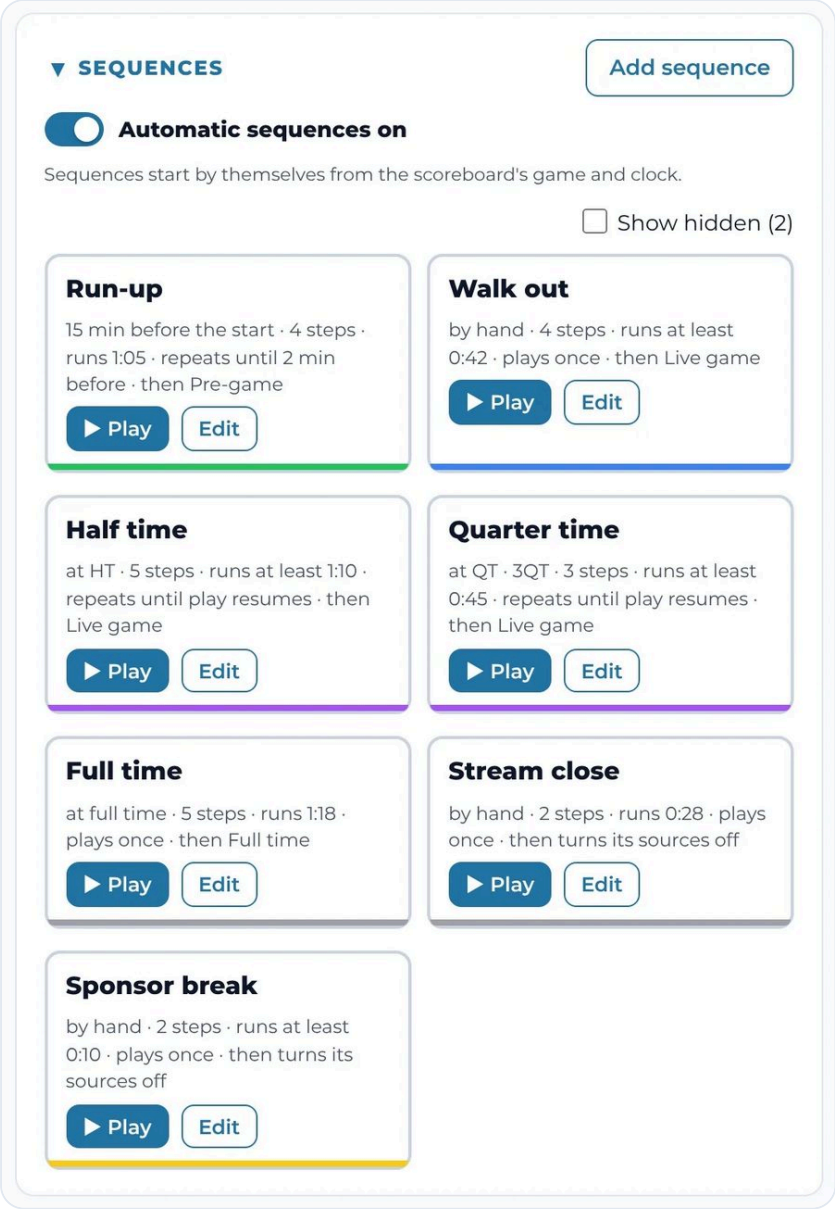
Edit scenes: every source against every scene. "Leave" keeps a source as it is, so a lower third can stay up through a scene change.

QUIET SCENES

Tick **Quiet the music when this starts** on a scene for an interview on court: whatever music is playing ducks and keeps playing until the next scene or sequence without the tick.

Sequences

A sequence plays the scene's sources one after another. Each step turns on the sources you pick and stays for a set time, until it finishes (its videos end or its animation animates away), until the game moves on (the clock starts, or the next break or full time), or until you press Next. As a step ends, sources the next step doesn't use go off with their own fades.

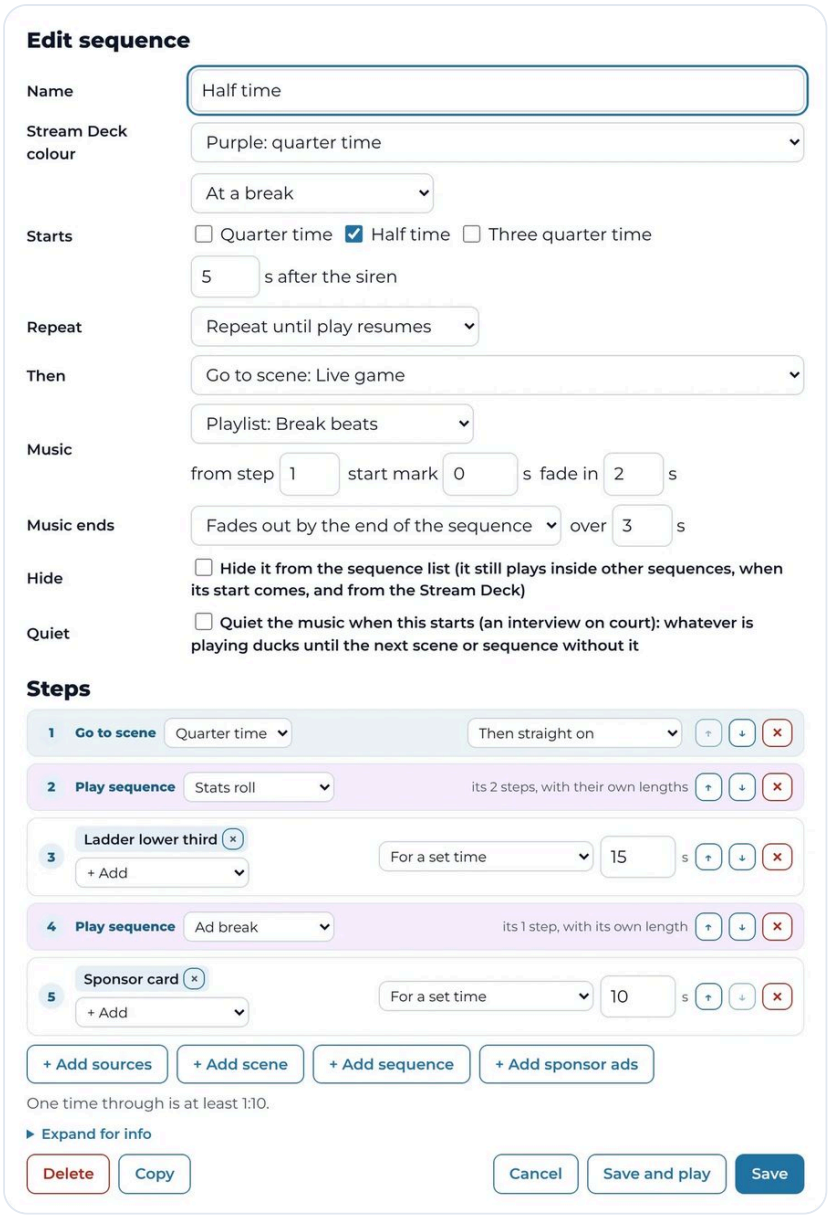


The sample league's sequences. Each card says when it starts, how many steps it has, how long it runs, whether it repeats, and what it does when it ends.

THREE KINDS OF STEP

- **Sources** (**+ Add sources**): one or more sources on together, for a length.
- **A scene** (**+ Add scene**): a starting point or a reset point. What the scene turns on stays on under the next steps and after the sequence. Start a break sequence with the Live game scene and the scoreboard stays up under the stats.
- **Another sequence** (**+ Add sequence**): its steps play in place. Build a mini sequence once (the sample league's **Break stats** and **Sponsor break**) and use it inside Quarter time, Half time and Full time. A sequence made only of sequences is marked **Master**. Tick **Hide** on a mini sequence to keep it off the list; it still plays inside the others and from the Stream Deck.

+ Add sponsor ads adds a step that plays the next one to five sponsor ads in rotation (Chapter 7).

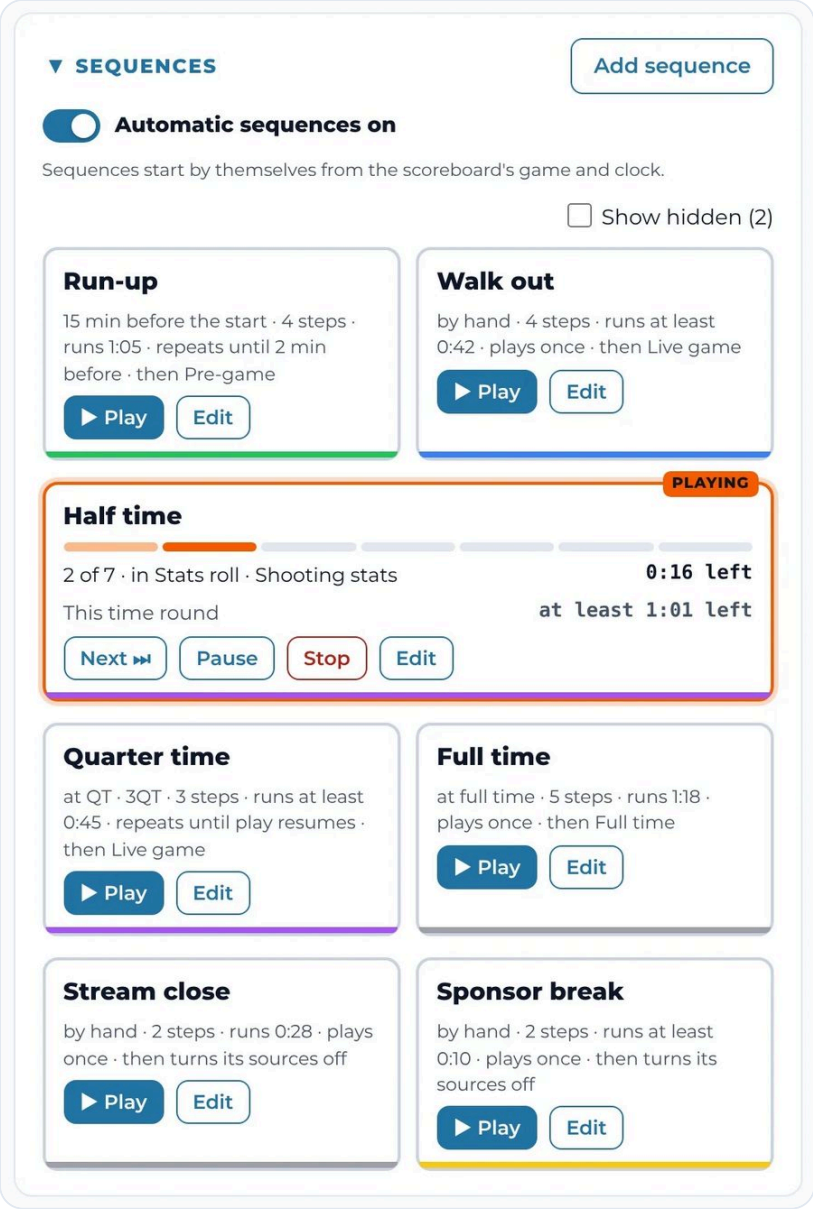


The Half time sequence: starts by itself at half time, repeats until play resumes, plays Break beats, and goes back to the Live game scene when it ends. Its steps are a scene step, two mini sequences and two source steps.

WHEN A SEQUENCE STARTS, REPEATS AND ENDS

SETTING	CHOICES	USED FOR
Starts	By hand · N minutes before the picked game's start (only if Studio was running then; opened later, play it by hand) · at the breaks you tick (QT, HT, 3QT) · at full time, a few seconds after the siren	Run-up 15 minutes out; the breaks and full time from the scoreboard's clock (the PlayHQ scorer's or yours).
Repeat	Play once · until stopped · until play resumes · until N minutes before the start	A break sequence loops until the clock starts again, finishing the step on screen.
Then	Go to a scene · turn off what it turned on · leave everything · play another sequence (one for each output it plays on)	Breaks go back to Live game; Run-up hands over to Walk out.
Music	None · a song or playlist from a step, from a start mark, with a fade in; ends by fading, cutting or carrying on into the next sequence	Warm-up music through the run-up that carries into the walk out.

A sequence that starts by itself waits a few seconds as **Starting**, with **Start now** and **Cancel**: press **Cancel** to hold the break back for a shot after the siren and start it by hand when the ball is dead, or **Stop** (or a scene) to come back from a break early. **Automatic sequences on** above the cards is the one switch for all of that, and it is remembered. Only one sequence plays at a time on each output; going to a scene, playing another sequence or hiding everything stops it without its Then. A sequence only ever turns off sources it turned on itself.



While a sequence plays its card shows a dot per step, the time left on the step and for the whole sequence, with Next, Pause and Stop.

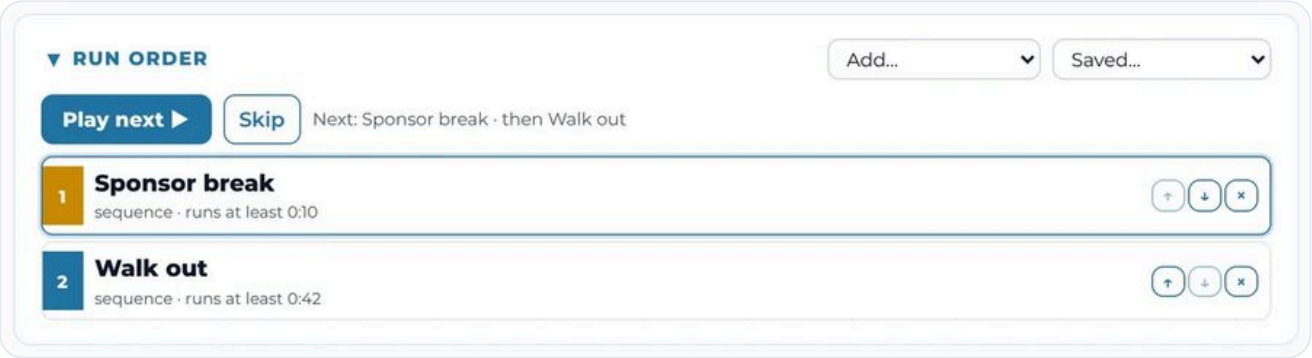
Sequences belong to the competition in use and change with **Use this one**; the sources and scenes are shared. The break lengths on the setup page (quarter time, half time, three quarter time) let the editor say how many times a break sequence fits in each break.

THE RUN ORDER

Above Scenes and Sequences sits the **Run order**: tonight's plan as a numbered list of sequences and scenes, in the order they play. **Add…** puts a scene or sequence at the end (Run-up, Walk out, Live, Quarter time, Live, Half time…), rows drag or move with their arrows, and × takes one out. **Play next** plays the head now, a sequence or a scene, and **Skip** drops it without playing it (stepping through a sequence is the Sequences panel's own Next). A sequence that ends looks at the run order first and plays what is queued after it; its own **Then** runs only when nothing is queued, so Then is the fallback and a sequence built for one night still works alone on another. Whatever is at the head

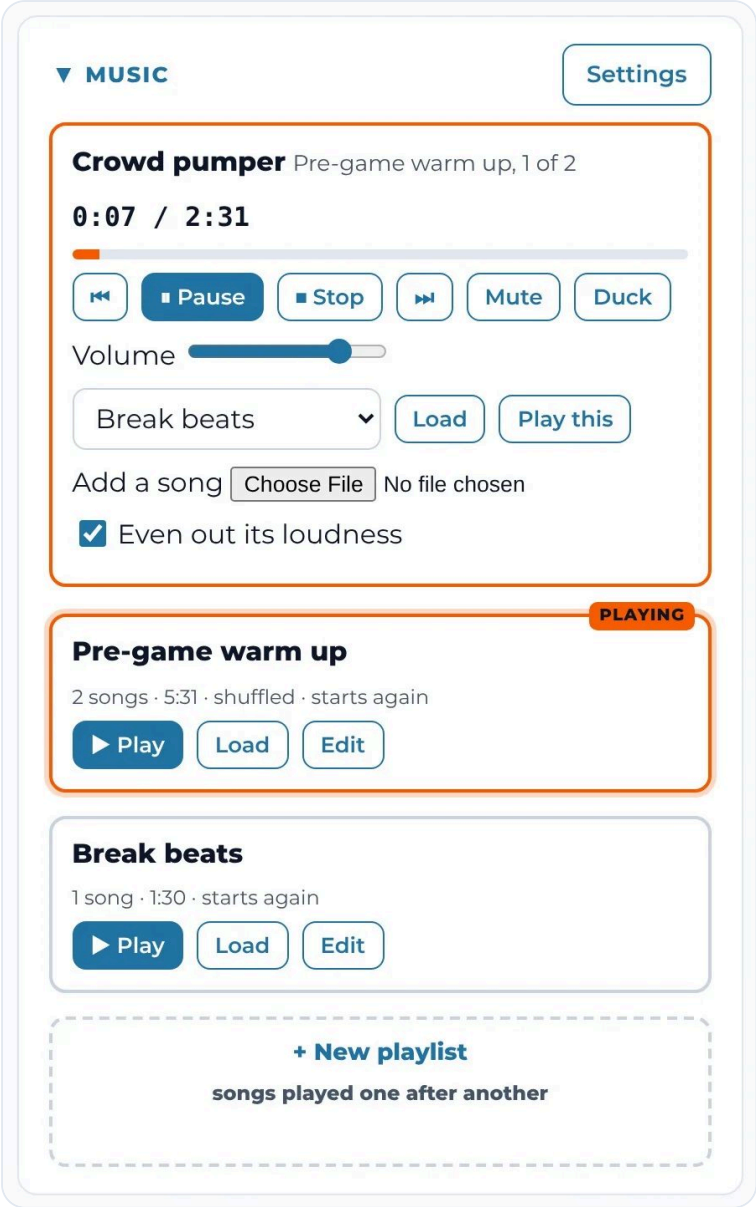
leaves the list as it starts, however it was started. A sequence that starts by itself from the game is put at the head with its few seconds' countdown rather than fired blind: **Cancel** takes it out again and **Play next** plays it when the ball is dead. Playing a scene or sequence by hand leaves the list alone unless it was the head, and nothing in the list starts by itself. The status board's **Next** line and the chip in the bar say what Play next will do. Each output has a run order of its own (pick its tab). A sequence that plays on more than one output says so on its row, "also on Commentary · then Commentary loop": **Play next** starts it on all of them, as its card does, and after it each output follows its own run order, or its own **Then** when that's empty. When the run order moves on by itself, another output joins the next sequence only if nothing is playing there; one that was playing the sequence that just ended does its own Then.

Saved… keeps tonight's order under a name for next week, loads one in place of what is queued, deletes one or clears the list. The run order and the saved orders live with the competition's sequences, so a relaunch picks the night up where it was. On the Stream Deck, the plugin's **Run order** key plays what's next, skips it, loads a saved order or clears the list, and names what plays next in that scene's or sequence's colour so you can see what a press will do; without the plugin, `/deck/runorder/next` plays what's next and `/deck/runorder/skip` drops it. The Sequence key's own Next still ends the step on screen.



Tonight's run order: Play next plays the Sponsor break, then the Walk out is next. The head is framed in blue while it waits its turn.

- **Music, playlists and mics**

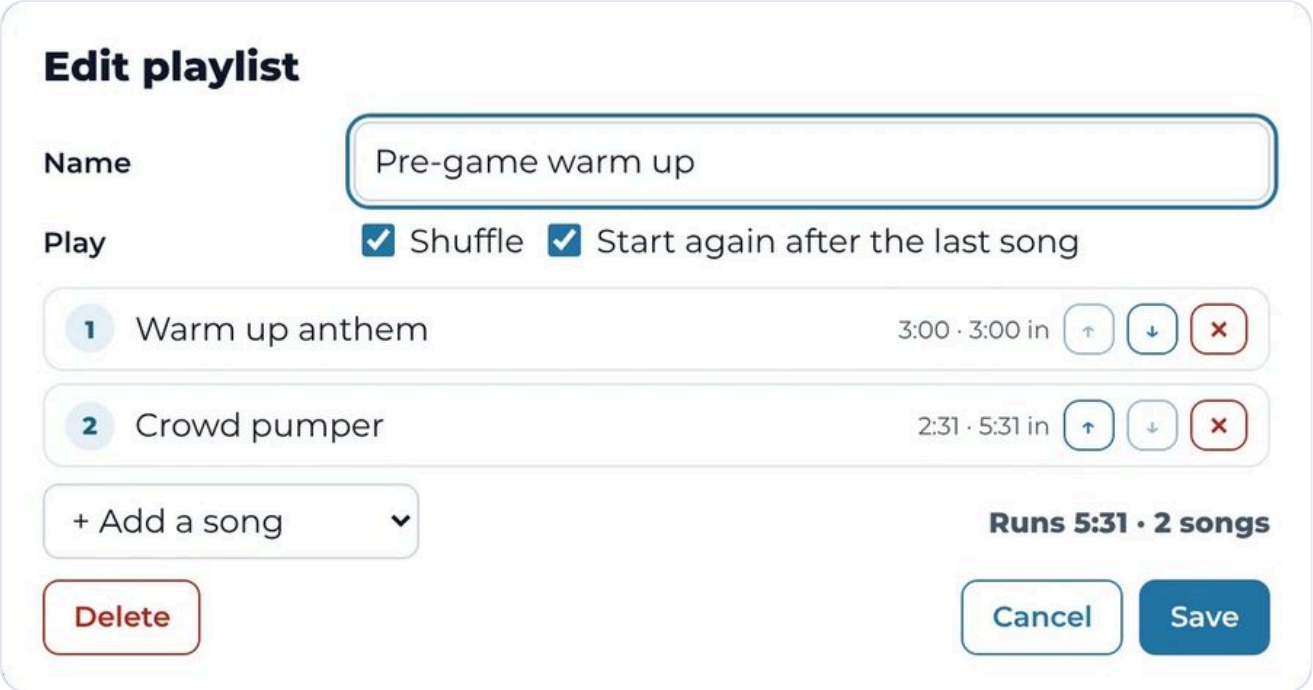


The Music player, two playlists, and the settings for fades, ducking and what happens under a video with sound.

The Music player plays songs from the media folder (MP3, M4A, AAC, WAV or AIFF; use music you're licensed to stream), one song or a playlist, shuffled or not. Add a song with **Even out its loudness** ticked and it is levelled to match the others. Music plays wherever the video sound plays: through the output, or in the Scene View while Video output is off, never twice.

- **Play** fades in; **Pause** and **Stop** fade out (slower by default). **Duck** lowers it and keeps it running; **Mute** keeps it running silently.
- While a video with its sound on plays, the music fades out and pauses, then fades back in from where it was (or lowers, or carries on: a setting). A themed intro or outro does the same.

- A sequence can bring its own music, and a sequence inside it plays its own while it runs. Going to a scene fades a sequence's music out.
- The **super shot song** cuts in when the super shot period starts in a super shot division, loops, and fades out when the period ends.



A playlist: songs in order, shuffle, and whether it starts again after the last.

Mics. Wireless effects mics on court and crowd mics go out with the music: plug their receivers into the Mac (directly or through an audio interface) and add each one under **Mics**, with its input and channel (two receivers on one stereo input are two mics, left and right). Each mic's **Goes to** picks the outputs it's mixed into (Output 1 to start), heard while they're on, never in the Scene View, so a speaker near a mic can't feed back. The first time, the Mac asks whether FirstPass Studio may use the microphone.

- A mic can be plugged into the switcher instead (an ATEM Mini's Mic 1 or Mic 2): the switcher mixes it and Studio moves its fader for the same level, mutes and ducks. Move the fader on the switcher and that becomes the mic's level; Studio never moves it back.
- Each mic has a level, a mute, a **Low cut** for wind and rumble and a meter; **Level** and **Mute all** cover every mic, and a limiter stops a shout clipping the stream.
- The mics **duck** while music plays, while a video with its sound plays and during a scene that quiets the music (each a tick box), and by hand with **Duck**. A mic marked **Has an external mixer** never ducks: the mixer does the fading.
- Stream Deck: `/deck/mics/mute/toggle`, `/deck/mics/duck/toggle`, `/deck/mics/up`, `/deck/mics/down`, and one mic with `/deck/mic/court-mic/mute/toggle`.

Video output

Settings over the picture opens the **Output settings**, which send the scene to your switcher. Pick one of three routes, then press **Output on** in the row over the picture. While the output is on its settings are locked; turn it off to change them. `/deck/output/on`, `off` and `toggle` do the same from a Stream Deck, and the outputs go dark when Studio goes off air.

ROUTE	WHAT IT NEEDS	WHAT THE SWITCHER GETS
Display	One or two extra displays on the Mac (HDMI into the switcher), set to exactly 1920 × 1080, not HiDPI or scaled	**Chroma**: the scene over a solid colour on one display (green by default, blue or any colour). **Fill and key**: fill on one display, key (white where the scene shows) on the other. A plain M1 drives one extra display; Pro and Max chips drive more.
Blackmagic	An UltraStudio or DeckLink with Blackmagic Desktop Video installed, and the device free	**Fill and key** through the device's external keyer (fill on one SDI out, key on the other), or fill only, at the switcher's 1080p format (1080p50 is picked first). The fill is premultiplied: turn the switcher's Pre Multiplied Key on for clean edges, and leave Straight off (it leaves a light fringe round logos and text). **Sound** goes in the SDI with the fill (48 kHz stereo, in time with the picture): turn that switcher input's sound on. It can go on the Mac's sound output instead, or both.
NDI®	NDI's free runtime for Mac, and a wired network	Fill and key as two sources, chroma as one, or one source with transparency for receivers that key on NDI's alpha. They appear as this Mac's name with the output's name in brackets (or a name of your own).

HD OR 4K

The **HD** and **4K** buttons over the picture set the picture for every output at once. In 4K each NDI and Blackmagic output sends 2160p at the frame rate you picked for it (2160p50 for 1080p50). Nothing else changes: scenes, sources, Stream Deck keys, links, NDI names and the switcher key stay as they are, and pressing HD puts everything back exactly as it was.

- **Sharp where Studio draws.** The graphics keep their 1920 × 1080 layout and are drawn at twice the density, so text, shapes, logos drawn by Studio and the animations' sparks and confetti are sharp. Pictures and videos are only as sharp as their files: a 1080p video or a small logo looks softer.
- **Only for 4K gear.** An HD switcher can't take 2160p. A Blackmagic device without the 2160p format (an HD UltraStudio) stays at 1080p and its card says so; a display is left alone, since a 4K screen set to look like 1920 × 1080 already draws at 4K.
- **Pick a 4K format directly.** In 4K a Blackmagic output's video format list also has the device's own 4K formats, such as 2160p50. In HD they're hidden, and the output sends the 1080p format at the same rate.
- **Any graphic's link works at 4K too.** Opened in a browser source or window set to 3840 × 2160, a graphic draws itself at twice the size, with nothing added to its link.
- **Check the Mac keeps up.** A 4K output is about four times the work of an HD one; the performance line under the output says whether it's keeping up.

MORE THAN ONE OUTPUT

One output is all most nights need. To give a camera its own graphics, press **Add an output** beside Output on (up to four). Each output is its own picture with its own route and its own on/off, so a second output can feed an upstream keyer tied to a camera and its graphics cut with that camera, while Output 1 stays the downstream key over everything. Name an output in its card ("Wide camera", say) and that name is used everywhere, including its Stream Deck links.

- **Sources are shared, on/off is per output.** Each source in the list has a coloured pill per output (orange, pink, blue, green for Outputs 1 to 4). Press a pill to show or hide that source on that output only. A new output starts with everything off. The **Show** filter above the list narrows it to one output.
- **Every output plays its own sound.** Each plays the sound of the videos and animations on it. The music plays on the outputs ticked in the music player's **Music plays on** (Output 1 to start), and a sequence's music on every output playing the sequence. Each mic's **Goes to** picks its outputs, so a commentators' feed can have the effects mics without the commentators' own voices.
- **Scenes and sequences are shared; each output plays its own.** Pick an output's tab over the Scene View and clicking a scene or sequence plays it on that output (the sections say which). **Hide all** clears that output only (or the one the Sources list's **Show** is set to), and says which, so clearing a monitor never blanks the stream. Each output has its own current scene and plays one sequence at a time, so the same scene or sequence can be on two outputs at once; coloured dots on scene cards, and on sequence cards each output's tick with how the sequence is there (Playing, Paused, Starting or Stopped (it ended there, played out or cut short); a hollow dot and "off" while that output isn't sending), show where each one is on. A sequence's **Plays on** ticks (on its card and in its editor) pick the outputs it plays on together: started by itself, or played on any of them, it starts on all of them at once, and each output has its own **Then** for when it ends there, so a quarter-time sequence can go back to the live scene on Output 1 while the commentary monitor goes on to its stats loop. Played on an

output it isn't ticked for, it plays there alone (`/deck/output/2/sequence/<name>/only` does that on a ticked one). Stop on a card stops it on that output only. Remove every output it plays on and the sequence stops starting by itself until you tick another, so it never lands on air by surprise. A Sponsor ads step off Output 1 plays its ads in that output's own ad sources. `/deck/scene/<name>` and `/deck/sequence/<name>` play on Output 1; `/deck/output/2/scene/<name>` , `/deck/output/2/sequence/<name>` , `/deck/output/2/scene/next` and `/deck/output/2/sequence/stop` (and the rest) play on Output 2.

- **Sponsors count on Output 1, and on outputs you tick.** An extra output's card has **Counts for sponsors**, off to start. Tick it only when everything on that output is seen on the stream, like a second downstream key. Leave it off for a camera's upstream key or the commentary monitor: Studio can't tell when they're on air, and the report shouldn't claim time nobody saw. Ads there show whose turn it is without using it up, so the order on air stays fair.
- **The Scene View has a tab per output**, coloured to match, so you can check what each one is sending.
- **Routes can't be shared.** Two outputs can't use the same display, the Blackmagic device or the same NDI name; the earlier output keeps it and the later one says so on its card. Two HDMI displays give either two chroma outputs or one fill and key output. Several NDI outputs can run at once, each with its own name, which suits feeding a switcher's inputs over the network.
- **Stream Deck.** `/deck/output/2/toggle` (or the output's name) turns Output 2 on or off, `/deck/output/all/off` turns them all off, and `/deck/output/2/layer/<source>/toggle` shows or hides a source on Output 2. Source settings lists these links for each output.

EACH OUTPUT COSTS

Every extra output runs its pages once more on the Mac. Try the number of outputs you plan to use on your own Mac before a stream night: while an output is on, its card shows its CPU, memory and frame rate (amber when it can't keep up), with Studio's total under the status, and the log notes them once a minute.

A COMMENTARY MONITOR

Commentators can have graphics of their own: a monitor at the commentary desk with the live stats they talk to, never seen on the stream. The **Commentator stats** page is made for it: one full screen with both team lists on the left (each team's time since its last goal, and each athlete's games played this season), the game stats big in the middle (each team's logo with its ladder position, record and form under it, head to head under the names, then shooting %, quarters won, best quarter, super shots, runs, biggest lead and lead changes), down the right the game clock with the super shot window, who is on the mic (from the Production Planner) and up next, then a list of the latest moments (lead changes, comebacks, super shots, hot streaks), newest at the top, each with its quarter, game time and how long ago it happened, and across the bottom the last 10 goals, the current run in big numbers and the score worm. Its stats

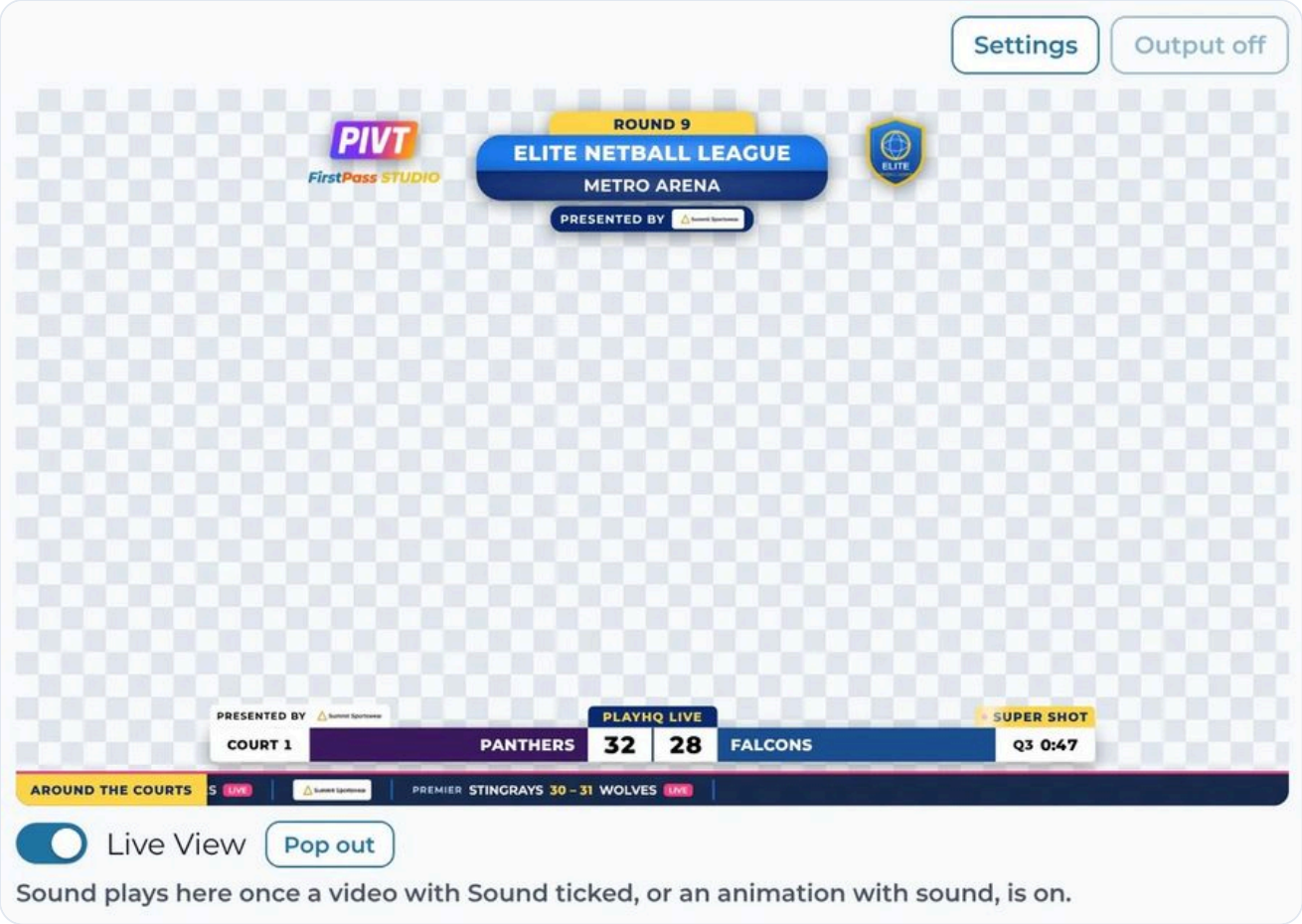
refresh themselves every 15 seconds and when the score changes (untick **Auto-refresh** in its options, or pick another frequency), and the clock, moments and goals follow the game as it happens. Other graphics can go on the commentary feed too: the last 10 goals and the scoreboard follow the game live, and the rest show what they read each time they come on.

- **On the same Mac.** Add an output for the commentary monitor (a display, or NDI to a screen on the network) and turn sources on for it with its pills or its Stream Deck keys, such as `/deck/output/commentary/layer/ladder/toggle`. Play scenes and sequences on it from its Scene View tab or its own Stream Deck links while the stream carries on as normal.
- **On a second Mac.** Run Studio on another Mac with the same competition and game, and Follow PlayHQ on. It reads PlayHQ's live scoring for itself, so it needs nothing from the stream Mac. Its Output 1 goes to the commentary monitor, and everything on it is for the commentators.

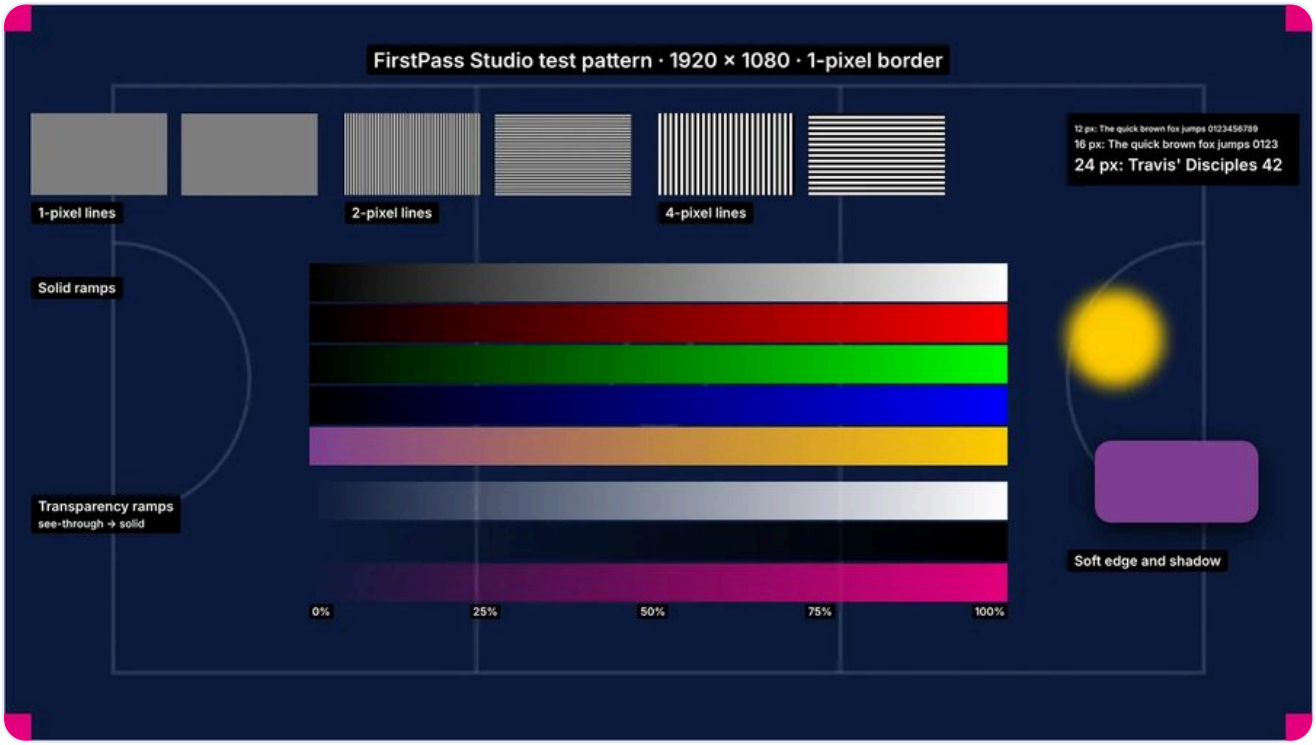
On the commentary output (or the second Mac), make a **commentaryLoop** sequence that steps through what commentators find useful (game stats, shooting stats, the last 10 goals, the ladder) with **Repeat** set to until stopped, and give the commentators a Stream Deck of their own:

- A key per graphic they want on demand, each going to a scene with that graphic (on the same Mac, `/deck/output/commentary/scene/ladder`). Going to a scene stops the loop on that output only, so their pick stays up and the stream is untouched.
- A **Loop** key to go back to looping when they're done: `/deck/output/commentary/sequence/commentaryloop` on the same Mac, or `/deck/sequence/commentaryloop` on a second Mac. With the FirstPass Studio plugin, use Scene and Sequence keys with **Output** set to the commentary output; they light pink while it's on.

Each output's sound plays through the Mac's audio output, except on a Blackmagic output, where it goes in the SDI with the fill (or on the Mac too, as its **Sound** says). The **Scene View** on the Output page shows the scene as it is right now, on a checkerboard so see-through parts show; **Pop out** opens it in its own window to move to another screen, with a floating list of On/Off buttons. The Scene View plays the sound of the output it shows only while that output is off, so it never plays twice.



The Scene View: the scene right now. Untick Live View on a busy night, since it runs the scene's pages a second time.



The **test pattern** (Add page) checks an output and the switcher's key end to end: a 1-pixel border, fine lines that shimmer if anything is resizing the picture, ramps for banding and soft shapes for the key. Not for air.

CHROMA KEY SAFE

A chroma key cuts out one colour, so anything half see-through fringes. With a chroma output picked, **Chroma key safe** is on: every graphic goes out with solid panels, no glows or soft shadows, and fades become cuts. Pick another output and everything is back as it was. The Output page also lists any competition or club colour close to the key colour.

THE SWITCHER

Studio can also drive a Blackmagic ATEM switcher over the network, so the camera cuts can be pressed from the Output page or a Stream Deck beside the graphics. Open the **Switcher** card on the Output page, press **Set up** and **Look for switchers**: Studio finds the ATEMs on the network the way ATEM Software Control does (macOS asks the first time to let it use the local network) and remembers the one you pick by name. Typing its address works too.

- **Each output's key, checked.** In each output's setup, **Into the switcher as** says which switcher key its graphics come into (usually downstream key 1 for Output 1, upstream keys 2 to 4 for the others, an upstream key for chroma) and which inputs its fill and key are plugged into. Studio reads that key's settings and lists anything that doesn't suit the output, such as the wrong fill source, Pre Multiplied Key off, or Straight ticked on a Blackmagic output. **Set up this key for me** fixes just those, after asking, and leaves the key on or off air as it was.
- **Graphics that go with a camera.** An output on an upstream key can tick the cameras it **goes with**, say a bench graphic with the bench camera. Studio's cuts, autos and stingers then bring its key on with those cameras and take it off with the others, on the same frame. With **Keys follow the camera** on, a cut on the switcher's own panel brings the key a frame or two later. A key turned off by hand stays off until the camera changes. The output's card says **On air with Bench** or **Waiting for Bench**.
- **Inputs in tally colours.** Each input shows by its name from ATEM Software Control, red on program and green on preview. Click one to put it on preview, then **Cut** or **Auto**. Fade to black and each upstream and downstream key are a press away too.
- **Only what your switcher has.** An ATEM Mini or Mini Pro has 4 inputs, one upstream key and one downstream key, so fill and key into the downstream key leaves two cameras; a Mini Extreme has 8 inputs, four upstream keys and two downstream keys. Studio offers what it finds, and says so when a key or link asks for something the switcher doesn't have.
- **Your hands stay on the switcher.** Its own buttons, ATEM Software Control and other controllers keep working, and the card follows them. Studio sends nothing by itself, not even on connecting.
- **If it drops off the network, or is off when Studio connects,** the card turns red within about five seconds and Studio keeps trying. Presses meanwhile are refused rather than saved up, so a cut never lands late.

- **Stinger cuts.** Add the Stinger page as a source and the card's **Stinger** button plays it and cuts the switcher underneath, right in the middle of the bands. It plays on the output on a downstream key, which is Output 1 unless another output that's switched on is. If the cut shows before the bands do (an output's picture can reach the switcher a few frames late, more over NDI), set the **line-up** in frames and press **Test the line-up** before you go live. Each output keeps its own line-up.
- **Camera cuts in sequences and scenes.** A sequence step's 🎥 button, or a scene's 🎥 **Switcher** under Edit scenes, adds switcher actions: cut, auto or stinger to a camera, put one on preview, a key on or off, fade to black, or run a macro, as the step starts or a set number of seconds in. So the walk-out sequence can cut to the bench camera as the line-ups come up and bring the scoreboard key back as it ends, with nobody on the switcher. They happen only on Output 1, and if the switcher isn't answering at that moment the action is skipped and logged rather than done late. You can always cut by hand on the switcher itself.
- **Stream Deck.** The plugin's **Switcher** key puts a camera on preview or program, plays a stinger, cuts, autos, turns a key on or off or runs a macro, and lights like the switcher's own buttons: red on program, green on preview. While the switcher isn't answering it goes dark with a yellow dot. Without the plugin, the links are `/deck/switcher/cut`, `auto` and `ftb`, `/deck/switcher/preview/2` (or `preview/Camera%202`), `/deck/switcher/usk/1/toggle`, `/deck/switcher/dsk/1/auto` and `/deck/switcher/macro/1`.

A night, start to finish

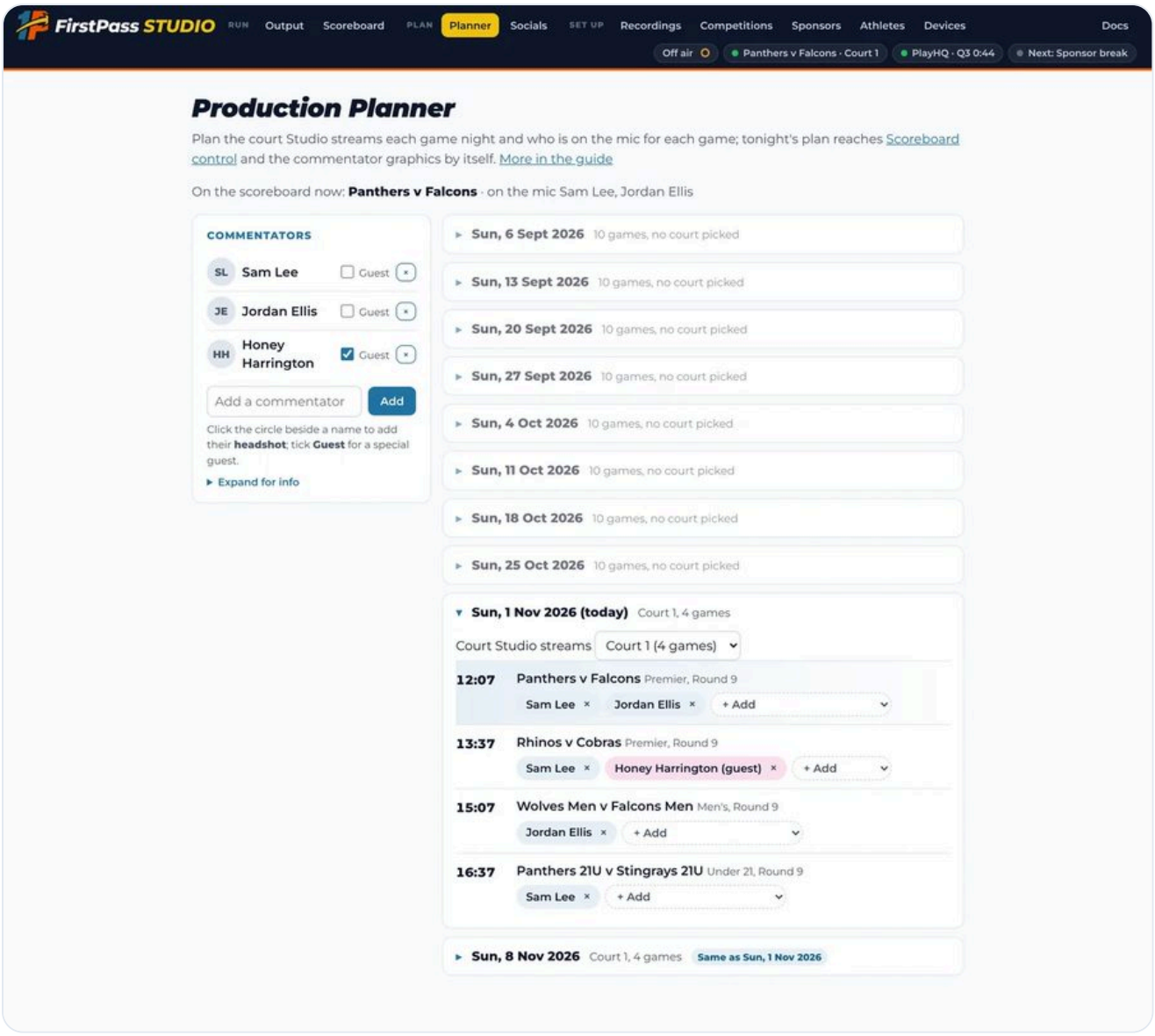
With the sample league's scenes and sequences, a solo operator's evening looks like this. Everything in the right-hand column happens by itself.

ARRIVE	Open the app. Studio goes on air, picks tonight's first game on Court 1 with Follow PlayHQ on. Turn the video output on and check the test pattern on the switcher.
15 MIN OUT	**Run-up** starts by itself: the Pre-game scene, then the sponsor wall, match up and ladder in turn over the warm-up playlist, repeating until 2 minutes before the start.
2 MIN OUT	Run-up hands over to **Walk out**: The Clash smashes the two logos together, the position match ups run through GS to GK, the countdown holds until the clock starts, then the Live game scene comes up and the music fades.

Q1 TO Q4	The scoreboard follows the PlayHQ scorer's score, quarter and clock. Super shots, hot streaks and moments play by themselves. The super shot tab and song mark the last five minutes of each quarter. The operator watches the stream and, if they like, keys up a ladder lower third or a lower third for a commentator.
QUARTER TIME	**Quarter time** starts a few seconds after the siren: shooting stats, the last 10 goals and the ladder, then two sponsor ads and the sponsor card, over the break music, round and round until play resumes, then back to Live game.
HALF TIME	**Half time** does the same with Up next and the ladder lower third added, and the house ad if no sponsor has one due.
FULL TIME	**Full time** starts after the final siren: the winner celebration, the full time card, game stats, the score worm, a sponsor break and Up next (which becomes the end-of-night card after the last game).
NEXT GAME	Automatic selection moves the board to the next game on Court 1 as its electronic scoring opens, with the quarter back to Q1 and a full clock. Run-up starts again 15 minutes out.
CLOSE	Press **Stream close**: the Clear scene, then the outro with the night's teams. Turn the output off. The sponsor report has the night's exposure ready to send.

The Production Planner

The **Production Planner** tab plans the season ahead: which court Studio streams each game night, and who is on the mic for each game. Studio streams one court a night; a second court is a second Mac running its own Studio, with its own plan.



The Production Planner: the commentators down the side, and the season's game nights from PlayHQ, the next one open, with each game on the night's court and who calls it.

COMMENTATORS

Add the people who call your games to the list on the left. Tick **Guest** for a special guest. The list belongs to the competition; as you type a name, people from your other competitions are suggested, so someone who calls in two leagues is added in a tap. Click the circle beside a name, or drop a photo on the row, to add their **headshot** for the Socials tab's Commentary team picture; the small × on it takes it off again.

A NIGHT

Every game night in the season is listed, read from PlayHQ's fixture, by date rather than round. Open one, pick the **Court Studio streams**, and its games on that court appear in time order. Tap **+ Add** on a game to put up to three people on the mic; tap × to take someone off. Everything saves as you go. A night you haven't planned takes the plan of the night before it (the same court, and the same people game by game), marked **Same as** that night, so a season with a regular team is planned once and only the

nights that differ need changing. **Same as the night before** drops a night's own plan again.

WHEN THE FIXTURE CHANGES

PlayHQ's fixture changes through a season: a game moves court for availability, or is refixtured to another date. The commentators belong to the time on the court, because that is where the desk and cameras are set up, so they stay put and the game is flagged **Fixture changed** with what was there before. A game that has gone from the court keeps its row, so its commentators can be moved to wherever it went.

ON THE NIGHT

When the night arrives, the plan's court goes to Scoreboard control, which follows it game by game as usual. The **Commentators** graphic, with no names typed in its options, names whoever is planned for the game on the scoreboard, and says **Special guest commentators** when everyone on the mic is a guest. **Commentator stats** shows them under the clock, and the **Fixture board** full page puts the whole night's fixture on a screen in the competition room or backstage, with a camera on the streamed court's games and who is on the mic for each: their headshots (or a silhouette) with their names, under an **On the mic** tab beside the teams. Manual control always wins: if you change the court or pick a game by hand on Scoreboard control, the Planner says so at the top, so the commentary team can be moved to match. Swapping someone in the Planner takes effect the next time the graphic comes on.

Socials

The **Socials** tab turns a game night into pictures for Instagram and the league's other channels, from the same fixture the stream uses, so the posts and the broadcast always agree. Pick the **Game night** (by date, so every division playing that night is in it), **Upcoming** or **Results** (a night with results opens on Results), and the sizes: **Square** 1080 × 1080, **Portrait** 1080 × 1350 (the most room in the feed) and **Story** 1080 × 1920. Every picture shows as a preview in each size, in the competition's colours, font and logos.

UPCOMING PICTURES

The **Game night fixture** lists every game of the night on every court, by time. A **Match up** for each game puts the two logos side by side with the date, time, court and venue, one for each club to share. **On the stream** shows the court Studio streams, with each game's division and who calls it, from the Production Planner. **Commentary team** is a picture for each game someone calls, with that game's commentators side by side with their headshots and the game's time and teams once underneath, to share before the livestream. Someone who calls more than one game is on each of their games' pictures, so the social team can post them one at a time or grouped as they like. Long names shrink together so the row stays balanced, breaking only between words or after a hyphen; anyone without a headshot shows their initials, and a guest gets a

Special guest pill. The **Polaroid** is a tilted white print for a match photo: drop a photo on it, drag the preview to frame it and set the zoom. With no photo, its window is see-through, so the social team can put their own photo under it in Canva or a phone app.

RESULTS PICTURES

After the night, **Results** shows every finished game's final score, division by division, with the winner in the highlight colour. Each game gets a **Full time** card, with the logos and score big and who won by how much, and a **Polaroid** with the score as its caption and the division and round under it. Each division that played gets its **Ladder**, with the finals line, labelled "after" the round when it is that division's latest night with results. Each division also gets its own **Top shooters**, naming its best six that night from each game's line ups (with super shots only where that division plays them), and its **Tournament top shooters**: the best eight over every finished game so far, after this night's round. It's called **Season top shooters** when the fixture runs 15 days or more; pick either name under Game day on Competitions setup to override. On a carnival day with several rounds, set **As at** to an earlier round to count only the games up to it. Tick **Highlight movers** and each shooter shows the places they climbed or fell since the round before, even when both were played that day, with NEW for anyone new to the top eight; the tick is remembered for the competition. A game appears once PlayHQ has its final score.

Finals get their own pictures. After a division's last home and away round, Results adds **Finals bound**, the teams that made the finals with what each one's ladder place earned (a double chance, a week off, an elimination final), and the **Finals draw**, the finals week by week. On each finals night the draw is there again, under Upcoming to post before the games and under Results with the winners filled in, and the Match up, Full time and Results pictures name each final, such as Major semi final or Grand final. Once the grand final has a winner, Results adds **Champions**: the winners' logo, their name and the grand final's score. Call them **Premiers** instead under Finals on Competitions setup. A division gets these when it has a finals format on Competitions setup.

SPONSORS, THE LOGOS AND SAVING

Each picture has a **Sponsor strip** tick, which puts your naming, elite and major sponsors along the bottom and is remembered for next week. Under **Options**, **FirstPass Studio mark** (bottom right) and **PIVT logo** (bottom left) turn those on or off for every picture. **Template** sets the look of every picture: **Default** is the stream's own, and Dynamic, Junior, Retro, Professional, Thunder, Gradient, High contrast, Neon, Poster, Splash, Game Day, Hero and Final each bring their own shapes and lettering, and three netball sets draw on the game itself: Court (the court's lines and circles), Bib (a bib with its position letters) and Goal (the ring, net and post), and five sporting sets bring the kit around it: Bottle, Cones, Sneakers, Stadium and Locker room. Star (☆) the templates you like and tick **Favourites only** to list just those, and the one in use; it's remembered for the competition. Any template works with any colours, and every title stays on one line beside the competition logo, shrinking a little where it would otherwise wrap. A short ladder or a couple of games grow to fill the picture, and a Story

keeps its words and logos out of the top and bottom, where Instagram puts its own bar and reply box. A long team name shrinks to stay on one line in its row. Athletes' photos come from the Athletes page, and only for athletes ticked **Approved for photo use**. Top shooters shows each shooter's photo automatically. On a Match up, Full time or Polaroid card, pick an athlete by name, narrowing the list by division and team. Hero and Final stand the picked athletes tall, in front of the title; a cut-out PNG with a transparent background looks best. **Colours** gives every picture a different look: **Competition colours** is the stream's own, and the others swap the competition's main, highlight and accent colours around (Accent first, Highlight ground, Accent ground), or go darker (Night), paler (Day) or down to two colours. **New colour set** makes your own from three colours, for a finals night or a sponsor round. A set marked **low contrast** has titles that may be hard to read. Your choice is remembered for the competition. Untick anything you don't want (**Select all** and **Select none** tick or untick every picture at once), then **Save pictures**: they are saved as PNGs in a folder for the night in **Pictures › FirstPass Studio**, which opens in Finder. **Share…** saves them too and opens the Mac's Share menu, to AirDrop them to your phone or send them by Messages or Mail. Tick **As a zip** to also save them as one .zip beside the folder; Share then sends that one file, which is easier to email or upload.

Recordings

For each PlayHQ game on the scoreboard while on air, Studio keeps a recording of what PlayHQ said and what went to air, timed: the live score, clock and quarter, the line ups, every scoreboard and Stream Deck press, and every change to the scene and the graphics. There's no video or sound in it, only what the graphics need, so a game is a megabyte or so. A recording starts when its game is due on the scoreboard, 10 minutes before the scheduled start unless automatic game selection on the competition setup page says otherwise, and stops 10 minutes after PlayHQ says the game is finished (change it with **Keep recording for … minutes after full time**), when the scoreboard moves on to the next game, or when Studio goes off air. Being on air between games, or while you set things up, records nothing. The **Recordings** tab lists each game's recording, with **Download** to keep a copy and **Remove**, and says when the next one starts. Recording is on unless you untick **Record each PlayHQ game on air**; the Output page's status board has the same switch as **Record** and **Stop**. A recording older than 60 days is tidied away. Recordings stay on this Mac and hold nothing secret.

PRACTICE GAMES

Press **Practise** beside a game to play it back as if it were live. Studio doesn't ask PlayHQ anything while it runs: the recording answers, with the game's date and times moved to now, so the game starts a few minutes from now (5 to begin with: set **A practice game starts in** above the games, from 0 to 60 minutes, for time to run a pre-game sequence) and everything works as on a real night. The scoreboard follows the score and the clock, sequences and quarter-break triggers fire, and the outputs can be on. At the top of the Recordings tab, **Jump to** goes to the start, any quarter or full time, **Pause** stops the clock where it is, and **Stop** ends it and puts the scoreboard back exactly as it

was. A red strip on every Studio window says a practice game is running. Nothing in it counts for your sponsors, and it isn't recorded as a game of its own. Use it to rehearse before the season, try a new sequence, or show Studio to someone.

4K REPLAY

Streamed a night at 1080p and recorded the camera in 4K? Press **Replay** beside the game and Studio puts that game's graphics back on every output, exactly as they went to air. Turn on **4K** on the Output page and they go out at 2160p, sharp, to key over the 4K recording in a 4K switcher or recorder, ready for a YouTube upload. Studio doesn't play the video. The replay starts paused at the game: pause the video on a moment you can see, such as the first centre pass, pick the same moment from the list of what went on air and press **Jump there**, then press Play on both together. **Nudge** moves the graphics a frame or a second if they drift. The live scene and scoreboard are kept as they were while it runs, and **Stop** puts them back. Sound comes from the HyperDeck's 1080p programme recording.

With the 4K video on a HyperDeck, set **Follow** to **A HyperDeck's timecode** and type its IP address. The graphics then go wherever the HyperDeck is: play, stop or scrub it and they follow, frame for frame, with nothing to line up by hand. If its clock was a little off on the night, stop it on a moment you can see, pick the same moment and press **Jump there** once. Studio remembers that sync for the recording. It only reads the HyperDeck's timecode and never controls it.

Any other player works through the UltraStudio: send its SDI into the UltraStudio's SDI input and set **Follow** to **Timecode into the UltraStudio (SDI)**. Studio reads the timecode that comes with the picture and follows it the same way. HDMI doesn't carry timecode, so it has to be SDI.

On a Stream Deck, the plugin's **Replay** key plays or pauses, jumps to the next moment, nudges or stops the replay, or a practice game, whichever is running. It lights while one runs.

DEMO GAMES

Showing Studio to another league? Set up their competition, then press **Demo** beside one of your recorded games. Pick one of their divisions and a finished game: Studio plays your recorded game as theirs, with their competition, team names, logos, colours and athletes, while the score, clock, quarters, moments and timing come from your recording. Athletes are matched by position, so your GS's goals become their GS's; anyone without a match is left unnamed and their goals go to the team. If their game has no line up, Studio uses each team's season squad, by the position each athlete played most. While the demo runs, the **Athletes** table shows who each of your athletes is shown as; change any of them there. **Save demo** keeps the whole setup by name, so before the meeting it's one click under **Saved demos**. **Swap home and away** lets either team win. The other teams in the division take the places of yours on the ladder and fixture. Play the recorded game's video wherever you like and lay Studio's output over it: the pictures won't show their teams, but the graphics will. Their competition's setup is read from PlayHQ, so a demo needs the internet.

More in the Reference: **Socials in detail**.

More in the Reference: **Recordings in detail**.

Running it by hand

When nobody is live-scoring the game on PlayHQ, or PlayHQ stops answering mid-game, every automatic part of Chapter 3 has a button. The board, the breaks and the celebrations can all be driven from Scoreboard control and a Stream Deck.

IN THIS CHAPTER

1 Two people, not one
2 Switching to manual
3 The score
4 The clock and the quarter
5 Breaks and celebrations by hand
6 Finals

Studio still needs the competition on PlayHQ for its fixture, ladder, squads and results; what changes in manual mode is who runs the score and the clock. The same competition can be followed one week and run by hand the next, and nothing on the Output page changes between the two.

Two people, not one

A followed game can be run by one person because the score, the clock and the breaks take care of themselves. A game run by hand is a different job: someone has to press a key for every goal within a second or two of it, start and stop the clock with the umpire's whistle, and still call the breaks and the graphics. We recommend two people share it:

Camera operator

- Frames and follows play on the cameras, and cuts between them on the switcher.
- Runs the game clock from the whistle, with **Start / Stop** and **Reset clock** keys beside the camera controls.
- Scores the goals as they go in, **+1** or **+2**, because they are already watching the ring.
- Nothing else: no graphics, no breaks, no music.

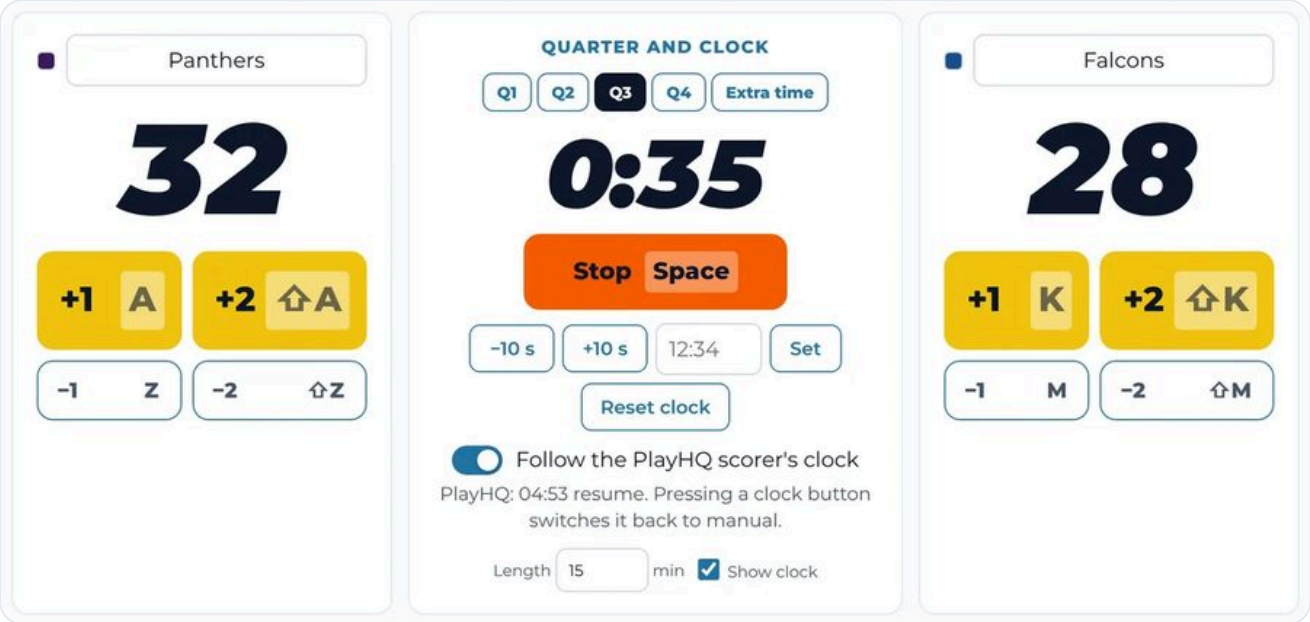
Production manager

- Keeps the Output page open and watches the stream itself, not the court.
- Corrects the score (**−1**, **−2**) and the quarter, and sets the clock when it drifts from the venue's.
- Plays the run-up, break and full time sequences from their keys, and the lower thirds, ladders and stats during play.
- Runs the music, the sponsor ads and the end of night, and picks the finals winners.

If one person must do it alone, keep the Stream Deck to the score and clock keys and let the sequences do the breaks, exactly as they do in automated mode: the game triggers follow the scoreboard's clock whoever runs it.

Switching to manual

Press **Manual** on Scoreboard control, or just press any score key: either switches the board to manual scoring and stops following the PlayHQ clock as well. Press **Follow PlayHQ** to go back; the score catches up with PlayHQ on the next check and the clock follows the scorer's again.



Score and clock on Scoreboard control. Keyboard: A and Z for the home team, K and M for away, Shift for a super shot, Space for the clock.

WHEN PLAYHQ DROPS OUT

If PlayHQ stops answering during a followed game, the scoreboard keeps its last score, the ladder, fixtures and game list keep what they last read, and the debug log records each failed read. Switch to Manual and run the board by hand; Follow PlayHQ picks it up again when PlayHQ is back, and corrects the score to the scorer's.

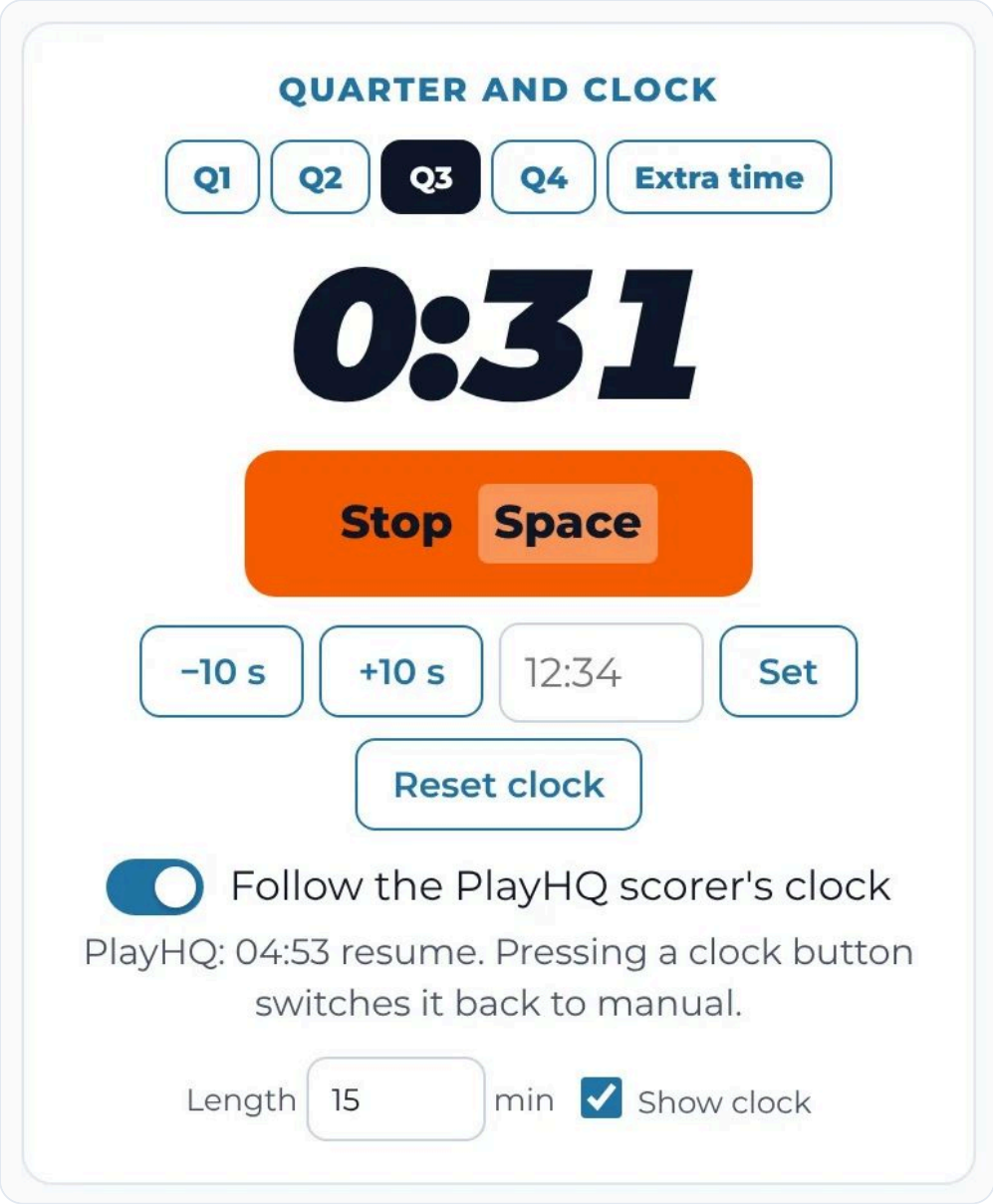
The score

Each team has **+1**, **+2**, **−1** and **−2**. A +2 is a super shot: it scores two and plays the super shot celebration, so a +2 pressed by mistake is fixed with −2 (which plays nothing). **Reset scores to 0** clears both. The hot streaks and game moments still play in manual mode, because Studio works out the goal order from the presses just as it does from PlayHQ's score.

KEY	WHAT IT DOES
Home +1 / Away +1	A goal.
Home +2 / Away +2	A super shot: two points and the celebration.
−1, −2	Take a goal or a super shot back, quietly.
Reset scores to 0	Both scores to zero, for the next game.
Super shot (home / away)	The celebration only, score unchanged: for a test, or a super shot that was scored as +1 +1.

The clock and the quarter

Start and **Stop** are one key on the Stream Deck. **Reset clock** puts a full quarter back (the competition's quarter length; a key can set any time). **+10 s** and **−10 s** nudge it, and a key can add or take off any amount. Type the venue's clock in **Match venue clock** to line the two up, and untick **Show clock** for a game with no reliable clock at all. Quarters are **Q1** to **Q4** plus extra time, or H1 and H2 for a competition played in halves; **Next quarter** steps through them. Changing the quarter doesn't touch the clock, so make the quarter key a Stream Deck Multi Action that also resets the clock.



Quarter and clock. The super shot tab, the super shot song and the break sequences all read this clock, so they work by hand as they do when following PlayHQ.

Breaks and celebrations by hand

A sequence that starts by itself at quarter time is started by its key too, so the production manager can run **Quarter time** the moment the siren goes instead of waiting for the clock, or run **Half time** early when a game is called. **Next**, **Pause** and **Stop** work on whatever is playing, and **Automatic sequences on** on the Output page can be unticked for a night where every break is called by hand.

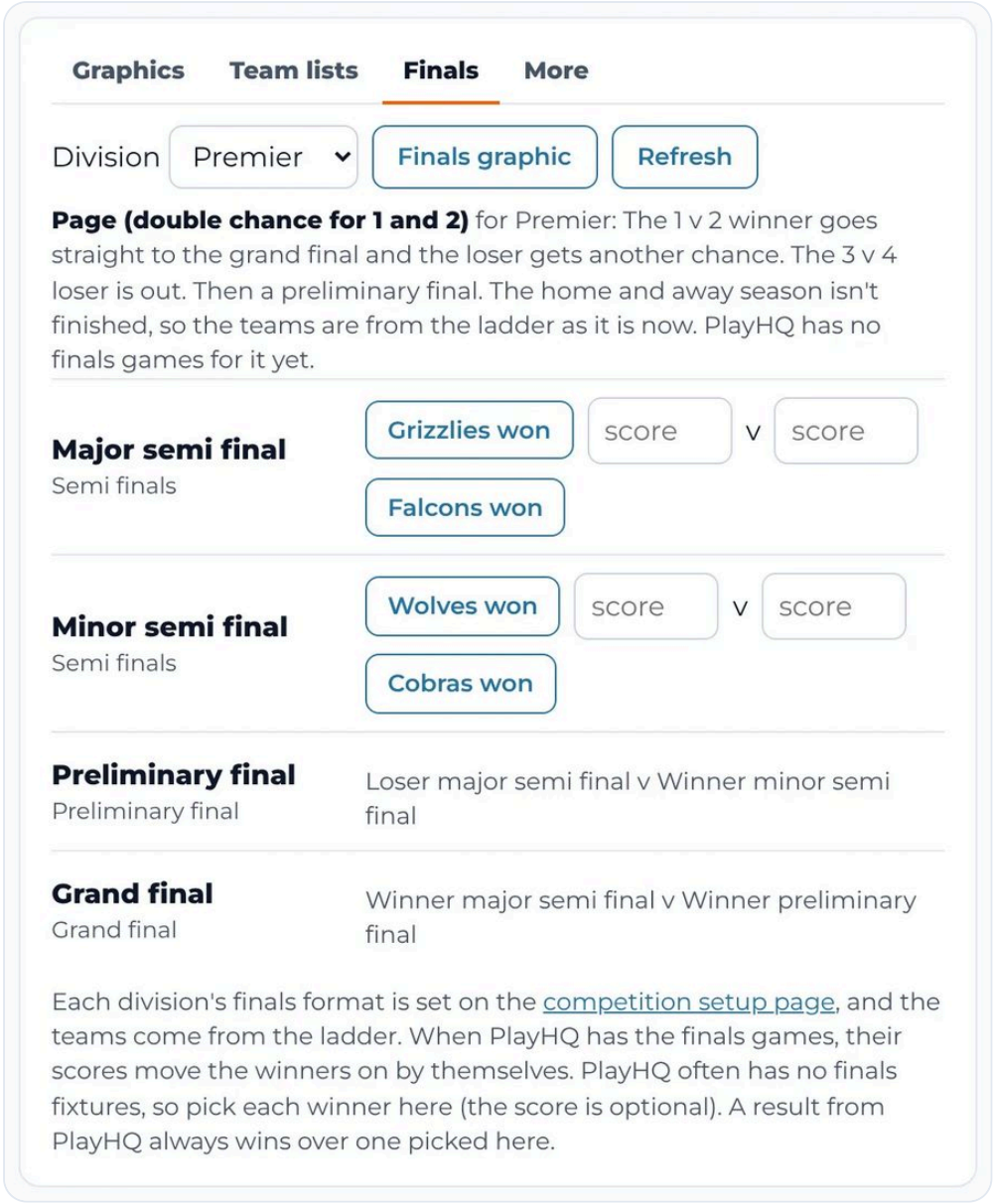
- **Winner.** One key. The celebration reads the board's score when it comes on, so it covers a home win, an away win and a draw. Check the score first. A source with **Result** set to home, away or draw forces it.
- **Super shot and streaks, once.** Besides the automatic sources, a Super shot source set to **Plays: home** or **away**, or a Hot streaks source set to one streak, plays once each time

it is turned on. Don't keep a manual source and the automatic one on together, or it plays twice.

- **The Clash.** A pre-game animation, played from its key or as a step in the walk-out sequence, with the two teams, their logos and club colours.
- **Lower thirds.** The person lower third's name box on the Output page suggests the competition's athletes; type a guest's name and title for a commentator or an official. Turn it on when they appear and off a few seconds later.
- **Stats during play.** The ladder lower third, shooting stats, last 10 goals and match up are all Source keys. Each loads fresh when shown, so it has the latest numbers.

Finals

PlayHQ often has no finals fixtures until the day. In **Finals** on Scoreboard control, pick the division, then pick each game's winner (the score is optional) as the finals are played; the finals graphic and the ladder's finals key move on with each pick. A result from PlayHQ, when it has one, always wins over one picked here.



Finals on Scoreboard control. The format for each division is set on the Competitions setup page, and the teams come from the ladder.

The graphics

Every graphic Studio draws, grouped by when it is used. Each one is added to the scene with **Add page** on the Output page, and its options appear as fields under its address; the options named here are the ones worth knowing.

IN THIS CHAPTER

1 Always on
2 Before the game
3 During play
4 After the game
5 The competition
6 The breaks
7 Checking the outputs

All of them follow the game night, game and court picked on Scoreboard control, take the competition's colours, font and logos from its setup, and fit long team names. Cards sit in the space between the title and the scoreboard and shrink to fit when they have more in them than fits. Every page, with its link, its options and a Preview button, is also listed under **Links** on the Devices tab.

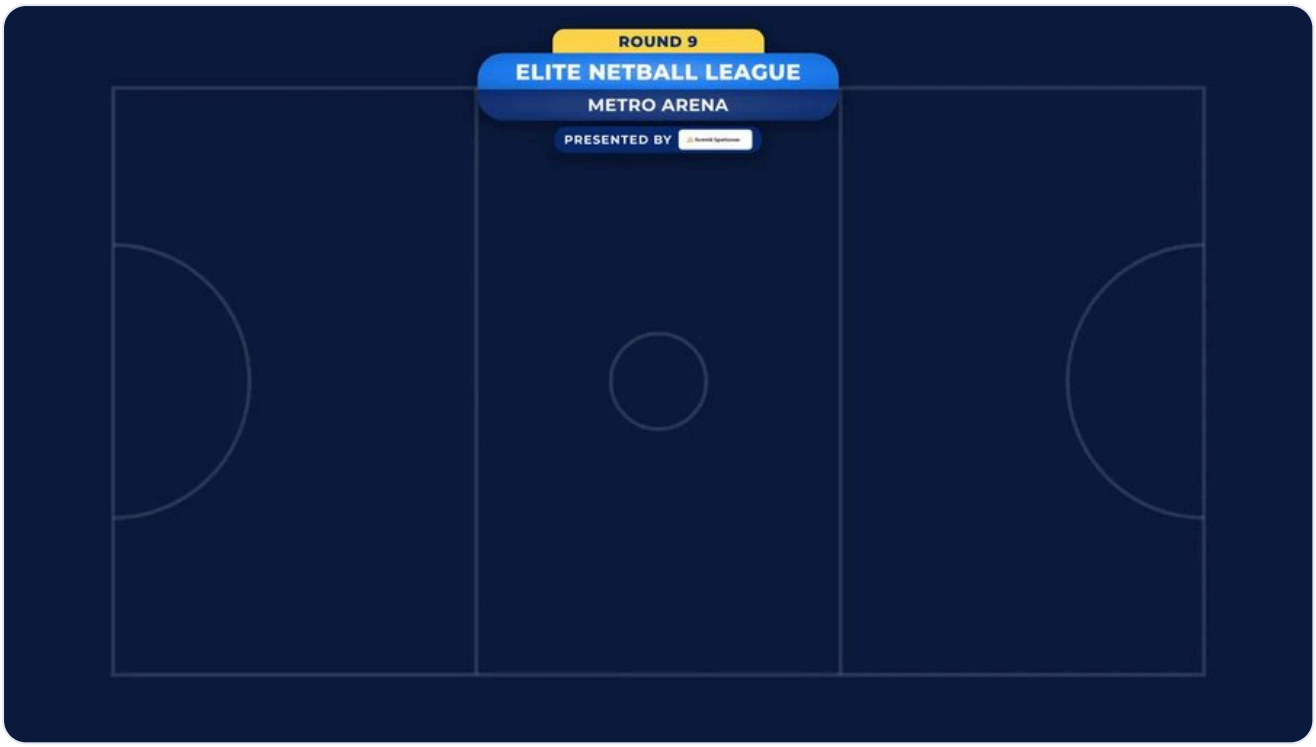
Always on



Scoreboard. Court, both teams with their logos and club colours, the score, the quarter and the clock, centred at the bottom above the ticker. Options: position (top centre, bottom left and so on), super shot tab on or off, and the naming-rights sponsor's tab.



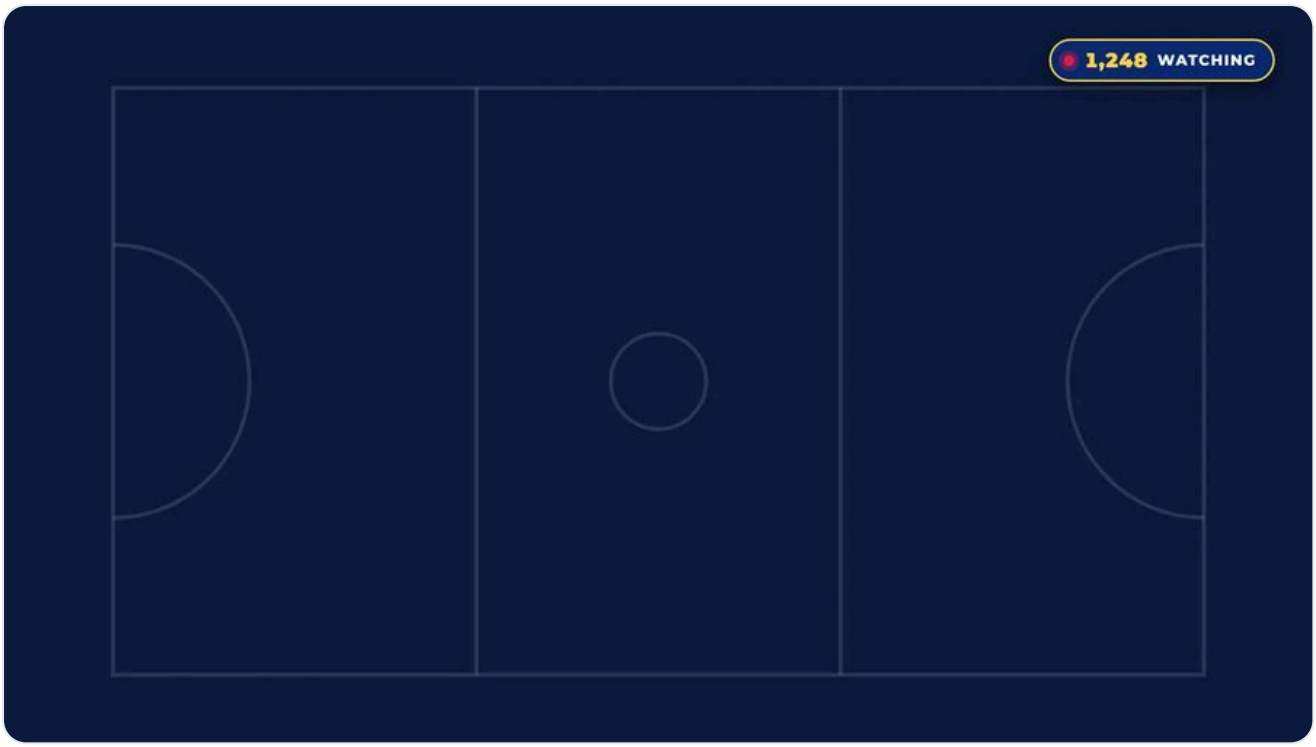
Score ticker. The scores from tonight's other games scrolling along the bottom, leaving out the game on the board, updated every 30 seconds. Speed from the control page or a Stream Deck key; a Partner sponsor's logo takes a slot in the line.



Title. Competition, round and venue in a banner at the top. Options: an emoji either side, the competition logo above it, top or bottom, and the naming-rights sponsor under it.



Branding logos. PIVT and FirstPass Studio left of the title, the competition logo right. Each can be hidden; gap, height and size adjust the placement.



Viewers. "N watching" from the competition's YouTube and Facebook streams, top right, shown only once the count passes the competition's minimum. **Shows: peak** makes it a card for the end of the night.

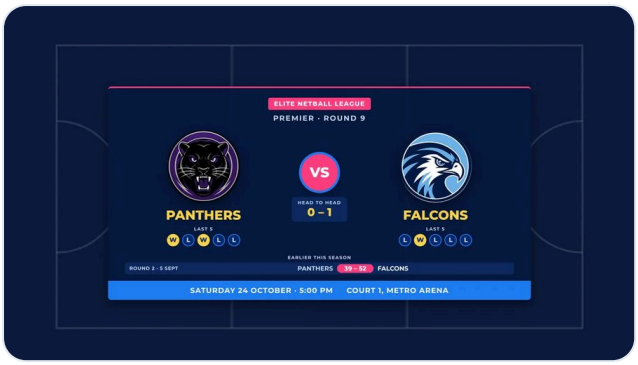
Before the game



Countdown. Counts down to the picked game's scheduled start, with the two teams. Options: heading, the words at zero, a fade over the last seconds, position and a sponsor.



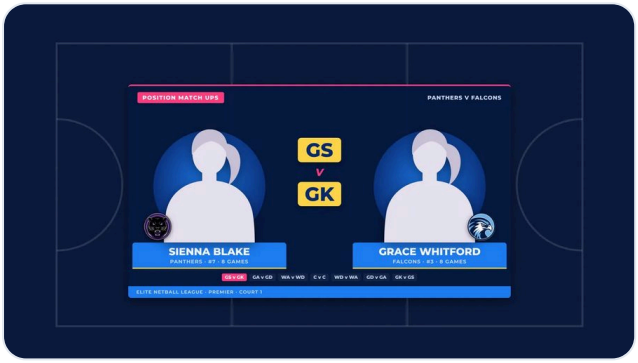
Up next. The next game on this court (or any court), with its time, teams and logos. After the last game it becomes the end-of-night card.



Match up. Both teams side by side with their last five results as form dots and this season's head to head. **Form** and **Head to head** can each be hidden.



The Clash. The two teams slam together behind a big word (IMPACT, CLASH, SHOWDOWN, your own) in their club colours. Options: the word, the bottom line, animate away after N seconds.



Position match ups. Who marks whom, position by position, with the athletes' photos from the Athletes page or a silhouette. Options: seconds per pairing, loop or stop, start with GS or GK, one pairing only, animate away at the end.



Position match ups, all seven. The same page can show every pairing at once as one card.

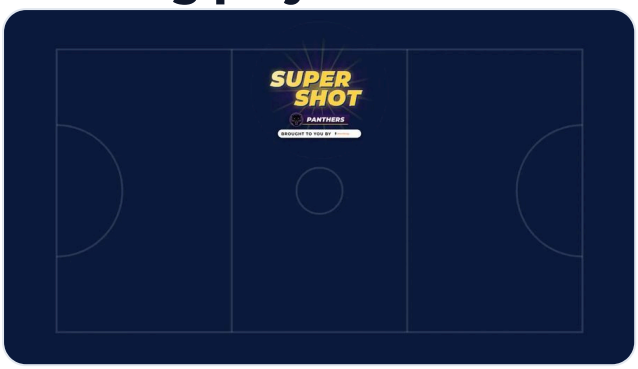


Team list. One team's line-up from PlayHQ with positions, captain and bench, warmed up before the game so it shows at once. Add one source per side.

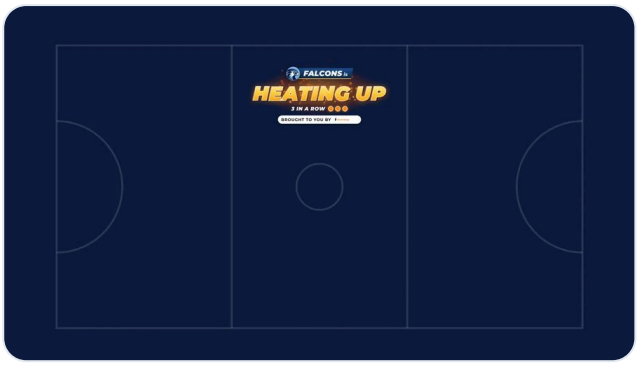


Intro and outro. Six full-screen styles, about 8 seconds, with the round's teams and a yellow message line. Options: style, teams from a division or round, time on screen, a theme song with its timing and volume, sound effects, a background.

During play



Super shot. Automatic, 3 seconds, the scoring team's logo and colour; the sponsored-moment tag names who brought it. **Plays: home / away / no team** makes a manual one.



Hot streaks and moments. Heating up, on fire, goal for goal, hits the front, all square, coming back, on the siren, back to back. Automatic from the goal order, or **Plays** picks one to play once.



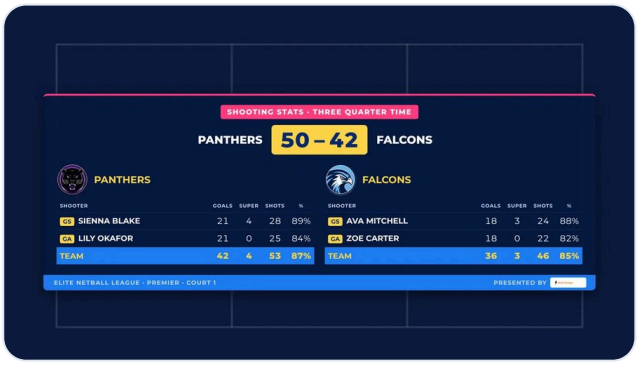
On the siren. A goal in the last seconds of a quarter. Moments are the only animations that need nothing from you; they also take the sponsor's "brought to you by" tag.



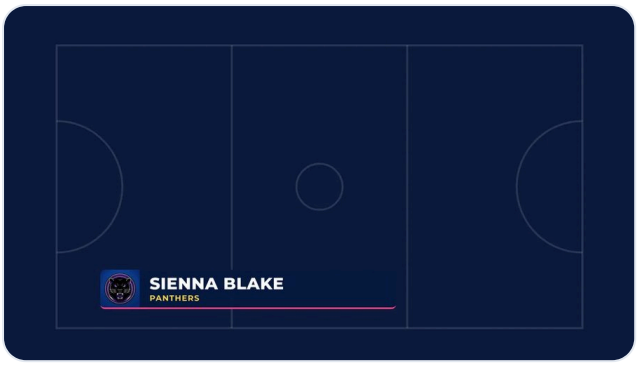
Ladder lower third. Both teams' ladder position and record as one strip right above the scoreboard, the same width as it, for a glance during play without covering the game.



Last 10 goals. The game's recent goals oldest to newest with a momentum line and the current run, worked out by Studio from the score, updating while it stays on. Options: across the top or down the side, how many goals, position and heading.



Shooting stats. Goals, super shots and attempts per shooter from PlayHQ's game page, for a break or a quarter-time sequence. Super columns are left off for a division without super shots.



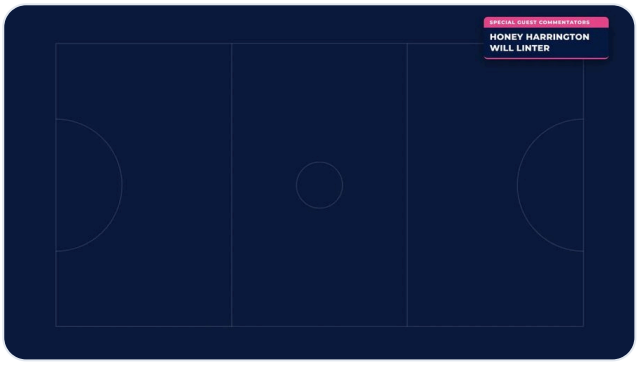
Person lower third, athlete. Name, team and club logo from the athlete register, with their photo if there is one. The Output page suggests names as you type.



Person lower third, guest. Name, title and the competition logo, for a commentator, an official or a sponsor's guest. Options: a tab above, position left or right, animate away.

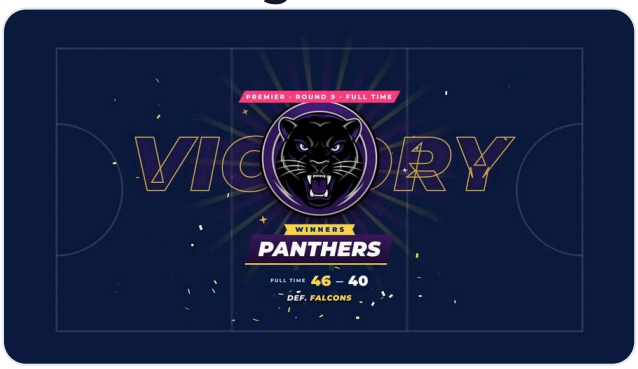


Commentators, thin line. The commentary team in one slim strip above the scoreboard: a pink tab, like Special guest commentators, and up to four names joined with "and". With no names typed, whoever the Production Planner has on the mic.



Commentators, box. The same names as an upper third in the top right corner beside the competition logo, a line per name. Options: under the title instead, the tab's words, animate away.

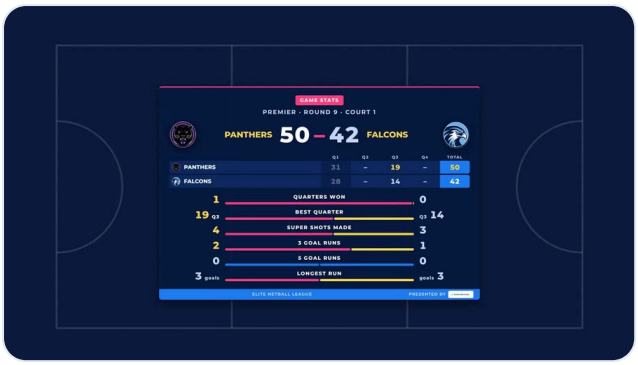
After the game



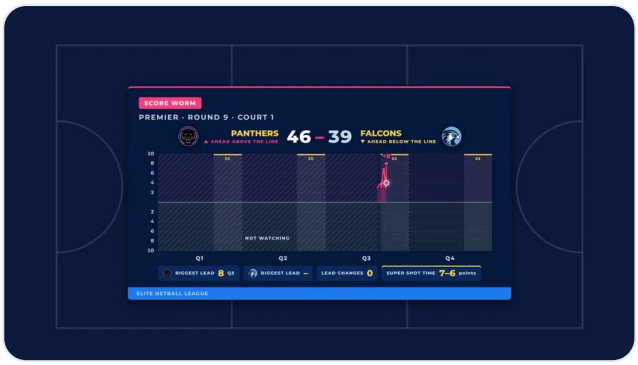
Winner. The winning team, logo, final score and a big word, worked out from the board each time it comes on. Options: force a result, the word, the line under the score, animate away.



Full time. The final score from the board with each team's top shooter and shooting accuracy from PlayHQ's game stats. **Heading** makes it a half time card.



Game stats. The score by quarter, tracked by Studio from the board, then team against team: quarters won, best quarter, super shots made, 3 and 5 goal runs and the longest run, with a sponsor's footer.



Score worm. The margin through the game as a line, drawn in over a few seconds. Options: drawing time, club colours or the competition's.



End of night. Up next after the last game: thanks for watching, next week's date or the finals, and the Supporter sponsors in a strip. Options: the invite line, sponsors on or off.

The competition



Ladder. The division's ladder with the finals places marked and a key under it. Options: division, rows, logo on or off, light or dark mode logo.



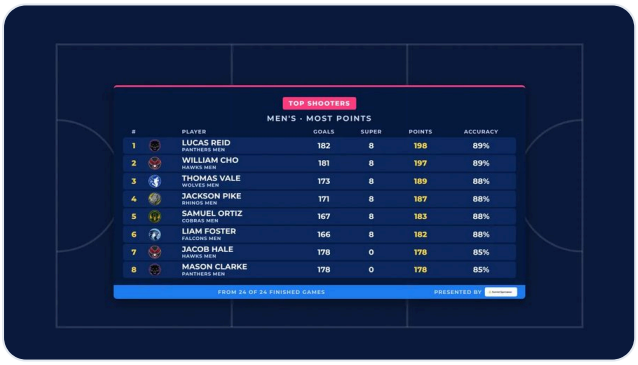
Round fixture. This round's games across all divisions with day, time and court. **Round** takes next, prev, +2, a number or a name like Final. When divisions are on different rounds, each game shows its own.



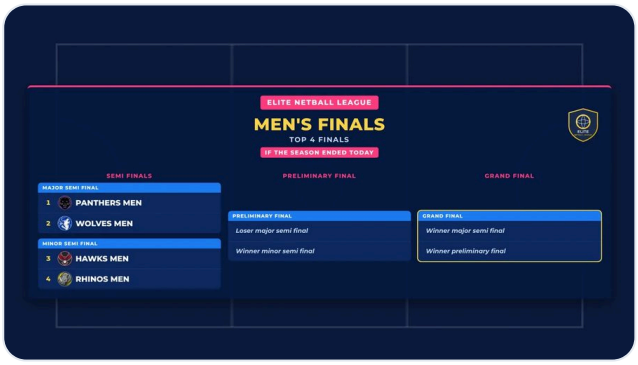
Fixture board (Full Page). The night's whole fixture, a card per court with its games by time, for a screen in the competition room or backstage, or a picture to share before the night (**Tuesday Night's Fixtures**). The streamed court has a camera and its games show the commentators from the Production Planner, **On the mic** with their headshots (a silhouette for anyone without one) where there is room, as name pills on a crowded night; games being played show their live score, and finished ones grey out with the winner in yellow. Not for the stream.



Results. The same round once its games are final, with scores. The same round option applies.



Top shooters. The season's leading shooters for a division. Options: order by goals or average, how many shooters, a sponsor footer.



Finals. Who plays whom week by week in the division's finals format, with ladder positions and scores; "If the season ended today" until the home-and-away rounds finish. Options: one week only, the heading and the line under it.

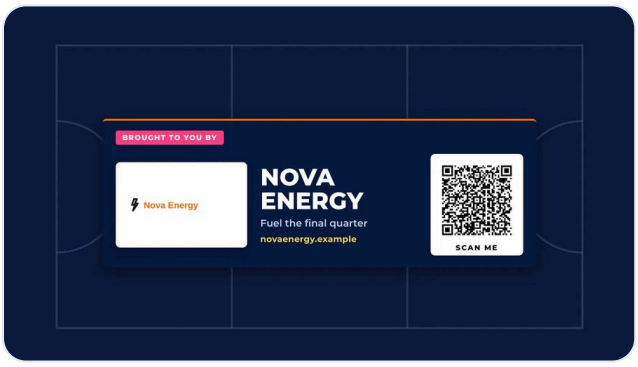


Competition logo animation. Seventeen templates (pro, coin, tiles, glitch, badge, lightning, court, bibs, net and more), about 5 seconds, with optional words and a smaller line.

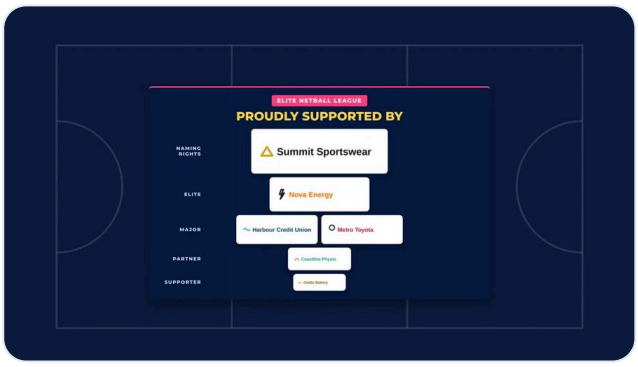


Stinger. A 1.5 second transition: bands in the competition's colours sweep across and cover the whole picture with the competition logo for half a second, so a cut can happen underneath, then carry on off the other side. Options: hold, length, logo on or off, light or dark mode logo, whoosh volume, and the music quieted for the next scene or ducked while it plays.

The breaks



Sponsor card. One sponsor's logo, tagline, offer and a QR code to their website, taking turns by tier. Options: one sponsor always, which tiers rotate, animate away.

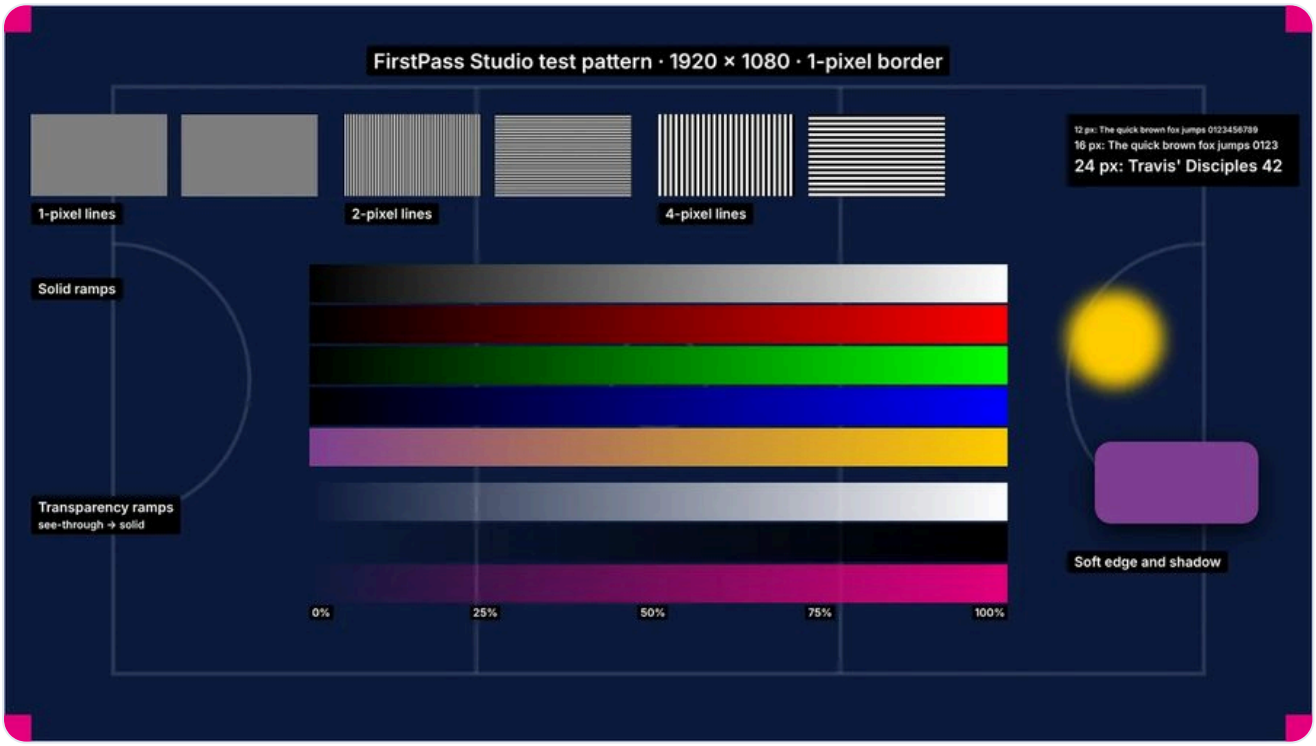


Sponsor wall. Every sponsor by tier on one card, for the run-up and the close. Options: which tiers, names under the logos.



PIVT and FirstPass Studio ad. A 10-second house ad for a break with nothing sold. A Sponsor ads step plays it when no sponsor's ad is due, if **House ad** is ticked.

Checking the outputs



Test pattern. Borders, fine lines, ramps and soft shapes to check an output, the switcher's scaling and its key end to end before the stream. Never for air.

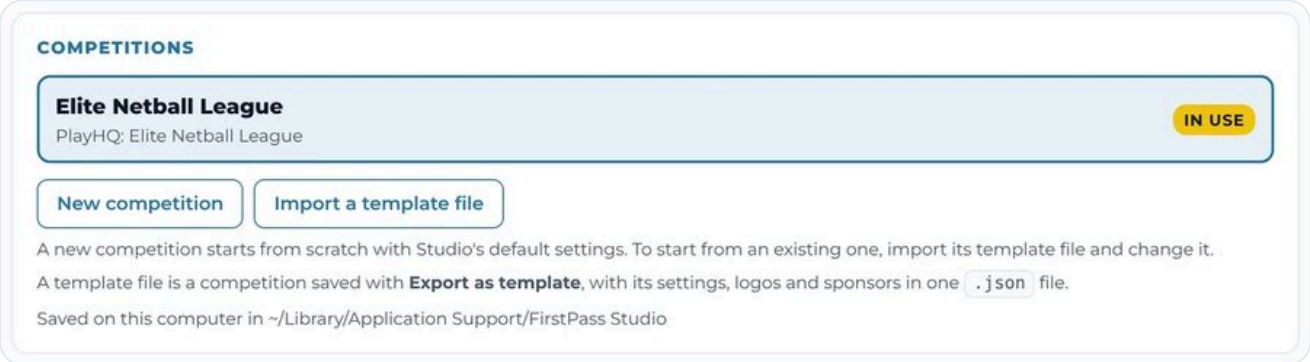
Competitions setup

Everything about a competition that doesn't come from PlayHQ lives on one page: its PlayHQ link, the names on the graphics, the game rules, finals, game day, colours, font and logos. Set it up once a season and every graphic follows.

IN THIS CHAPTER

1 PlayHQ
2 Names on the graphics
3 Game rules
4 Finals
5 Game night
6 Colours and font
7 Logos
8 Live viewers
9 Templates, export and more

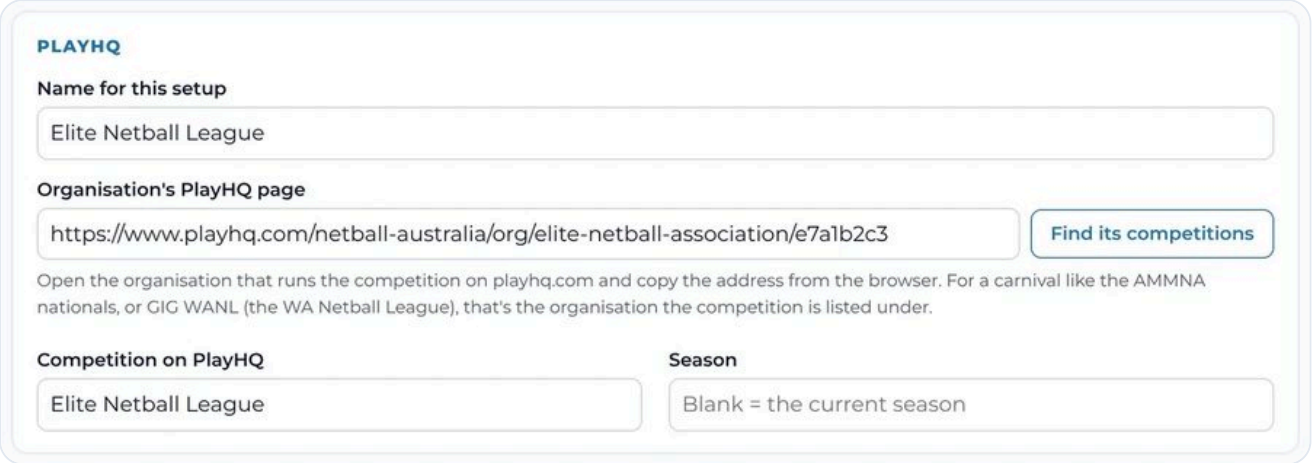
Studio keeps a setup per competition. The one marked **In use** is what every graphic, Scoreboard control, the Sponsors page and the Stream Deck use. Changes aren't kept until you press **Save** (or **Save and use this competition**), and leaving the page with unsaved changes asks first.



The competitions list. Start from a template file (the repository has Thunder League, AMMNA Nationals, GIG WANL and West Coast Thunder) or press New competition for Studio's defaults.

PlayHQ

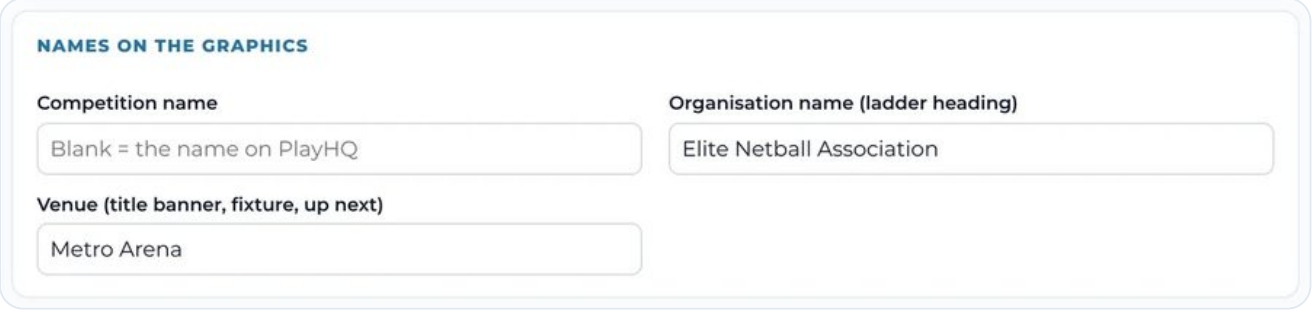
Paste the address of the organisation's page on playhq.com and press **Find its competitions**; then pick the competition and the season. For a carnival or a league run under another body, use the organisation the competition is listed under. Nothing else about PlayHQ is needed: no account, no key.



The PlayHQ section for the sample league. The season list comes from PlayHQ once the competition is picked.

Names on the graphics

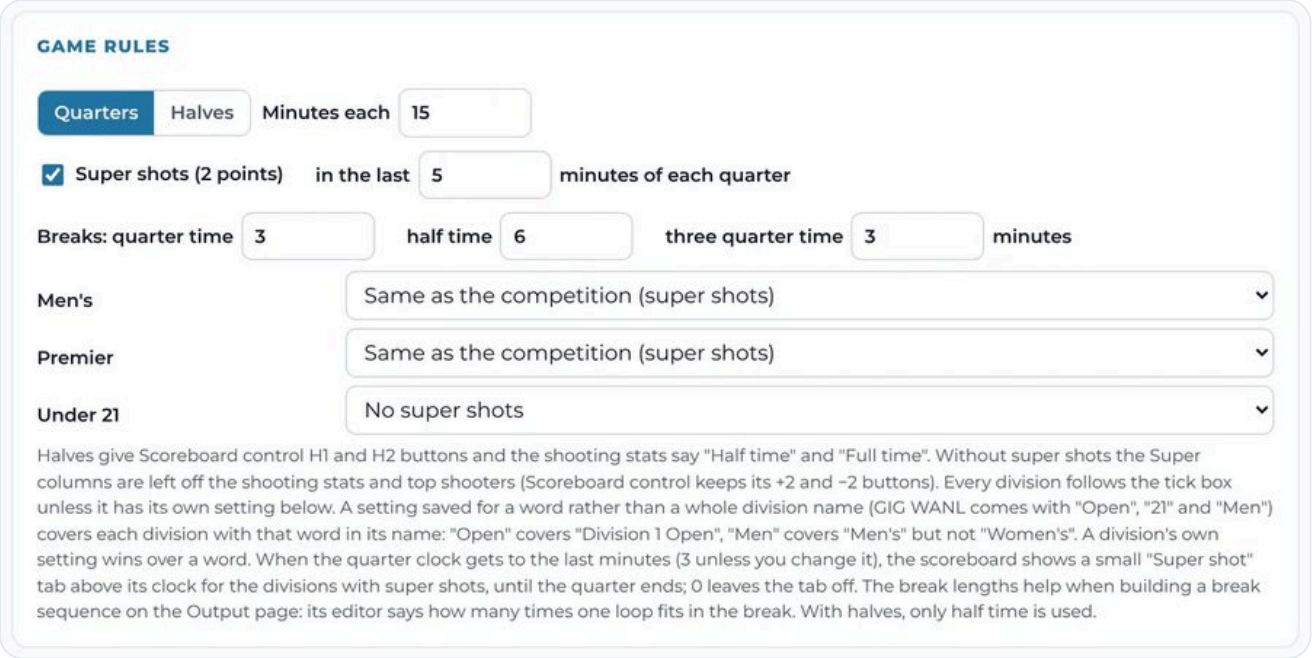
The competition name on the title banner, the organisation name as the ladder's heading, and the venue for the title, fixture and Up next. PlayHQ's names are often long or formal; these are the ones viewers see.



Names on the graphics.

Game rules

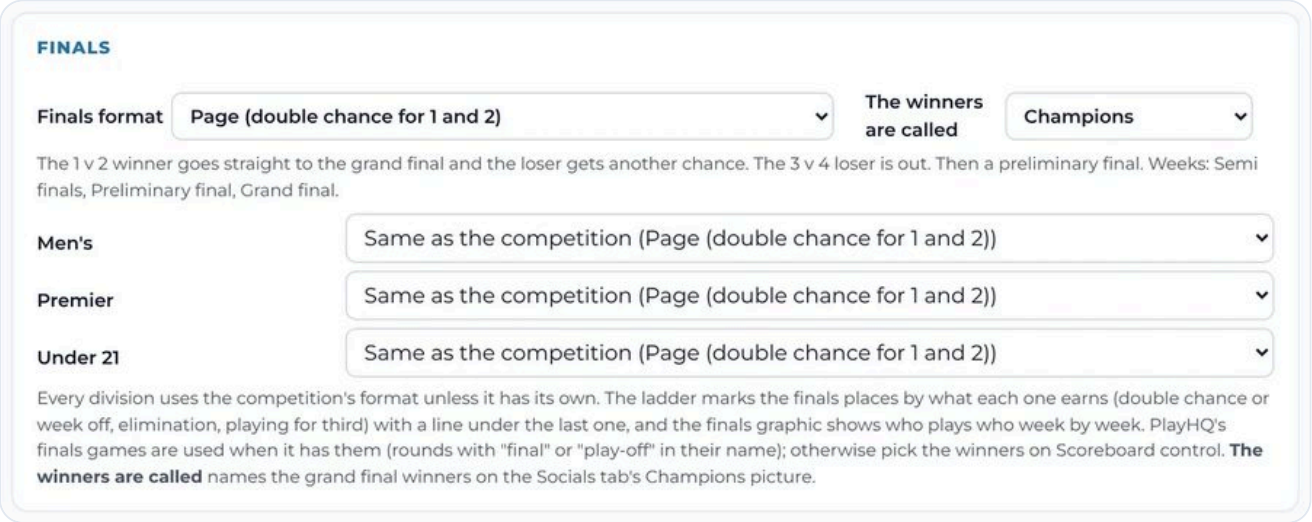
Quarters or halves and their length set the clock, the quarter buttons and the wording of the break cards. **Super shots** turns on the +2 scoring, the celebration, the super shot tab in the last minutes of each quarter and the Super columns on the shooting graphics; a division can have its own setting, by its name or by a word in it ("Open", "Men"). The break lengths let the sequence editor say how many loops of a break sequence fit in the break.



Game rules. The sample league plays 15-minute quarters with super shots in the last 5 minutes, except in its Under 21 division.

Finals

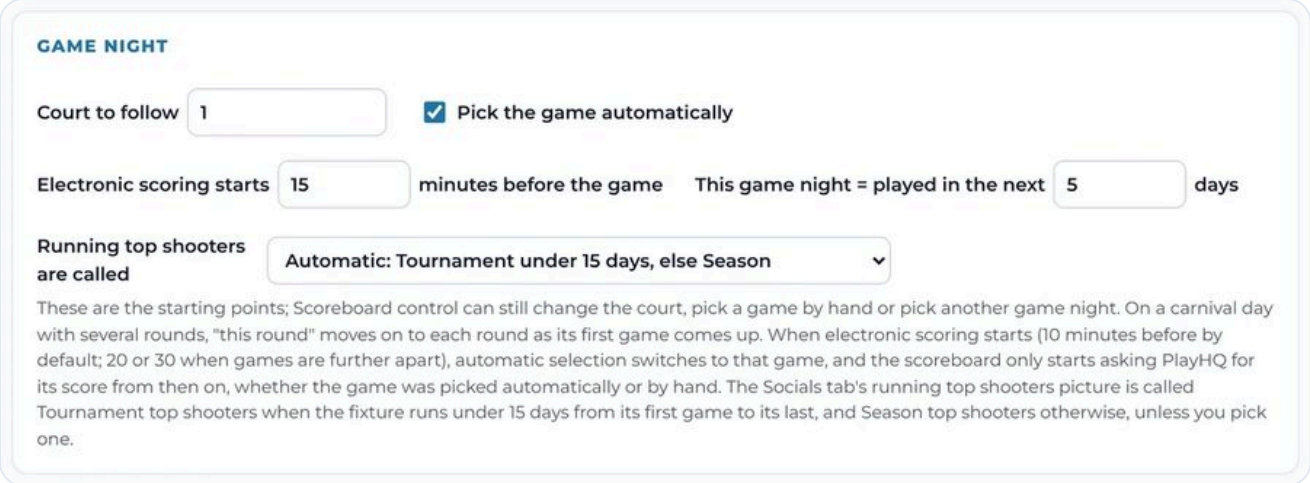
Pick the finals format for the competition, or per division: it decides the finals places the ladder marks (double chance or a week off, elimination, playing for third) and the bracket the finals graphic draws week by week. PlayHQ's finals games are used when it has them; otherwise the winners are picked on Scoreboard control. **The winners are called** picks Champions or Premiers for the Socials tab's picture of the grand final winners.



Finals formats, per division where they differ.

Game night

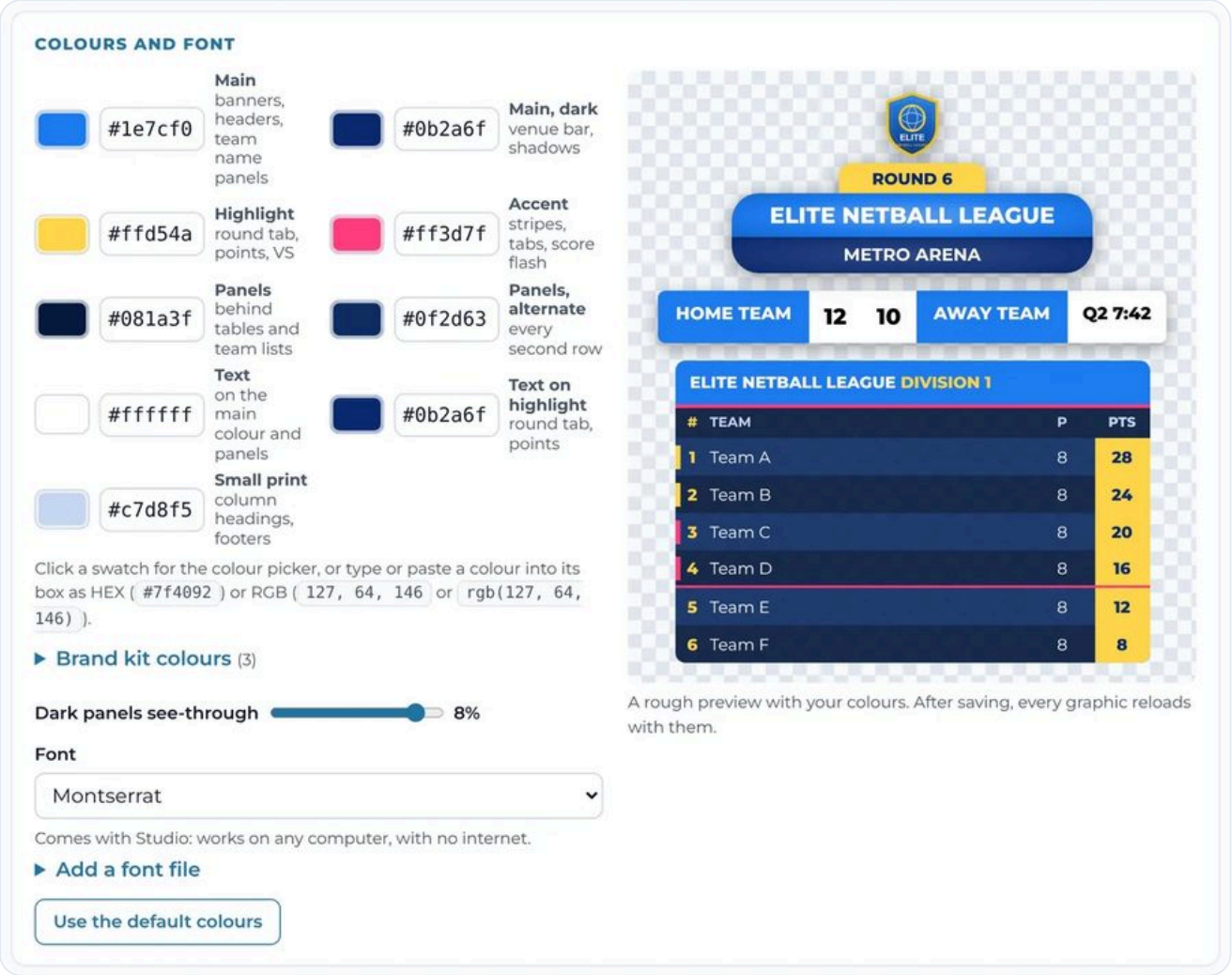
The court Studio follows, whether it picks the game automatically, when electronic scoring starts before each game (10 minutes by default; 20 or 30 when games are further apart) and how many days ahead tonight's games are found. These are starting points: Scoreboard control can change the court, pick a game by hand or pick another game night. On a carnival day with several rounds, this round moves on as each round's first game comes up. **Running top shooters are called** sets the Socials picture's name: Automatic calls it Tournament top shooters when the fixture runs under 15 days and Season top shooters otherwise.



Game night.

Colours and font

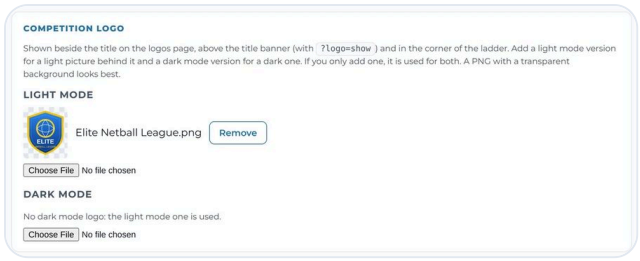
The competition's colours, see-through dark panels, and the font. Paste a brand kit's text into **Brand kit colours** and every HEX or RGB value in it becomes a swatch in each colour picker. Add a font file (TTF, OTF, WOFF or WOFF2, one per weight or a variable font) and it travels with the competition and its template, so the graphics look the same on any Mac; every font is sized to match the capitals of Montserrat, which the graphics were drawn with. The preview under the pickers shows a rough title and scoreboard in the colours before you save.



Colours and font, with the preview.

Logos

The **competition logo** has a light mode and a dark mode version (one is used for both if that's all you add); it shows beside the title in the branding logos, above the title with its option on, in the corner of the ladder and in the logo animations. **Team logos** are matched to PlayHQ team names by file name, so drop in a folder of files named after the teams. Logos and colours belong to clubs: every team of a club, in every division, gets the club's logo and colour, worked out from the team name without its division ("Falcons 21U" is Falcons). Fix a wrong match in the **Club** box. A club colour goes behind its teams' names on the scoreboard and behind their logo on The Clash; without one the competition's colours are used.



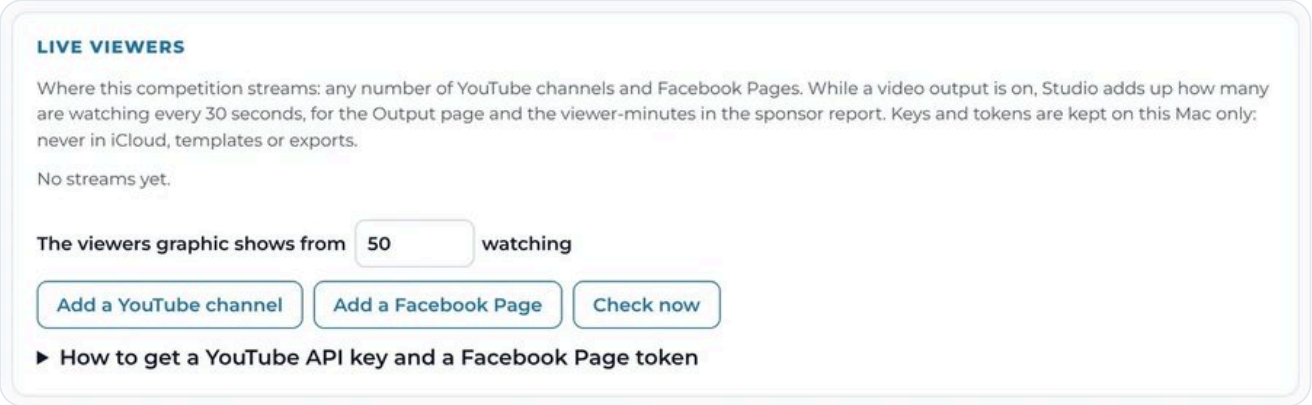
Competition logo, light and dark mode.



Team logos and the teams on PlayHQ, grouped by club with a club colour each.

Live viewers

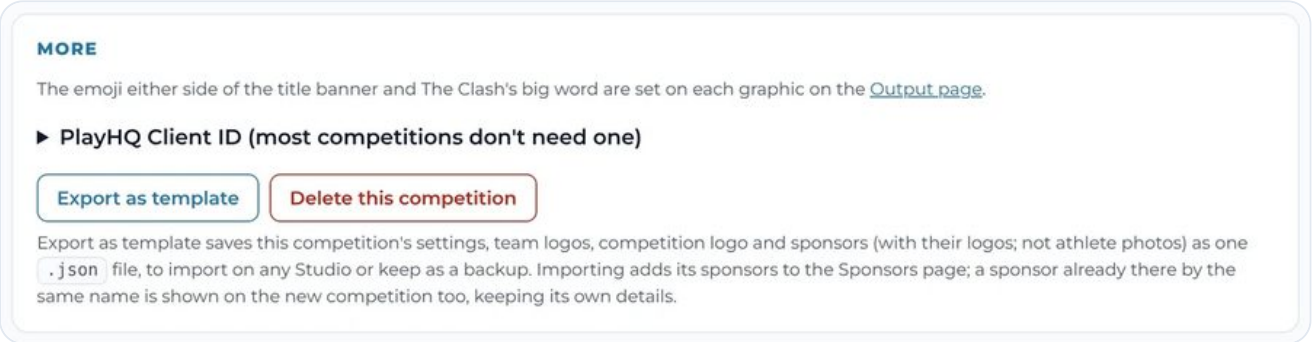
Add the YouTube channels and Facebook Pages the competition streams to. While a video output is on, Studio adds up how many are watching every 30 seconds, for the Output page, the viewers graphic (which shows only from the minimum set here) and the viewer-minutes in the sponsor report. The page explains how to get a YouTube API key and a Facebook Page token; both are kept on this Mac only, never in iCloud, templates or exports.



Live viewers.

Templates, export and more

Export as template saves the competition's settings, team logos, competition logo, font and sponsors (with their logos) as one .json file to import on any Studio or keep as a backup; **Import a template file** adds it. A PlayHQ Client ID is only needed if PlayHQ's game page has no line-ups and the association has given you one. **Delete this competition** removes it; deleting the one in use switches the graphics to another.



The More section: Client ID, export and delete.

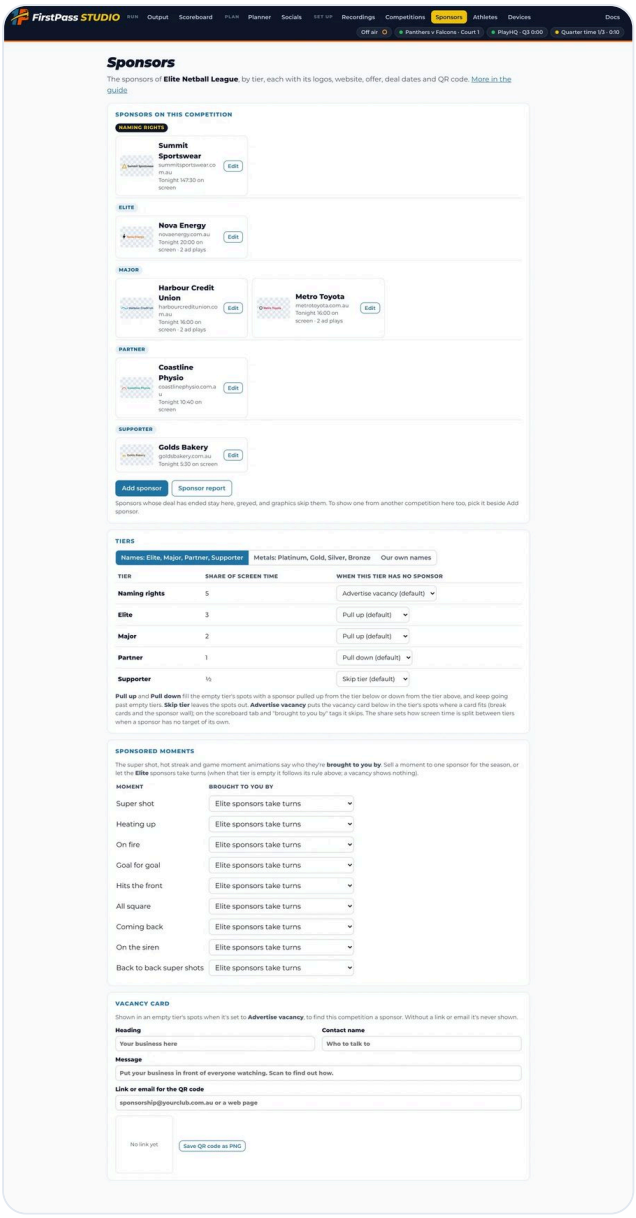
More in the Reference: **Competitions setup in detail**.

Sponsors

The Sponsors tab turns sponsorship into something you can sell, run and prove: a tiered library of sponsors, spots on the graphics that fill themselves, ad breaks that take turns fairly, and a report of what each sponsor got.

IN THIS CHAPTER

1 Tiers
2 A sponsor
3 Spots and moments
4 Proof of play



The Sponsors page for the sample league: sponsors by tier with tonight's exposure meter, the tier rules, sponsored moments and the vacancy card.

Tiers

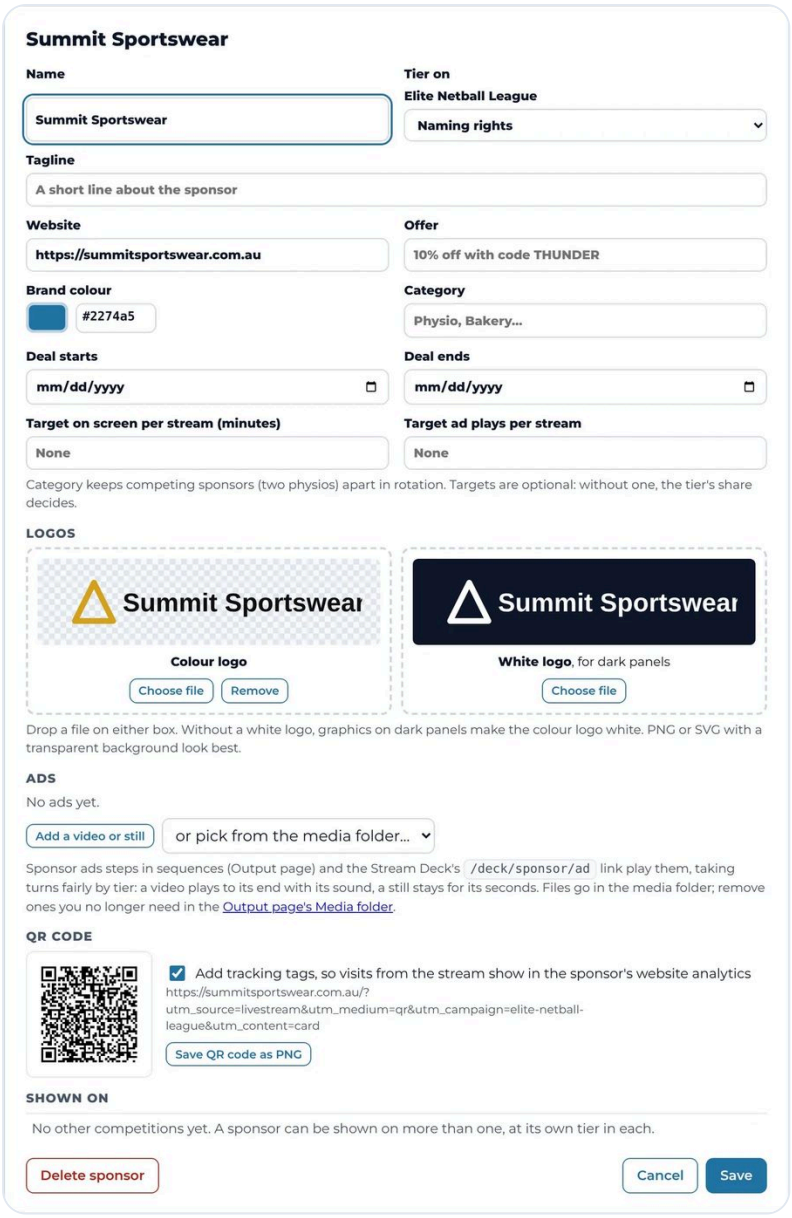
Five fixed ranks, named per competition: the **Names** preset (Naming rights, Elite, Major, Partner, Supporter), the **Metals** preset (Naming rights, Platinum, Gold, Silver, Bronze) or your own. Each tier's **share of screen time** decides how turns are split when a sponsor has no target of its own. Each tier gets different places:

TIER	WHERE IT SHOWS
Naming rights	A tab on the scoreboard, under the title banner, first on the sponsor wall, and everything below.
Elite	The sponsored moments (super shot, streaks), ads in the breaks, the break card, and everything below.
Major	Ads and break cards, "presented by" footers on the stats cards, countdown and top shooters.
Partner	A slot in the score ticker and the logo spots.
Supporter	The sponsor wall and the end-of-night strip.

When this tier has no sponsor says what a spot does when its tier is empty: **Pull up** fills it from the tier below, **Pull down** from the tier above, **Skip tier** leaves it out, and **Advertise vacancy** shows the competition's vacancy card wherever a card fits. The vacancy card has a heading, a contact, a message and a link or email for its QR code, so an empty tier finds its own sponsor.

A sponsor

A sponsor has a name, tagline, website, offer, brand colour, category, deal dates, and optional targets per stream (minutes on screen, ad plays). Category keeps competing sponsors apart in rotation. Add a colour logo and a white one for dark panels (without a white one, the colour logo is made white). A sponsor is kept once and can be **shown on** several competitions at its own tier in each; sponsors whose deal has ended stay greyed and the graphics skip them.



The sponsor editor: details, logos, ads, the QR code with tracking tags, and the competitions it is shown on.

Ads are videos or stills (up to 10 per sponsor) in the media folder. A **Sponsor ads** step in a sequence plays the next ads due, taking turns fairly by tier, each video to its end with sound and each still for its seconds; tick **House ad** and the PIVT and FirstPass Studio ad plays when nothing is due. The **QR code** points to the sponsor's website with tracking tags, so visits from the stream show in their analytics; save it as a PNG for the sponsor's own use.

Spots and moments



The sponsor card in a break, with its QR code.

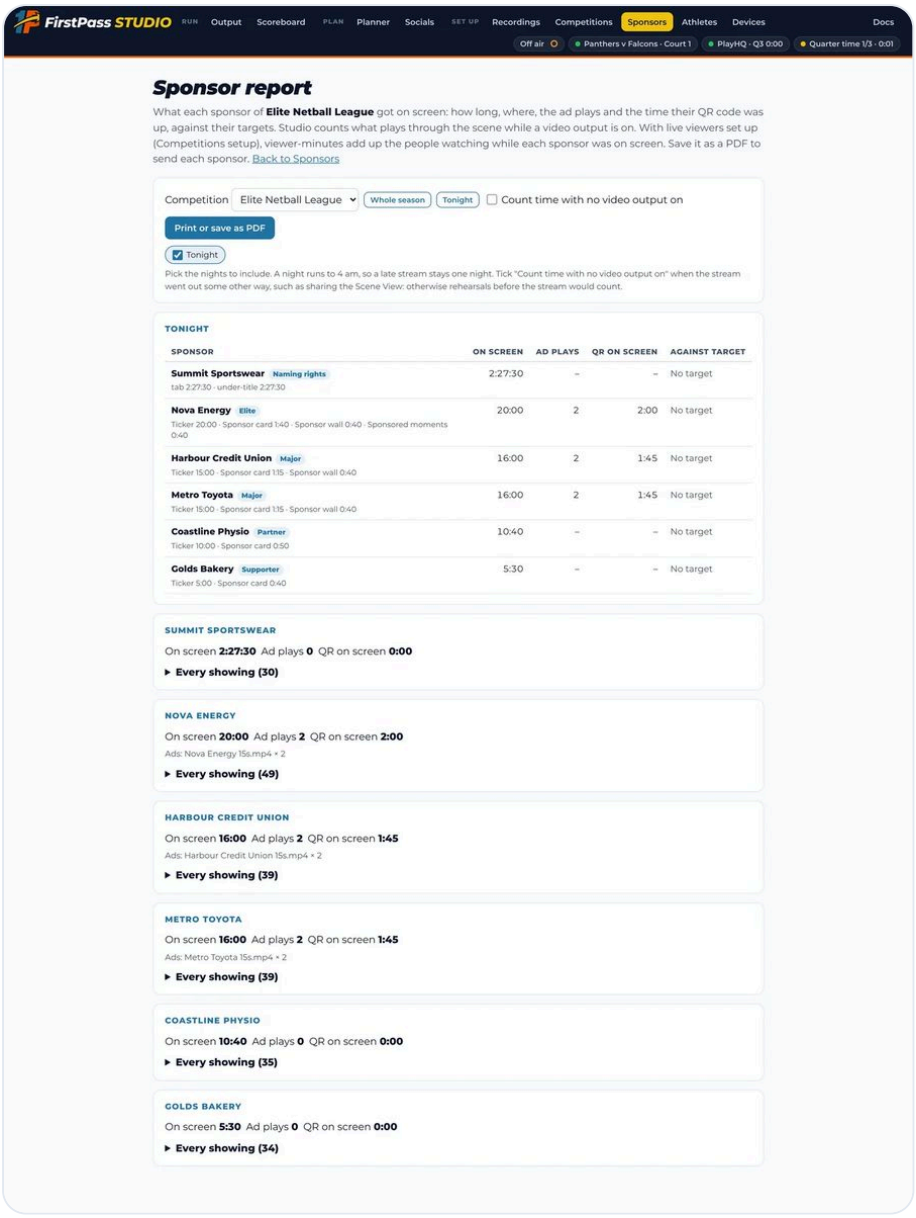


The sponsor wall before the game, sized by tier.

Every spot picks its sponsor from the Sponsors page by default. Each graphic's **Sponsor** option can pin one sponsor or turn the spot off. **Sponsored moments** sells the super shot, each hot streak and each game moment to one sponsor for the season, whose name appears in a "brought to you by" tag on the animation; by default the Elite sponsors take turns.

Proof of play

While a video output is on, Studio logs every second each sponsor is on screen and every ad played, matched to the scene's sources, and reads the live viewer count every 30 seconds. **Sponsor report** turns that into a page per night (or a run of nights) with minutes on screen, ad plays, viewer-minutes, peak and average viewers per sponsor, ready to print or save as a PDF for the sponsor. A tick box counts time with no video output on, for a rehearsal. With more than one output, only Output 1 and outputs with **Counts for sponsors** ticked are counted, and each counts as on air only while it's on: an output set to **Web links only** is off.



The sponsor report for the sample night.

More in the Reference: **Sponsors in detail**.

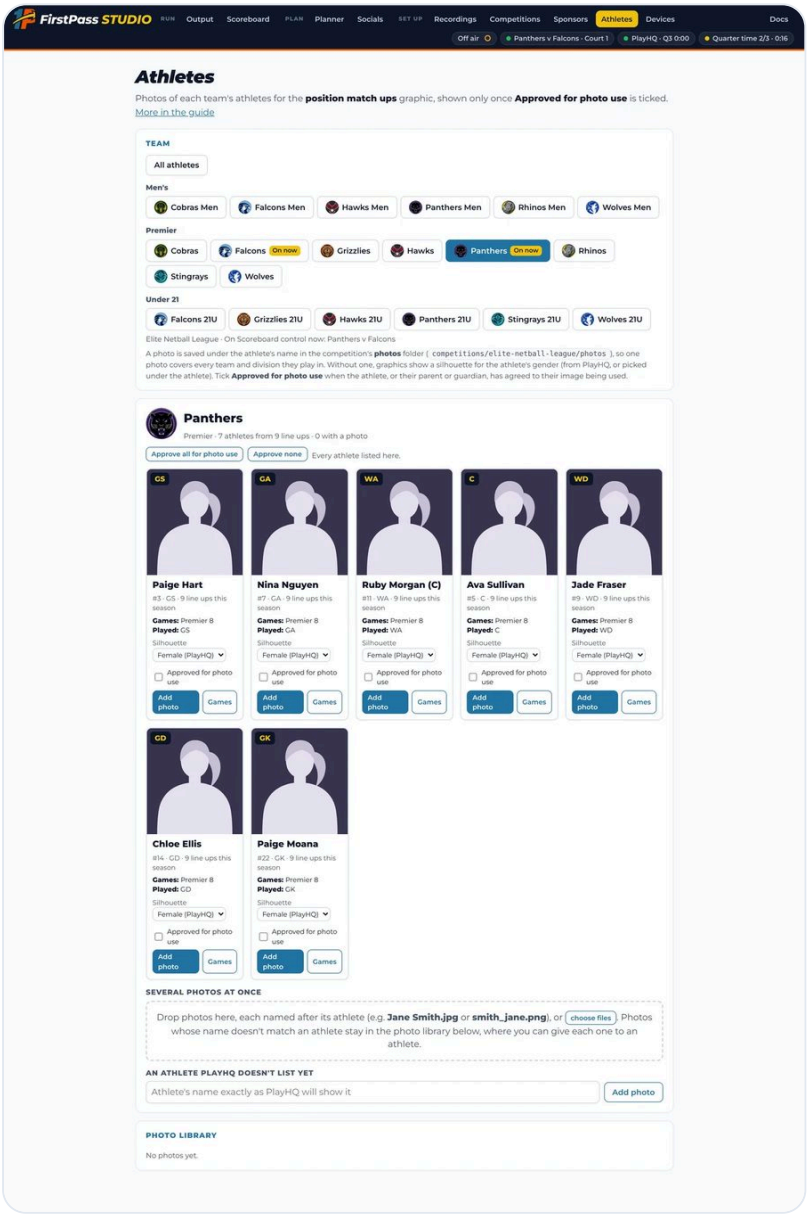
Athletes

PlayHQ gives Studio every athlete in a line-up, but not their faces. The Athletes tab holds a photo per athlete for the position match ups and the person lower third, and a record of positions and games played that PlayHQ's history doesn't cover.

IN THIS CHAPTER

1 Photos

2 Positions and games played

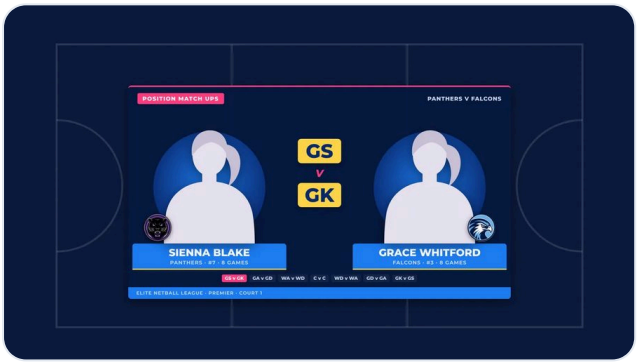


The Athletes page: teams grouped by division, every athlete PlayHQ has in a line-up this season, and the photo library.

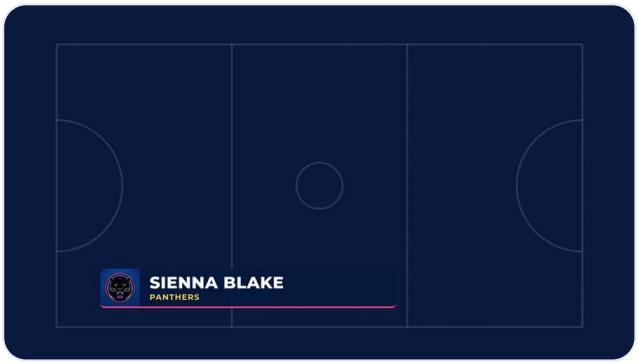
Photos

Pick a team and add a photo to each athlete, or drop a folder of photos named after the athletes ("Jane Smith.jpg" or "smith_jane.png") on **Several photos at once**; any that don't match wait in the photo library to be given to someone. A photo is saved under the athlete's name for the competition, so an athlete in two divisions needs one photo. Studio also remembers each athlete's PlayHQ id, so if PlayHQ changes a player's name during the season, their photo still finds them. Photos are made smaller when saved; cut-out PNGs with a transparent background look best. Without a photo the graphics show a silhouette for the athlete's gender, taken from the division (men's, women's; mixed and open are neutral) and changeable under the athlete. **An athlete PlayHQ doesn't list yet** adds one ahead of their first game.

A photo only goes on the stream and the Socials pictures once **Approved for photo use** is ticked under the athlete. Tick it when the athlete, or their parent or guardian, has agreed to their image being used. Until then the graphics show the silhouette. **Approve all for photo use** ticks every athlete listed for the team at once.



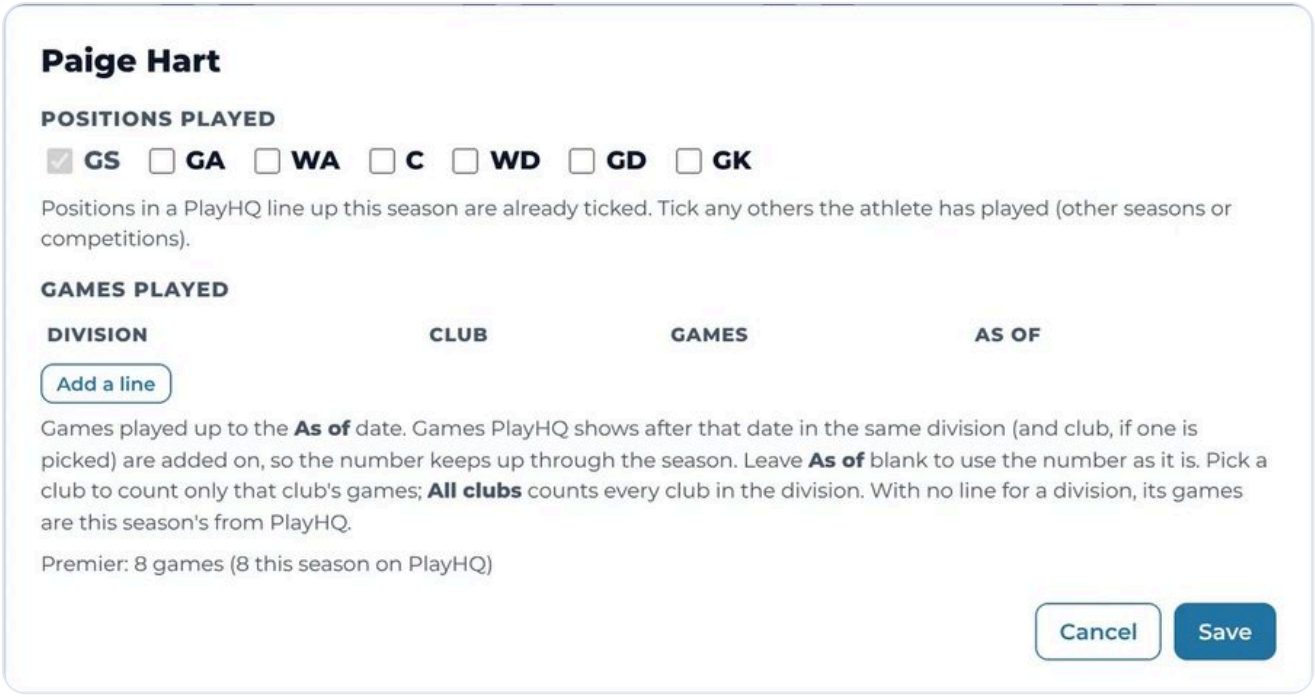
Position match ups use the photos, pairing by pairing.



The athlete lower third, named from the register as you type on the Output page.

Positions and games played

Games under an athlete opens their record. Positions played in a PlayHQ line-up this season are ticked already; tick any others from other seasons or competitions. Games played takes a line per division (and club): the number as of a date, to which Studio adds the games PlayHQ shows after it, so milestones keep up through the season. With no line, a division's games are this season's from PlayHQ.



An athlete's positions and games played.

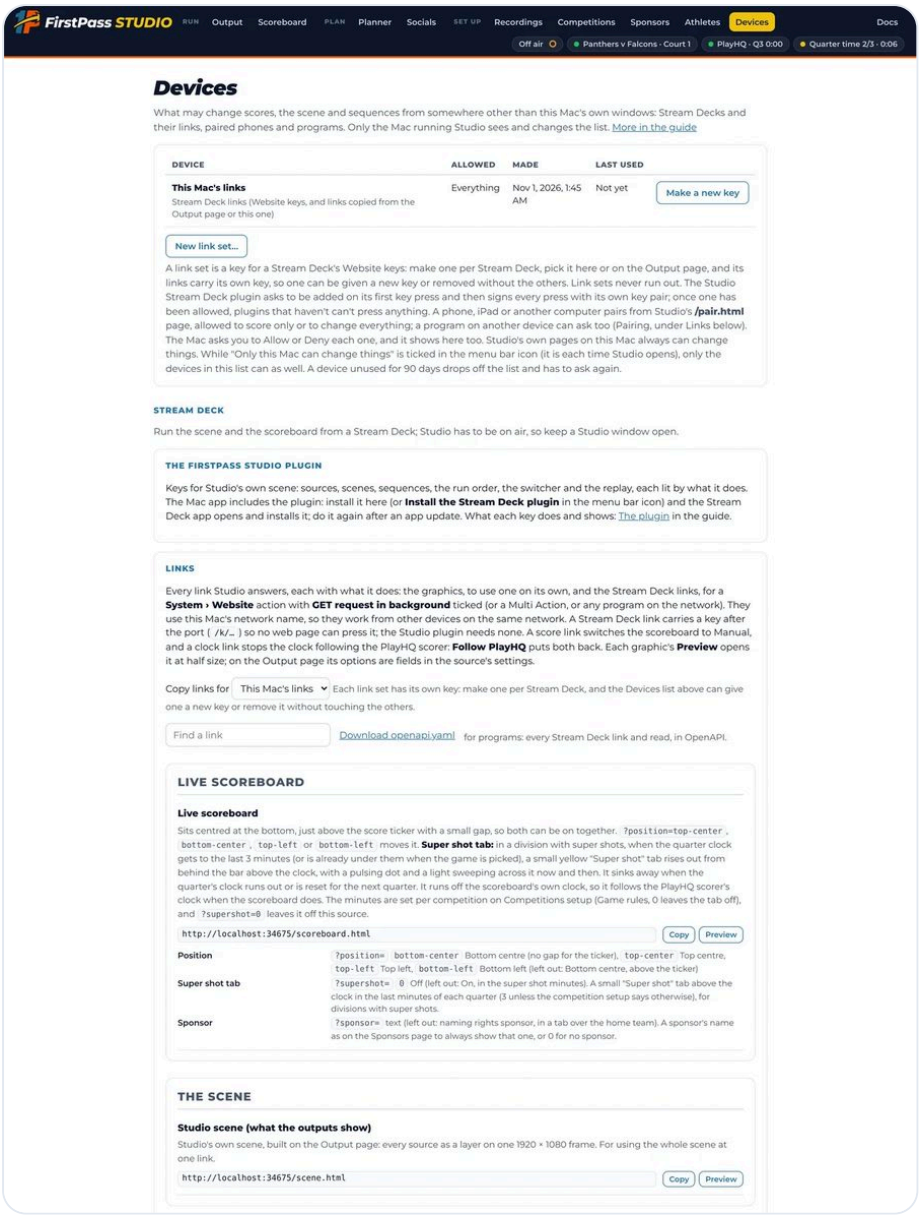
More in the Reference: **Athletes in detail**.

Stream Deck

A Stream Deck is the control surface for stream night: one key per source, scene and sequence, each lit while it's on, plus score, clock and quarter keys for a game run by hand. Studio ships its own plugin, and every control also has a plain link. It all lives on the **Devices** tab, under the list of devices allowed to change things (Stream Decks and their link sets, paired phones and programs), which is also where **Devices** in the menu bar icon goes.

IN THIS CHAPTER

1 The plugin
2 Links for everything else
3 Timed keys and the icon pack



The Devices tab: the devices allowed to change things at the top, then the plugin, then Links: every link Studio answers, each with what it does, Copy and its settings.

The plugin

Install it from the 1P menu (**Install the Stream Deck plugin**) or the button on the Output page; the Stream Deck app opens and installs it. Do the same after an app update. The **Studio** field in a key's settings stays empty when the Stream Deck is on the same Mac; for another Mac, type its address, like `http://studio-mac.local:8080`. Six kinds of key, each a different shape at a glance:

KEY	LOOKS	PRESS	SHOWS
Source	A faint eye and one strip along the bottom	Show or hide (default), fade in, fade out, refresh the page or restart the video, video sound on or off	The source's name; lit in its output's colour while it's on, however it was turned on
Scene	Stacked layers and a frame in its colour	Go to a scene, next or previous scene, or back to the current scene as saved	Lit while its scene is on, with a yellow dot when sources have changed since
Sequence	A play button and a strip in four steps	Play or stop a sequence, play from the first step, or Next, Next when waiting, Pause and Stop for whatever is playing	Step and time left, like **3/5 0:12**, the whole sequence's time, both or none; **in 0:04** when one is about to start by itself
Run order	A list with a play button at its head, and a strip whose first block stands taller than the ones behind it	Play next (default), skip what's next, load a saved order, clear the list	What plays next, like **Play next ▶ / Quarter break / +2 more**, in that scene's or sequence's colour; lit while something is queued, **empty** when nothing is; a Load key names its order and how many items it holds
Switcher	A camera and a strip in two halves	A camera to preview (default) or program, stinger to a camera, Cut, Auto, an upstream or downstream key, Fade to black, run a macro	The camera's name from the switcher; red on program and green on preview, a key or Fade to black red while on air; a yellow dot while the switcher isn't answering

Replay	Play and pause with a timeline and its playhead	Play or pause, next or previous moment, nudge, stop the 4K replay or practice game running	What it acts on (**4K replay**, **Practice**, **Demo**); lit while one runs

Keys follow their source, scene or sequence by id, so renaming on the Output page doesn't break them; a removed one says **gone**, and while Studio is off air they say **off air**. Colour groups keys by the part of the stream: green pre-game, blue live, purple quarter time, grey post-game, yellow ads, or the scene's or sequence's own colour.

With more than one output, each key has an **Output** setting: the output it shows a source on, plays a scene on or plays a sequence on (a Switcher key has none: there is one switcher). A lit key takes its output's colour (Output 1 orange, 2 pink, 3 blue, 4 green) with a black title, so a commentators' Stream Deck of Commentary keys lights pink while the stream's keys light orange. A key for a removed output says **no output**.

Links for everything else

Every control is also a link, for a **System > Website** action (tick **GET request in background**), a Multi Action, or any program on the network. Scoreboard control, the Output page and the Devices tab copy them, with this Mac's Stream Deck key after the port (`…:8080/k/key/deck/…`) so no web page can press them. The Studio plugin needs no key: the first time you press one of its keys the Mac asks you to **Allow** it, and from then on it signs every press with its own key pair. A link without it is refused, so copy any older Website keys' links again from Studio. With more than one Stream Deck, give each its own link set: pick **New link set…** in **Copy links for** on the Devices tab, name it after the Stream Deck, and copy its links from there. If a link gets out, choose **Devices** in the menu bar icon and press **Make a new key** on that set: its links stop working and you copy them again, while the other Stream Decks carry on. A program on another computer can ask for a key of its own: the Mac asks you to **Allow** or **Deny** it by name and address, and Devices can take that one away without touching your other keys. The Devices tab lists every one under **Links**, with what it does, the settings it takes and **Copy**; **Find** narrows the list, a graphic has **Preview** and a status read has **Try**. The useful ones:

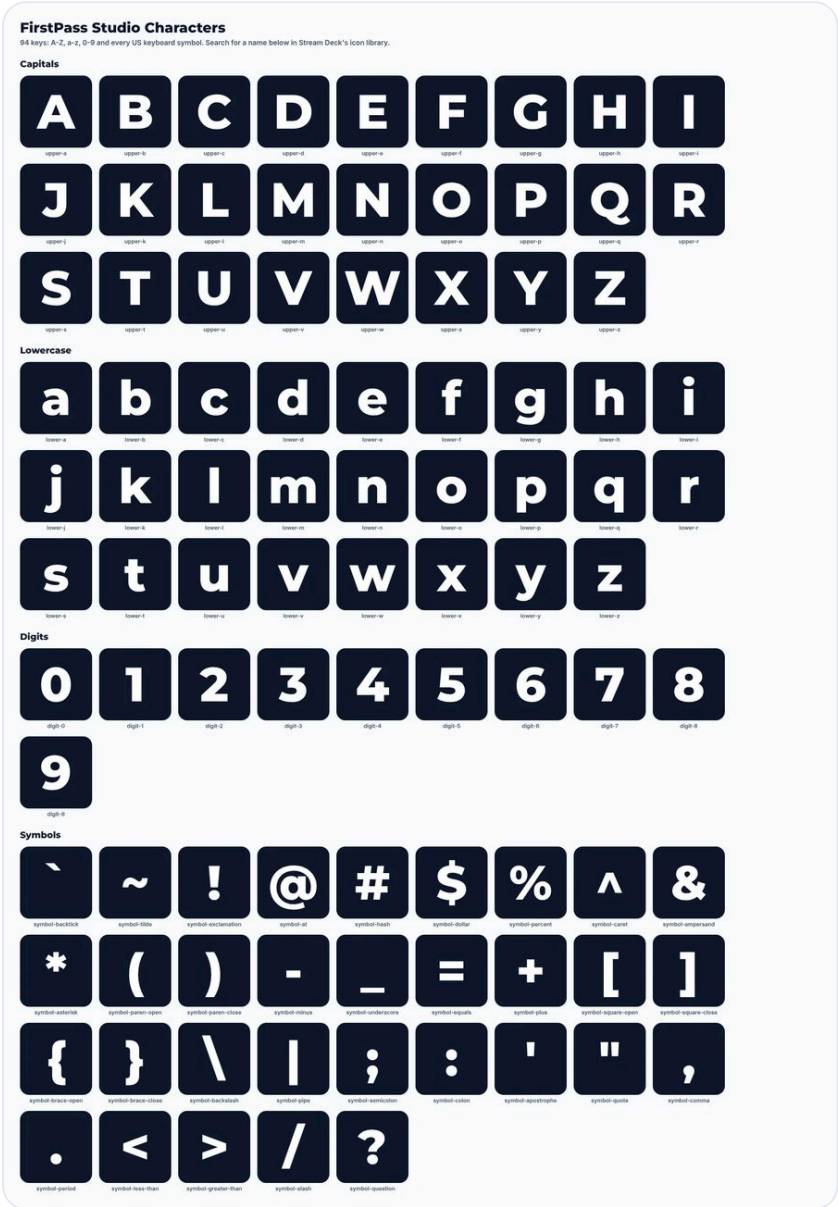
GROUP	LINKS
Scores	Home and away +1, +2, −1, −2; reset to 0; play a super shot or any streak without changing the score.
Scoring mode	Follow PlayHQ, Manual, toggle. Any score key switches to Manual.
Clock	Start, stop, toggle, reset, set to a time, add or take off any amount, follow the PlayHQ clock on or off, quarter length.
Quarter	Q1 to Q4, extra time, next, previous. The clock is separate; pair them in a Multi Action.
Game	Automatic selection on, off, toggle; the court to follow.
Ticker	Faster, slower, normal, a set speed.
Sources	Show, hide, toggle, restart, video sound; every source out; video output on or off.
Scenes	Go to a scene, next, previous, back to the scene as saved.
Sequences	Play, toggle, next, advance (only when waiting), pause, stop, start the one about to start now, cancel it, automatic sequences on or off.
Run order	Play what's next, skip it, clear the list, add a sequence or scene, load a saved order.
Music	Play, pause, resume, stop, fade out, next, previous, load a song or playlist, mute, duck; add `?fade=2` for a 2-second fade.
Sponsors	Play the next sponsor ad due.
Switcher	Cut, auto, stinger (to a camera or what's on preview), fade to black, a camera to preview or program, an upstream or downstream key on, off or toggled, run a macro.

A LAYOUT THAT WORKS

Put the scene keys in a row across the top (Pre-game, Live game, Clear), the sequence keys under them with Next when waiting and Stop at the end, and a page of Source keys for the lower thirds and stats. A game run by hand gets a second page: home scores left, away scores right, the clock and quarter keys down the middle.

Timed keys and the icon pack

A second plugin, **Timed Sequence**, arms a key for a time of day and then presses a list of links with delays between them: reset the board at 18:55 and start the clock at 19:00, say. It is built from the repository. The **FirstPass Studio Characters** icon pack gives Stream Deck keys letters, numbers and symbols in the Studio style, for labelling keys by hand.



The Characters icon pack: A to Z, a to z, 0 to 9 and the usual symbols on navy.

CHAPTER 10 OF 11

Settings and care

Where Studio keeps its settings, how to carry them between Macs and back them up, how updates arrive, and what to do when something isn't right.

IN THIS CHAPTER

1 The 1P menu
2 The data folder
3 Settings in iCloud Drive
4 Backup and restore
5 Updates
6 Links and other devices
7 When something isn't right
8 Data and bandwidth

The 1P menu

The menu bar icon is Studio's home. From top to bottom: the version and whether Studio is on air; **Stop Studio** (while on air); **Open Studio**, Scoreboard control, Competition setup and **Copy the Studio link**; **Devices**; **User guide**; **Install the Stream Deck plugin**; **Open the settings folder**, **Settings folder** (iCloud), **Back up...** and **Restore...**; **Open the debug log**, **Open the app log**, **Restart Studio**, **Check for updates**; **Open at login**, **Appearance** (Follow the Mac, Light, Dark), **Only this Mac can change things**, **What's new...** and Quit.

This guide comes with Studio: **User guide** in the menu bar icon, **Help > FirstPass Studio User Guide** while a Studio window is in front open it in a window of its own, and it is the **Docs** tab (⌘0) in any Studio window, with no internet needed. Phones and other computers on the network can read it too, at the Studio link with `/guide/` on the end.

The data folder

Everything Studio saves lives in one folder outside the app, `~/Library/Application Support/FirstPass Studio`, so updates never touch it. In it: **competitions/** (a folder per competition with its settings, logos, font, athlete photos and register), **sponsors/**, the scene, sequences, music and playlists, the **media** folder, **control.json** (Scoreboard control's choices), **scoreboard-state.json** (the game, scores and clock, saved every few seconds so a restart mid-game loses nothing), **config.json** for this Mac's own settings, **logs/** and **backups/**.

Every file and folder in it: **The data folder** in the Reference.

Settings in iCloud Drive

Settings folder > Move to iCloud Drive… moves the settings into a folder you choose in iCloud Drive, in a subfolder named after this Mac, so each Mac on the same Apple account keeps its own and none saves over another's. The debug log, scores and clock, caches and window positions always stay on the Mac. **Restore from another Mac…** copies another Mac's settings over this one's (competitions, scenes, sequences, music, control choices), backing this Mac's up first. **Move back to this Mac…** undoes the move. Studio goes off air for a moment for each and comes straight back.

In detail: **Settings in iCloud Drive** in the Reference.

Backup and restore

Back up… saves every setting in one zip named with the date and time, with notes you type and, if you choose, the media folder; beside it goes a text summary listing the competitions, sequences, playlists, sources, scenes and sponsors inside. **Restore…** shows when and where a backup was made and what it holds, backs up this Mac's settings first, and replaces them only once the whole zip has unpacked and checked out. This Mac keeps its own **Only this Mac can change things** choice, port and link name whatever the backup says. Make a backup after setting up a season and before trying anything new on the Output page.

In detail, with the link for scripts: **Backup and restore** in the Reference.

Updates

Updates download by themselves and install when the app quits, never mid-stream; **Check for updates** looks now, and **Restart to update** appears when one is ready. If a newer version has come out since that one downloaded, it asks whether to get the newer one first or install the one it named. The first launch after an update shows **What's new**, the changes since the version you last opened. When an update changes how settings are saved, Studio backs the data folder up to **backups/** first; settings saved by a newer Studio open read-only in an older one.

Links and other devices

Studio's links use this Mac's Bonjour name (the Local hostname in System Settings > General > Sharing), so the same link works from a phone, another computer or a Stream Deck on the network; `localhost` still works on the Mac. Anything on the network can open the graphics. Only this Mac can change scores, the scene, sequences and settings: **Only this Mac can change things** is ticked every time Studio opens. To score from a phone or let another computer drive, pair it instead: open `/pair.html` on it (the Studio link with `/pair.html` on the end), type a name and press **Ask the Mac**, then choose **Allow scoring only** or **Allow everything** on the Mac. A paired device

works with the box ticked, and **Devices** in the menu bar icon lists it, changes what it may do, or removes it. Unticking the box lets every device on the network in until Studio quits; keep that for emergencies. On a network you don't control, leave it ticked, or block incoming connections to FirstPass Studio in the Mac's firewall.

The link name, pairing and what other devices may do: **Links and access** in the Reference.

When something isn't right

WHAT YOU SEE	WHAT TO DO
The board doesn't move before the game	Normal: nothing is read before the electronic scoring window opens. Scoreboard control says when it will start checking.
The board stops mid-game	PlayHQ isn't answering. Switch to Manual and run it by hand; Follow PlayHQ picks it up again when PlayHQ is back.
The wrong night or game is on	Pick the game night on Scoreboard control and pick the game; tick automatic selection again afterwards.
A key says off air	No Studio window is open. Open Studio from the 1P menu.
A graphic looks stale	Press its Restart key or turn it off and on; with **Load the page fresh** on it reloads each time.
Studio itself stops	**Restart Studio** in the 1P menu brings it back with the scores and clock it had.
Anything else	**Open the debug log** in the 1P menu. Everything Studio does and every PlayHQ answer is in it, 15 MB at most. Send it with a note of the time.

Data and bandwidth

The graphics only ever talk to Studio on the Mac. Studio alone contacts PlayHQ: one small request every 5 seconds for the followed game (or as set, down to one a second), the ticker's other courts every 30 seconds, the fixture about once a minute for automatic selection, and the rest once when a graphic loads. The Production Planner and the Socials tab read the same fixture, results and ladders through the same cache, and finished games' line ups once, so they add nothing; the Fixture board reads other courts' live scores the ticker's way, so it adds nothing while the ticker is on and at most

one request a minute per live game when it isn't. Commentator stats' auto-refresh asks Studio only; its team lists are read at the start and at each quarter break. Finished games are read once and kept. Extra outputs read the same data from Studio, so they add nothing here. It all comes to well under a megabyte an hour, so a phone hotspot is enough for Studio; the stream itself is the bandwidth.

Page by page: **Data and bandwidth** in the Reference.

Quick reference

IN THIS CHAPTER

1 Keyboard
2 Stream night in nine lines
3 Where things are set

Keyboard

WHERE	KEYS
Scoreboard control	A / Z home +1 / −1, K / M away +1 / −1, Shift for +2 / −2, Space starts or stops the clock.
Any Studio window	⌘1 to ⌘9 and ⌘0 go to Output, Production Planner, Scoreboard control, Recordings, Socials, Competitions setup, Sponsors, Athletes, Devices, Docs. ⌘N new window, ⇧⌘O open in the browser, ⇧⌘C copy the page link, ⌘[back, ⌥⌘W close every Studio window.

Stream night in nine lines

1. Open Studio from the 1P menu and keep the Output page open all night.
2. Check the competition in use and the court on Competitions setup.
3. On Scoreboard control, the first game on your court is picked with Follow PlayHQ on.
4. Turn on the video output your switcher takes and show one graphic to see it land.
5. Play a few seconds of music and an animation with sound.
6. Press a Source key and watch the Output page follow it; put scoring back to Follow PlayHQ after testing.
7. Go to the Live game scene and let the sequences run the breaks.
8. Pick finals winners as they happen; turn on lower thirds when people appear.

9 Press Stream close, turn the output off, and send the sponsor report.

Where things are set

TO CHANGE	GO TO
Colours, font, logos, rules, finals format, court	Competitions setup
Which game, scoring mode, score, clock, finals winners	Scoreboard control
What's on screen, scenes, sequences, playlists, music, mics, outputs	Output
A graphic's options (heading, position, division, animate away)	Output, under the source's address
Sponsors, tiers, ads, moments, the report	Sponsors
Photos, positions, games played	Athletes
Each night's court, the commentators and who calls each game	Production Planner
Keys and links, and the devices allowed to change things	Devices
Pictures of a game night for social media	Socials
Backups, iCloud, updates, the debug log	The 1P menu

FirstPass Studio is made by PIVT. The Elite Netball League, its clubs, athletes and sponsors are fictional.

REFERENCE

Reference

Everything in detail, for when the chapters above aren't enough: what the Mac app needs and keeps, every setting on every page, and what Studio asks of the network.

IN THIS CHAPTER

1 The Mac app
2 Links and access
3 The data folder
4 Settings in iCloud Drive
5 Backup and restore
6 Unsaved changes
7 Data and bandwidth
8 Sponsors in detail
9 Competitions setup in detail
10 Athletes in detail
11 The Output page in detail
12 Scoreboard control in detail
13 Socials in detail
14 Recordings in detail

The Mac app

FirstPass Studio needs an Apple silicon Mac (M1 or later), and fits a 13" MacBook's screen. A plain M1 drives one extra display; Pro and Max chips drive more, which fill and key over two displays needs. A Blackmagic output needs Blackmagic Desktop Video, and NDI needs NDI's free runtime (both under The Output page in detail); the app brings everything else with it, ffmpeg for video conversion included.

The app always starts in the menu bar off air, whether opened at login (tick **Open at login**) or by hand, so nothing talks to PlayHQ until you open Studio. On air it serves the graphics and the Stream Deck links, talks to PlayHQ, and restarts by itself if it stops. **Stop Studio** in the 1P menu closes the Studio windows and goes off air.

The app window is like any Mac app's: no address bar, tabs or bookmarks, the Studio header is the title bar (drag it to move the window), and it shows in the Dock and ⌘Tab while a window is open. Links that open a new tab on Studio (a preview, the control or setup page) open another window; links elsewhere (PlayHQ, GitHub) open in your browser. The menu at the top of the screen has **File > Open in Browser** (⇧⌘O) and **Copy Page Link** (⇧⌘C) for the page in front, **Go** for each Studio page (⌘1 to ⌘9, and ⌘0 for Docs), **View** for reload, back and page zoom (in steps, the level shown in the menu; **Zoom In** ⌘+, **Zoom Out** ⌘−, **Actual Size** ⌘⌥0, or pinch on the trackpad), **Help > FirstPass Studio User Guide** for this guide, **New Window** (⌘N), and **Close Studio Windows** (⌥⌘W) in the FirstPass Studio menu.

Links and access

Studio's links (on the Devices tab, Scoreboard control and in the Mac app's menu, **Copy the Studio link** too) start with this Mac's Bonjour name, such as `http://studio-mac.local:8080`, instead of `localhost`. It is the **Local hostname** in System Settings > General > Sharing, so the same link works on this Mac and from a phone, another computer or a Stream Deck on the same network. The port stays 8080, and `http://localhost:8080` still works on this Mac, so links and Stream Deck buttons you already have carry on. To change it, set `"linkHost"` in `config.json` in the data folder: `"bonjour"` (the default), `"localhost"`, or any name or IP address, then choose **Restart Studio** in the 1P menu.

Only this Mac can change things in the menu bar icon sets `remoteChanges`, and is ticked again every time the app opens: ticked, phones, other computers and Stream Decks on the network that aren't in **Devices** can still open every graphic but can't change scores, the scene, sequences, settings or media, or read private data (the settings and scene backups, the athlete register, the debug log, the sponsor reports, the competition setup); Studio answers those with a 403 that says so. Either way a change must come from Studio's own pages or a program (the Stream Deck plugin, a Website key): a request a browser marks as coming from a web page on another site, or that asks for Studio by a name this computer doesn't answer to, gets a 403 too, and so does a read of private data asked for by such a name. A Stream Deck link must carry this Mac's key (`/k/<key>/deck/…`, as Studio copies them; kept in `devices.json`, never backed up) or be signed by a paired Studio plugin, or come from Studio's own pages, so a web page can't press one. A phone, iPad or another computer pairs from `/pair.html`: the Mac offers **Allow scoring only** (the scores, clock, quarter and game) or **Allow everything**, and the device then works while the box is ticked; Devices can change its level or remove it. (Until a plugin has paired, the plugin's older `x-studio-deck` header still counts, so plugins installed before pairing keep working until a new one is allowed.)

On a network you don't control, leave **Only this Mac can change things** ticked, or block incoming connections to FirstPass Studio in the Mac's firewall.

The data folder

Everything Studio saves lives in one data folder on this computer, outside the app, so updates never touch it: `~/Library/Application Support/FirstPass Studio` (the setup page shows where, and **Open the settings folder** in the 1P menu opens it).

- **competitions/**: a folder per competition with its setup (`settings.json`), team logos, competition logo, athlete photos and athlete register, its sequences, run orders and the Production Planner's nights (`sequences.json`, `planner.json`), and **active.json**, the one in use.
- **sponsors/**: every sponsor once (`sponsors.json`), a folder of logos each, and **exposure/**, the log the sponsor report is made from.

- **scene.json**, **outputs.json**, **music.json** and **mics.json**: the sources and saved scenes, the video outputs, the music player and playlists, and the mics, shared by every competition; **media/**: the images, videos and songs they use.
- **control.json**: Scoreboard control's choices (the game night picked by hand, automatic game selection and its court, following the PlayHQ clock, how often to check PlayHQ, showing the clock, the ticker speed, finals winners picked by hand).
- **scoreboard-state.json**: the game on the board, the scores and the clock.
- **config.json** (optional): changes this computer's settings (port, log size, how often to check PlayHQ, `linkHost`, `remoteChanges`) from the app's defaults (see Links and access).
- **devices.json**: the Stream Deck key, link sets and the devices allowed to change things, on this Mac only and never backed up.
- **switcher.json**: the ATEM switcher Studio controls (its name on the network, or the address typed) and the stinger's line-up in frames, one for the Mac whichever competition is in use.
- **app-settings.json**: the Mac app's own choices (where the window was).
- **recordings.json**, **demos.json** (saved demo games), **replay.json** (what a 4K replay follows, and each recording's sync) and **Recordings/** (on this Mac, not in iCloud Drive or a backup): whether Studio records each PlayHQ game on air and for how long after full time, and the recordings themselves.
- **logs/**: the debug log. **backups/**: a copy of the folder made before an update changes how settings are saved.

Settings in iCloud Drive

In the 1P menu, **Settings folder > Move to iCloud Drive...** asks for a folder in iCloud Drive (make one called FirstPass Studio, say) and moves the settings into a subfolder named after this Mac, its **Local hostname** in System Settings > General > Sharing: `iCloud Drive/FirstPass Studio/Studio-MacBook-Pro`. Each Mac signed in to the same Apple account gets its own subfolder, so two Studios never save over each other's settings, and a second Mac with the same name is stopped from using the first one's. Studio goes off air for the move and comes straight back. Before anything moves, a copy of the settings is saved in **backups/** in the folder on this Mac, and the originals are removed only once every file has copied across. If the folder already has settings for this Mac (from before), Studio asks whether to use those or replace them.

Some files always stay on this Mac: the debug log, the scores and clock (`scoreboard-state.json`, saved every few seconds during a game), PlayHQ caches, the app's window positions and `data-location.json`, which says where the settings are. If iCloud has kept only a placeholder of a file (Optimise Mac Storage), Studio downloads it before going on air; to keep it all on the Mac, right-click the folder in Finder and choose **Keep Downloaded**. If the iCloud folder can't be found (iCloud Drive turned off, signed out), Studio says so and doesn't start, rather than starting with no settings. **Settings folder > Move back to this Mac...** moves them back.

Settings folder > Restore from another Mac... copies another Mac's settings (competitions, scenes, sequences, music, Scoreboard control's choices) over this Mac's: pick the Mac from the list of folders beside this one, or choose its folder. The other Mac keeps its own; this Mac's settings are backed up in **backups/** on this Mac first, and are only replaced once the copy has checked out. If this Mac's settings are on the Mac, they go into iCloud Drive too, in a new folder for this Mac beside the other Mac's, and stay there from then on. Studio goes off air for the copy and comes straight back. **Open the settings folder** opens wherever they are now.

Backup and restore

Back up... in the 1P menu saves every setting in one zip: competitions with their logos and athletes, sponsors, scenes, sequences, music and playlists, the media folder, Scoreboard control's choices and `config.json`, wherever they are kept (on the Mac or in iCloud Drive). The debug log, the scores and clock, PlayHQ caches and stream keys stay out. First it asks for notes (what the backup is for, say) and whether to include the media folder (videos, songs), which can make the zip big. The zip is named `FirstPassStudioBackup-2026-10-24_10-46-02.zip`, the date and time it was made, so each one is new and they sort oldest to newest. Beside it goes a summary with the same name, `FirstPassStudioBackup-2026-10-24_10-46-02.txt`: the notes, then the names of the competitions (and which is in use), sequences, playlists, sources, scenes and sponsors. Inside the zip is one folder of the same name holding the settings as they are in the settings folder, the same `summary.txt`, and `studio-export.json` saying which Mac and Studio version made it. Files iCloud hasn't downloaded yet are downloaded first. Studio stays on air while it backs up, and works off air too. Link for scripts: `/api/settings/export` downloads the same zip (add `?media=0` to leave media out, `&notes=…` for notes).

Restore... in the 1P menu puts a backup back: pick a `FirstPassStudioBackup-….zip`, and Studio shows when and on which Mac it was made, its notes and how many competitions, sequences, playlists, sources, scenes and sponsors it holds, then asks before replacing this Mac's settings with it (on the Mac or in iCloud Drive, wherever they are kept now). This Mac's settings are first copied to **backups/** in the folder on this Mac, and only replaced once the whole backup has unpacked and checked out, so a damaged zip changes nothing. Either way this Mac keeps its own **Only this Mac can change things**, port and link name, and the ffmpeg and PlayHQ addresses in its config.json. A backup made without the media folder keeps this Mac's media. The debug log, scores and clock aren't touched. Studio goes off air while the settings swap and comes straight back. A backup from an older Studio is brought up to date as Studio starts; one from a newer Studio opens with competitions read-only until this Mac's Studio is updated.

Unsaved changes

Changes on the Competitions setup page aren't kept until you press **Save**. With changes not saved yet, leaving the page (a tab, or the Go menu), opening another

competition, **Use this one**, **New competition** or **Import a template file** first asks **Save**, **Don't save** or **Keep editing**. **Export as template** asks to save first, since the file is made from the saved settings. Closing the window, reloading or quitting asks whether to leave without saving.

The editors that open over a page (a sponsor, a sequence, a playlist, a mic or a scene's **Switcher** actions on the Output page, an athlete's **Games**) ask the same question: with changes not saved yet, pressing Escape or leaving the page from the Go menu asks **Save**, **Don't save** or **Keep editing**, and closing the window asks whether to leave without saving. Their **Cancel** button still just cancels.

Data and bandwidth

scoreboard.html, **goals.html** and **countdown.html** (and Scoreboard control) follow the scoreboard over the one connection Studio's pages keep open for the scene, so a score press or a PlayHQ reading reaches them the moment it happens, with no asking in between. **ticker.html**, **viewers.html** (every 15 seconds, Studio only) and the automatic **supershot.html** and **streak.html** keep asking for new data; **ticker.html** also asks for its speed every 3 seconds. None of that reaches PlayHQ. **commentator.html** follows the scoreboard live for its clock, moments and run, and with Auto-refresh (on by default there) asks Studio for its stats every 15 seconds and for who is on the mic each minute, neither of which reaches PlayHQ; its team lists load at the start and each quarter break (see below). The super shot and hot streak pages ask Studio every second for the scoreboard and nothing else, so they add no PlayHQ requests. Every other graphic gets its data once when it loads, and again each time it's shown on the Output page (with **Load the page fresh each time it's shown**, the default). The Clash and the winner celebration only retry if Studio isn't running.

Pages only ever talk to Studio on this computer, which uses no internet. Studio is the only thing that contacts PlayHQ, and it caches what it gets:

- Scoreboard following PlayHQ: one small request every 5 seconds by default (**Check PlayHQ every** on Scoreboard control, 1 to 120 seconds), starting when the game's electronic scoring starts (set on the setup page). Nothing is asked before then. Studio makes this request itself whatever windows are open, so the score, the goal order and the game triggers (quarter breaks, full time) keep up even while the scoreboard is hidden and Scoreboard control is closed; the pages share that reading rather than adding to it.
- Studio scene (scene.html and the Output page): nothing. They keep one connection open to Studio, which only sends when the scene changes; the pages in the scene ask for what they normally do.
- Display output (Mac app): nothing. The output windows show the scene page, so they ask for what the scene's pages normally ask for, once per display (fill and key means the pages run twice). The Blackmagic and NDI outputs run them once more, off screen, and each extra output runs the pages turned on for it once more. NDI's video goes over your local network only, never the internet. In 4K nothing more is asked for, but each NDI source carries about four times the pixels (roughly 250 to 400 Mbit/s at

50 or 60 fps, and fill and key is two sources), so use one source with transparency on a gigabit network, and the Mac does about four times the work per 4K output: the Output page's performance line shows whether it keeps up.

- Mics: nothing over the internet. Each output window mixing mics tells Studio what it hears twice a second, and the Output page asks for it twice a second while its Mics section is open; the sound itself stays inside the Mac. A mic on the switcher is a few small fader messages over the switcher's connection when its level changes (about 20 a second while a dip fades).
- Switcher (an ATEM, if one is set up): nothing over the internet. Studio keeps one connection to the switcher on your local network, a few kilobytes a second of small packets each way; looking for switchers asks the local network once.
- User guide (/guide/): nothing over the internet. It comes with Studio, fonts and pictures included, about 4 MB when it opens and nothing after.
- Stream Deck plugin's keys: nothing. The plugin keeps one connection open to Studio while any of its keys is on screen (and one more while a Switcher key is), and Studio only sends when the scene or the switcher changes.
- Recordings: nothing over the internet. Studio writes what it already has to a file on this Mac every 5 seconds while a game is recording, a megabyte or so a game. A practice game asks PlayHQ nothing at all: the recording answers.
- 4K replay following a HyperDeck: nothing over the internet. While a replay follows it, Studio keeps one connection to the HyperDeck on your local network and asks where it is four times a second, a few hundred bytes each time.
- Countdown: nothing. The clock ticks in the page and the start time comes from the game already picked.
- Live viewers (Competitions setup): only while a video output is on, one small request per stream every 30 seconds to YouTube or Facebook, plus a YouTube search when a channel's live video isn't known yet (at most every 5 minutes). Nothing when no streams are set up.
- Last 10 goals: nothing extra. It shares the scoreboard's 5 second reading, and Studio works out the goal order from it.
- Commentator stats: the stats, shooting % included, share the scoreboard's reading, so auto-refresh adds nothing. Its team lists ask for the game's line up when the page comes on and at each quarter break, about four small requests a game; games played, ladder, form, head to head and up next are read once when the page comes on and for each new game (from the season Studio already keeps); games played reads the division's finished games once (kept in `game-cache.json`, as for the Athletes page).
- Ticker: other courts' scores at most every 30 seconds.
- Automatic game selection: the fixture about once a minute for the current round, every 10 minutes for the rest.
- Socials tab: nothing extra. Its previews and pictures read the season's fixture, results and ladders through the same cache, and the Production Planner, sponsors and logos from Studio. Top shooters and Tournament top shooters read each finished game's line up once and keep it, the same as the Top shooters graphic. Finals bound and the Finals

draw (and a finals night's other pictures) read the division's finals the way the Finals graphic does, through the same cache.

- Production Planner: nothing extra. It reads the season's fixture through the same cache as automatic game selection (the current round about once a minute, the rest every 10 minutes), and the commentators graphics ask Studio only. The fixture board's live scores for other courts go through the ticker's reader and cache: nothing extra while the ticker is on, at most one request a minute per live game when it isn't.
- Athletes page: when you open a team, the line up of each of its games this season, a few at a time. Finished games are only ever read once: Studio keeps their line ups in `game-cache.json` in the data folder, so they're kept through restarts too.
- Finals: the division's ladder and its season's rounds when the page loads (the rounds are kept for 10 minutes, the current one for a minute).
- Team lists: warmed up ahead, so they show straight away. When a game is picked (by hand or automatically) and when Studio starts, it reads the game's line up and both teams' season squads, and the next game's teams on the same court too. Finished games come from `game-cache.json`, so after the first time that's only a few small requests per game picked. Squads are kept for 30 minutes; the team list itself asks for the game's line up and score when it loads.

Each answer is a few kilobytes, well under 1 MB an hour in total. Two options make a page poll, both off unless you add them: `?refresh=` on the ladder, and `&reload=` on the scoreboard, scene and Scoreboard control.

Animations run at the same speed at any frame rate, 50p or 60p.

Sponsors in detail

A sponsor's QR code is made on this Mac, with tracking tags for the sponsor's website analytics unless they're turned off; **Save QR code as PNG** gives a copy. **Pull up** and **Pull down** take a sponsor from the nearest tier below or above that has one. The vacancy card is never shown without a link or email for its QR code. The Sponsor report covers any run of nights: time on screen and where, ad plays, time with their QR code up, and progress against their targets. What's on an output counts as on air only while that output is on: one that's turned off or set to Web links only isn't, even while another is. **Print or save as PDF** to send it to them.

Sponsors on the graphics: the naming rights sponsor (or what its empty-tier rule gives instead) is a **Presented by** tab over the home team on the scoreboard, a pill under the title banner and a line under the countdown. The game stats, shooting stats and top shooters cards carry the next sponsor in rotation from Major and up in their footer, the ticker leads with the next from Partner and up, and the end of night card (Up next after the last game) thanks the sponsors from Major and up with their logos (`?sponsors=all` every tier, `?sponsors=0` none). On any of them, `?sponsor=0` leaves the sponsor off and `?sponsor=Golds%20Bakery` always shows that one; set it in the source's options on the Output page. With no sponsors they look as they always did.

Competitions setup in detail

When an update improves one of the ready-made setups (AMMNA Nationals, GIG WANL), yours picks up only what you haven't changed yourself. Section by section:

- **Game rules:** quarters or halves and their length, and super shots, for the whole competition or per division. Under the super shots tick box each PlayHQ division can follow the competition, or have super shots or not. A setting can also be saved for a word instead of a whole division name, and then covers every division with that word in its name ("Open" covers "Division 1 Open"; "Men" covers "Men's" but not "Women's"); a division's own setting wins over a word. GIG WANL comes set for super shots in Open only, not 21&Under or Men's, by the words "Open", "21" and "Men". A division without super shots doesn't play the super shot animation for a +2 press or count goals as super shots, and its shooting stats and top shooters leave the Super column off. **Super shots in the last [x] minutes of each quarter** (default 3) is when the scoreboard shows its "Super shot" tab; 0 leaves it off.
- **Finals:** the competition's finals format, and a different one for any division that needs it. Top 2: grand final, or best of three. Top 4: semis then grand final, Page (double chance for 1 and 2), or a placings day (1 v 2 and 3 v 4). Top 6: byes for 1 and 2, or a double chance for 1 and 2. Top 8: AFL style, straight knockout, or carnival placings (a knockout over three weeks with play-offs, so every team gets a place 1 to 8). Every team: everyone plays on grand final day, 1 v 2 for the title, 3 v 4 for third, 5 v 6 for fifth and so on down the ladder, with nobody knocked out (an odd team out has no game). The ladder marks the finals places by what each earns and the finals graphic follows it.
- **Game night:** the court the scoreboard follows, whether it picks the game automatically, **Electronic scoring starts [x] minutes before the game** (default 10; set 20 or 30 when games are further apart), and how many days ahead counts as this game night. Automatic selection switches to the next game when its electronic scoring starts, and the scoreboard doesn't ask PlayHQ for a game's score before then, even for a game picked by hand: Scoreboard control says when it will start checking. That keeps PlayHQ's "game could not be found or was not electronically scored" error out of the debug log.
- **Colours and font:** nine colours, how see-through the dark panels are, and the font, with a preview. Click a colour's swatch for a colour picker (shade square, hue slider, HEX and RGB boxes, and on Chrome or Edge **Pick from screen**), or type or paste a colour into its box as HEX (#7f4092) or RGB (127, 64, 146). Under **Brand kit colours**, paste any text from a brand kit with HEX or RGB values in it; those colours are saved with the competition and show in every picker, one click to use.
- **Font:** the Font box lists the fonts that come with Studio first (they work on any computer, with no internet), then fonts added to the competition, then every font installed on this Mac (including any you add in Font Book), such as Helvetica Neue, Avenir Next, Futura and DIN Condensed. A Mac font shows only on this Mac: a graphic opened on another computer uses Montserrat instead. **Add a font file** takes TTF, OTF, WOFF or WOFF2 files (one per weight, or one variable font); they're saved with the competition and go in its template file, so it looks the same on any Mac. Add only fonts you're allowed to use, such as open licence fonts from Google Fonts. Any letter a font

lacks comes from Montserrat. Every font is sized and centred to match Montserrat, which the graphics were drawn with, so text fills its boxes the same whichever font you pick; narrow fonts stand a little taller so they don't look small. **Text size** (80 to 130%) makes a font other than Montserrat bigger or smaller on every graphic; keep an eye on long team names.

- **Logos:** the competition logo, in a light mode version (for a light picture behind it) and a dark mode version (for a dark one; if you add only one, it is used for both), and the team logos. The logos page and title use the light mode logo unless `?mode=dark` is added; the ladder's corner, on its coloured header, uses the dark mode one unless `?mode=light` is added. Drop team logo files on the page; they match clubs by file name, and the PlayHQ team list shows which logo each team gets, with a box to pick a different one.
- **Clubs:** a team's logo and colour belong to its club, so every team of a club in every division shares them, and changing one row changes them all. The club is the team's name without the division on the end ("Comets 21U", "Comets Men" and "Comets Open" are all Comets; "Stormz Div 2" is Stormz). The **Club** box in the team list links a team to a different club, or to a new one.
- **Club colours:** in the same PlayHQ team list, **Add colour** gives a club its own colour. The scoreboard puts it behind its teams' names and The Clash behind their logo and name, with dark or white text, whichever reads better. Clubs without one (or after ✖) use the competition's colours. The match up page always uses the competition's colours. Pick the colour with the colour picker or paste it as HEX or RGB.

Use this one switches every graphic, Scoreboard control and the Stream Deck to that competition: the game night goes back to automatic and the scoreboard starts on that competition's first game. Saving the competition in use reloads the graphics with the changes. Setups are kept in the **competitions** folder (see The data folder, above).

Athletes in detail

Teams are listed by division, so a club with a team in more than one division (Knights) has a button in each, showing that division's line ups; **All athletes** lists everyone once. A silhouette's gender is the one picked on the page if you've picked one, else PlayHQ's gender for the athlete, else the PlayHQ gender of their division when they play in a men's or women's division (mixed and open divisions leave it neutral). **Silhouette** on each athlete shows where it came from and can override it. **Approve none** takes photo approval away from every athlete listed. Games played can be kept per club as well as per division.

The Output page in detail

- **Sources: Add page** adds any Studio graphic or web address; **Add image** adds a PNG, JPG, GIF, WebP or SVG, and **Add video** an MP4, WebM or MOV (saved in the data folder's **media** folder). Each opens a box to pick it. A new source goes above the selected one (on top if none is selected), hidden, with its settings open under its row so

you can set it up straight away. Pages not in the scene yet are listed first, and **Add all unused pages** adds every one of them, hidden. Each row has the source's On / Off (a pill per output once there's more than one), an eye to preview it on its own and a pencil for its settings (name, address and page options, size, fade, position, its Stream Deck links, **Move up**, **Move down** and **Remove**); **Done** closes them. The list is top layer first: drag a source up or down to move it between layers.

- **Export sources** saves the scene's sources as a .zip, with the images and videos they use (untick **Export with their images and videos** to leave them out), to keep as a backup or move to another Mac. **Import sources** replaces every source in the scene with the ones in an exported .zip (or its `scene.json`), after asking. Imported sources start hidden and keep their Stream Deck keys. Their images and videos go in the media folder; a file already there with the same name but different contents isn't overwritten, the imported one is kept as ***name (2)*** instead. Exported without media, the files have to be in the other Studio's media folder already; the Output page lists any that are missing.
- **Page options:** a Studio graphic's link options listed on the Devices tab (position, heading, division, round, animate away and so on) show as fields under its **Address**, and are saved in the address.
- **On / Off** fades a source in or out with its own **Fade** time in milliseconds (0 is a cut).
- **Position and scale** place it on the frame; **Full frame** fits it to 1920 × 1080. **Page size** is the size a page is drawn at (1920 × 1080 for full-frame graphics).
- **Load the page fresh each time it's shown** (on by default) reloads a page as it fades in so its animation replays, and unloads it when hidden. Untick it to keep a page running so it shows instantly. **Refresh page** reloads it now.
- **Animations that animate away switch themselves off:** once a graphic has animated away and left the frame clear, its source is switched off in the scene (the Sources list and its Stream Deck key show it off), rather than staying on as an empty layer. That's The Clash and the winner animation with **Animate away after** set, a person lower third with **Animate away after**, positions with **At the end: Animate away**, the logo animation and the break ad with **At the end: Animates away** (their default), and the super shot and hot streak animations when they play once. Pages that stay on screen or wait for something to happen (super shots and hot streaks playing by themselves, the countdown, the scoreboard) stay on. This needs a scene window open (the Output page's Live View, its pop-out or a display or DeckLink output) to see the animation finish.
- **Preview** opens the selected source on its own in a new window, the way the Preview buttons on the Devices tab do; the eye beside a source's On / Off does the same from the list. A preview opens at half size (960 x 540): the page is still laid out at 1080p, just scaled down, on a checkerboard so the see-through parts show, so a few fit on one screen. Resize the window to make it bigger.
- **Videos** start from the beginning each time they're shown (untick to carry on), can loop, and fade out when they finish unless looping. A video that fades out when it finishes is switched off in the scene, so the Sources list and its Stream Deck key show it off and the next On plays it again. **Sound** turns a video's sound on or off, with a volume; it plays

through the Mac's audio output, or in the SDI of a Blackmagic output. **Restart video** starts it again.

- **ProRes:** the pages can't play ProRes, so a .mov added to the media folder is converted to a WebM beside it (keeping a ProRes 4444 clip's transparency; the Output page warns when a .mov has none to keep, such as H.264), and sources using it switch over when it's done. The original is kept. The Mac app includes ffmpeg to do this (running from a copy of the code instead, run `vendor/ffmpeg/build.sh` once or `brew install ffmpeg`). MP4 and WebM play as they are.
- **Media folder:** the Output page's **Media folder** (at the bottom, folded until you open it) lists every song, video and image added (under Music, Add image or video, or on the Sponsors page), with its length, size and where it's used: sources, page themes (`?theme=`), sequence music (in any competition), playlists, the super shot song, the music player and sponsors' ads. **Remove** deletes a file for good (with a levelled song's original) after asking, and says where it's used first. A file on air (a source showing it, or the song playing) can't be removed until it's off. Playlists, the super shot song and sponsors' ads let go of it; sources, page themes and sequence music keep its name, shown as ***(missing)***, so their settings and Stream Deck keys stay until you pick another file. A .mov and the WebM converted from it are separate files.
- **Sequences:** play the scene's sources one after another, for the run-up to a game, a quarter break or full time. **+ New sequence** on the Output page opens a list of steps: each step turns on the sources you pick and stays **for a set time** (seconds), **until it finishes** (its videos end, or pages that switch themselves off when done; it moves on after the video's length plus 3 s, or a minute for a page, if the end never comes), **until the game moves on** (the clock starts, or the next break or full time) or **until I press Next**. As a step ends, sources the next step doesn't use go off with their own fades. A step can also be a **scene** (+ Add scene: a starting or reset point; what the scene turns on stays on under the next steps and after the sequence; it goes straight on or holds like any step) or another **sequence** (+ Add sequence: its steps play there, so a mini sequence is built once and used in many; a sequence can't go inside itself). A sequence can be all sources, all scenes, all sequences or a mix. Tick **Hide** to keep one off the list (a mini sequence used inside others); it still plays inside them, when its start comes and from the Stream Deck, and **Show hidden** brings the hidden ones back to the list. **Starts** is by hand, N minutes before the picked game's scheduled start (only if Studio was running at that moment: opened later, the sequence is left for you to play by hand), at the breaks you tick (quarter, half and three quarter time) or at full time, a few seconds after the siren (from the scoreboard's clock, the PlayHQ scorer's or yours; no extra PlayHQ calls). While it waits to start its card says **Starting** with **Start now** and **Cancel**. **Automatic sequences on** above the cards turns all of that off and on (remembered). **Repeat** plays it once, until stopped, until play resumes or until N minutes before the start (it finishes the step on screen, then ends), and **Then** goes to a scene, turns off what it turned on, or leaves everything as it is; a sequence that plays on more than one output has a Then for each. A sequence only turns off sources it turned on itself. While one plays, its card shows a dot per step, the step's time left and the whole sequence's **Total** (**at least** when part of it waits for Next, the game or a page; **This time round** when it repeats), with **Next** (end the step now), **Pause** and **Stop** (ends it and does its Then). Going to a scene, playing another sequence or hiding

everything stops it without its Then; turning the step's source off by hand moves to the next step. One plays at a time, it keeps going with the Output page closed, and nothing is playing after Studio restarts. Sequences belong to the competition in use (saved in its folder as `sequences.json`) and change with **Use this one**; the sources and scenes are shared. Stream Deck: `/deck/sequence/<name>` plays one (`/toggle` stops it on a second press, `/stop` stops it), `/deck/sequence/next`, `pause` and `stop` act on whatever is playing (`/deck/sequence/advance` is Next only when the step on now is **Until I press Next**, and does nothing otherwise), `/deck/sequence/now` and `cancel` on one about to start, and `/deck/sequence/auto/on`, `off` or `toggle` set the switch. The break lengths (Competitions setup, Game rules) let the editor say how many times a break sequence fits in each break.

- **Run order:** tonight's plan, above Scenes and Sequences: a numbered list of sequences and scenes for the picked output, in the order they play (Run-up, Walk out, Live, Quarter time, Live, Half time…). **Add…** puts a scene or sequence at the end; rows drag, or move with their arrows, and **×** takes one out. **Play next** plays the head now (a sequence starts, a scene is gone to) and **Skip** drops it without playing it; stepping through a sequence is the Sequences panel's Next. A sequence that ends looks at the run order first and plays what is queued after it; its own **Then** runs only when nothing is queued, so a sequence built for one night still works alone on another. Whatever is at the head leaves the list as it starts, however it was started. A game trigger puts its sequence at the head (with its few seconds and Start now / Cancel as usual) rather than starting it blind; Cancel takes it out again, and **Next** plays it when the ball is dead. Playing a scene or sequence by hand leaves the list alone unless it was the head, and nothing in the list starts by itself. The status board's **Next** line and the bar's chip say what Next will do. A sequence the run order starts plays on every output it plays on together when **Play next** is pressed; when the list moves on by itself, another of its outputs joins only if nothing is playing there and it didn't just play the sequence that ended (that one does its own Then). Its row says, for each other output, what follows it there. **Saved…** keeps tonight's order under a name (**Save tonight's order as…**), loads one in place of what is queued, deletes one or clears the list; the list and the saved orders live with the competition's sequences, so a relaunch picks the night up where it was. Stream Deck: `/deck/runorder/next`, `skip`, `clear`, `add/<sequence or scene>` and `load/<saved order>` (`/deck/output/<output>/runorder/…` for another output's list); the plugin's **Run order** key presses these, names what plays next in its colour and lights while something is queued. The existing `/deck/sequence/next` still ends the step on screen.
- **Music:** the Output page's **Music** player plays songs from the media folder (MP3, M4A, AAC, WAV or AIFF; use music you're licensed to stream, as Apple Music songs are copy-protected), one song or a **playlist** (shuffled or not, starting again after the last or not). **Add a song** with **Even out its loudness** ticked levels it to the same loudness as the others and keeps the original in the media folder's `originals`. It plays wherever video sound plays (the outputs, or the Scene View while Video output is off), never twice. Play fades in over the **fade in** time; Pause and Stop fade out over the **fade out** time (slower by default); **Duck** (lower it, keeping it running) goes down over the fade out time and back up over the fade in time (or half of each: a setting); **Mute** keeps it running. While a video with its sound on plays, the music fades out and pauses, then fades back in from

where it was (or lowers, or carries on: a setting). A **sequence** can bring its own music: a song or playlist starting at one of its steps, from a **start mark** (type it, or **Listen** and press **Mark here**), with a fade in; it fades out so it's gone as the sequence ends (when its last step has a set time; ended early, it fades from then), or cuts as it ends, or **carries on into the next sequence** when the sequence's Then plays another one (Run-up then Walk out share one song). A sequence with no music of its own plays the music of a sequence inside it while that one plays, from its step, ending as that sequence's music ends. Going to a scene or another sequence fades a sequence's music out over its **Music ends** time (or cuts it, if set to cut). Tick **Quiet the music** on a scene (Edit scenes) or a sequence for an interview on court: whatever music is playing (the player's or a sequence's) ducks and keeps playing until the next scene, sequence or sequence step without the tick. A **super shot song** (a music setting, with its own volume) cuts in, no fade, when the super shot period starts in a division with super shots and fades out when the period ends. Stream Deck: `/deck/music/toggle`, `play`, `pause`, `resume`, `stop`, `fade`, `next`, `prev`, `/deck/music/mute/toggle`, `/deck/music/duck/toggle`, `/deck/music/load/<song>` and `/deck/music/playlist/<name>` (`?play` starts it too); `?fade=2` on any of them fades over 2 seconds, `?fade=0` cuts.

- **Mics:** live mics plugged into the Mac (wireless receivers for effects mics on court or crowd mics, directly or through an audio interface) go out with the music and video sound. Add each one in the Output page's **Mics** section: a name, the input (**Find inputs** lists the Mac's, in the Studio app) and the channel (both mixed to one, or left or right, so two receivers on one stereo input are two mics). Each mic's **Goes to** ticks the outputs it's mixed into (Output 1 unless set, shown with more than one output), so a commentators' feed can take the effects mics and leave out the commentators' own; it's heard while those outputs are on, and never in the Scene View, so a speaker near a mic can't feed back. The first time, the Mac asks whether FirstPass Studio may use the microphone. Each mic has a level (100% as it comes in, up to 200%), a mute, a **Low cut** for wind and rumble (on by default) and a meter; **Level** and **Mute all** are over all of them, and a limiter stops a shout clipping the stream. Like the music's duck, the mics **duck** to the **Duck to** level (30% by default) while music plays, while a video with its sound on plays, and while a scene or sequence with **Quiet the music** is on (each a tick box), going down over 1 second and back up over 2; **Duck** ducks them by hand. A mic marked **Has an external mixer** never ducks, since the mixer does the fading, and stays at its level. A device that isn't plugged in says so on its card. A mic can be plugged into the **switcher** instead (an ATEM Mini's Mic 1 or Mic 2): pick **Plugged into: The switcher** and its input. The switcher mixes it, and Studio moves its fader for the same level, mutes and ducks (fading in small steps), changing nothing else on it: set its On, EQ and dynamics in ATEM Software Control. Moving its fader on the switcher or in ATEM Software Control makes that the mic's level, and Studio takes the faders as it finds them when it connects. Left and Right need the input split into two mono channels in ATEM Software Control. These mics are heard whatever Studio's outputs are doing. Stream Deck: `/deck/mics/mute/toggle` (`on`, `off`), `/deck/mics/duck/toggle`, `/deck/mics/up` and `/deck/mics/down` (10% a press, `?by=5` for another step), `/deck/mics/level/80`, and one mic by name or number: `/deck/mic/court-mic/mute/toggle`, `/deck/mic/1/up`, `/deck/mic/1/down`.

- **Scene View:** the Output page shows the scene as it is right now (your switcher decides whether it's on air). It runs the scene's pages a second time, so untick **Live View** when the Mac is busy. The Scene View plays the scene's sound only while **Video output** is off (for testing); with a display, Blackmagic or NDI output on, the output carries the sound and the Scene View is silent, so it never plays twice. **Pop out** shows the Scene View in its own window to resize, move to another screen or make full screen while testing (with the same sound), with a floating Sources list to show or hide each source from there; the Output page's own Scene View turns off while it's open and comes back when it closes.
- **Display output (Mac app):** sends the scene full screen to the Mac's extra displays (HDMI into a switcher). **Chroma** puts it over a solid colour on one display (green, blue or any colour) for a switcher that keys on that colour. **Fill and key** uses two extra displays: fill is the scene over black, key is white where the scene shows and black where it's see-through, for the switcher's fill and key inputs. It's only offered with two extra displays connected (a plain M1 drives one; Pro and Max chips drive more). Set those displays to exactly 1920 × 1080 (not HiDPI or scaled) in System Settings, Displays, or the scene looks soft; the Output page warns when they aren't. Turn the switcher's **Pre Multiplied Key** on for clean edges (with the switcher set up, Into the switcher as checks it and can set it). Each output opens full screen in its own window; leave full screen (green button or ^⌘F) to move it to another display or use other apps. While the output is on, its settings are locked: turn it off to change them. **Output on/off**, Esc on an output window, closing it, or `/deck/output/on`, `off` and `toggle` show or blank the displays without changing the setup.
- **Blackmagic output (Mac app):** sends the scene out of a Blackmagic UltraStudio or DeckLink: **fill and key** with the device's external keyer (fill on one SDI out, key on the other) or **fill only**, in the video format you pick (1080p only; pick the one the switcher runs at, usually 1080p50). A format marked ***(device says it can't)*** is one the device reports it can't send (older UltraStudios with HD-SDI rather than 3G-SDI stop at 1080p30); you can still try it, and the Output page says if it fails. The fill is premultiplied: turn the switcher's **Pre Multiplied Key** on for clean edges (clip and gain aren't used then), and leave **Straight** off: straight fill into a Pre Multiplied Key leaves a light fringe round logos and text. Tick Straight only for a switcher that can't do a premultiplied key (every ATEM can); ticking it asks if you're sure. **Sound** goes in the SDI as 48 kHz stereo, in time with the picture (with fill and key, on the fill), so on the switcher turn on the sound of the input the fill comes into (On, or AFV to follow the picture); an SDI to HDMI converter passes it on. It can go on the Mac's sound output instead, or both. It needs Blackmagic Desktop Video installed (the Output page says if it isn't, with where to get it; quit and reopen Studio after installing) and the device free, so quit any other app using it first. Output on/off and the `/deck/output` links work the same.
- **NDI® output (Mac app):** sends the scene over the network as NDI sources, in the video format you pick (1080p at 23.98 to 120 fps, the same rates as a Blackmagic device, or 720p at 25 to 60 fps): **fill and key** as two sources (***Source name Fill*** and ***Source name Key***) for a switcher's fill and key inputs, **chroma** as one source over a solid colour (the same colour setting as the chroma display), or **with transparency** as one source with alpha for receivers that key on NDI's own transparency. They show up on the network as this Mac's name with the **Source name** in brackets (the output's name, for

example ***Output 2 Fill***, and it follows the output when it's renamed; tick **Use my own name** to type another; a source that changes name has to be picked again on the receiver). Each source only sends while something is receiving it, and the Output page says how many are. It needs NDI's runtime for Mac installed (free, the libNDI installer from ndi.link/NDIRedistV6Apple; Studio also uses the copy inside NDI Tools' apps if it finds one, as long as that copy is signed by NDI's makers, Vizrt or NewTek: a library runs inside Studio, so an unsigned file by that name is left alone and the log says so); the Output page says if it isn't, with where to get it, and finds it as soon as it's installed. Use a wired network: each 1080p source is roughly 100 to 150 Mbit/s at 50 or 60 fps and about twice that at 100 or 120 (fill and key is two of them), more than Wi-Fi carries steadily, so over Wi-Fi the receiver loses frames and moving graphics such as the ticker stutter. Pick the frame rate your switcher or stream runs at (1080p30 for a 30 fps stream): a higher one costs more of the network, and the receiver drops the extra frames anyway. Once a minute while something is receiving, the app log says how many frames a second went out and how many new pictures the scene drew; both at the full rate with a choppy picture means the network lost them. Sound plays through the Mac's audio output, not over NDI. Output on/off and the `/deck/output` links work the same. NDI® is a registered trademark of Vizrt NDI AB.

- **Picture, HD or 4K (Mac app):** the **HD** and **4K** buttons at the top of Video output set the picture for every output at once. In 4K each NDI and Blackmagic output sends 2160p at the frame rate picked for it (2160p50 for 1080p50) and keeps everything else: its scenes, sources, NDI source names, Stream Deck keys and links, and its switcher key. Press HD and it's exactly as before. The graphics keep their 1920 × 1080 layout and are drawn at twice the density, so Studio's text, shapes and logos come out sharp; pictures and videos are only as sharp as their files (a 1080p video or a small logo looks softer). A Blackmagic device without that 2160p format (an HD UltraStudio), or a 720p NDI format, stays as it is and its card says so; displays are left alone, as a 4K screen set to look like 1920 × 1080 already draws at 4K. In 4K a Blackmagic output's video format list also has the device's own 4K formats (2160p50 and so on, as the device lists them) to pick directly; HD hides them again and sends the 1080p format at the same rate. Use 4K only with 4K gear: an HD switcher can't take 2160p. Any graphic's own link also draws sharp at 4K: in a browser source or window set to 3840 × 2160 it draws itself at twice the size, with nothing added to the link.
- **More than one output (Mac app): Add an output** sends different graphics to up to four outputs at once, for example the downstream key on the Blackmagic and an upstream key on a camera over NDI, so those graphics switch with the camera. Each output is set up like the first (a chroma display, fill and key displays, the Blackmagic device or NDI) and has a name, a number and a colour: 1 orange, 2 pink, 3 blue, 4 green. Two outputs can't share a display, the Blackmagic device (one output) or an NDI source name: the earlier output keeps it and the Output page says so. Several outputs can each send NDI, each under its own source name (its own output name unless one is typed), and two chroma outputs can go to two displays over HDMI, each with its own key colour. **Output 1 is the primary**: the plain `/deck/layer` links act on it. **Every output plays its own sound**: the videos and animations on it, the music when **Music plays on** ticks it (Output 1 unless set; a sequence's music plays on every output playing the sequence), and the mics whose **Goes to** ticks it. **Scenes and sequences are**

shared by every output, and each output plays its own: a click on a scene or sequence plays it on the output picked over the Scene View. Each output has its own current scene and plays one sequence at a time, so the same scene or sequence can be on two outputs at once; coloured dots on scene cards, and on sequence cards each output's tick in its colour with how it is there (Playing, Paused, Starting or Stopped (it ended there, played out or cut short); a hollow dot and "off" while that output isn't sending, as it still plays there for when it's turned on), show where. **Hide all** (under Sources) clears one output only and its label names it: the one the list shows when **Show** is set to one output, else the picked output. A sequence's **Plays on** ticks (on its card, in each output's colour, and in its editor) pick the outputs it plays on together: started by itself, or played on any of them (a click with that output's tab picked, or its Stream Deck key), it starts on all of them at once, and each output has its own **Then** for when it ends there (Output 1 back to the live scene, a camera's keyer output off, say). Played on an output it isn't ticked for, it plays there alone, and `/deck/output/<output>/sequence/<name>/only` plays it on that output alone even when it's ticked for more. Stop on a card, `/deck/…/sequence/stop` and Next act on one output; `/toggle` and `/deck/…/sequence/<name>/stop` stop it on every output it plays on, each doing its own Then. A sequence plays on its outputs whether or not they're sending, so turning one on mid-sequence shows it where it's up to; an output it wasn't started on doesn't join in until it's played again. An output it plays on that is removed is left out (stopping what played there); with all of them removed, the sequence doesn't start by itself (rather than going to air on Output 1) until another is ticked, and the log says so. A Sponsor ads step off Output 1 plays its ads in that output's own Sponsor ad sources (named after it, and removed with it). Only Output 1 and outputs with **Counts for sponsors** ticked (on the output's card, off to start) go in the sponsor report, as Studio can't tell whether a camera's keyer or a commentary monitor is on the stream; time on a ticked output is listed as "Ladder (Wide camera)", and a sponsor on two outputs at once counts once. A Sponsor ads step on an output that doesn't count shows whose turn it is without taking it, so the ads on air keep their order. `/deck/scene/<name>` and `/deck/sequence/<name>` play on Output 1; `/deck/output/<output>/scene/<name>`, `next`, `prev` and `again`, and `/deck/output/<output>/sequence/<name>` (with `/play`, `/toggle`, `/stop` or `/only`), `next`, `advance`, `pause`, `stop`, `now` and `cancel` play on that output. Sources are built once and shared; each output has its own on and off for every source, as coloured numbered buttons in the Sources list (a new output starts with everything off). **Show** above the list narrows it to one output, the tabs over the Scene View show each output, and **Pop out** opens the one picked. Stream Deck: `/deck/output/<output>/on`, `off` and `toggle` for one output (its name as a link, or its number), `/deck/output/all/…` for every one, and `/deck/output/<output>/layer/<source>/on`, `off`, `toggle` and `/deck/output/<output>/layers/off` for one output's sources (the Source settings list them). Each output with its own graphics runs its pages once more, so watch the Mac's load with three or four: while an output is on, its card shows its CPU (a percentage of one processor core), memory and the frame rate its pages draw at against what they should (in amber when it falls under 90%), with Studio's total and the graphics processor under the status, and the log has a ***Performance*** line once a minute.

- **Chroma key safe:** a chroma key cuts out one colour, so anything half see-through (see-through panels, soft shadows and glows, fades, soft edges) mixes with that colour

and keys with a fringe or holes. With **Chroma** (a display, or NDI's chroma signal) picked, **Chroma key safe** is ticked unless you untick it, and the Output page warns that the sources are switched to it: every graphic and animation goes out with solid panels, no soft shadows, glows, light rays or flashes (hard shadows stay, made solid), and nothing half see-through, so fades become cuts (sources cut instead of fading) and soft edges are hard. Options that only fade, like the countdown's fade and the ad's glow, are skipped, and the source's options say so. Titles, logos, light and dark modes and every other setting work as usual. It never changes a saved setting: pick another output, or untick it, and everything is back as it was. The Scene View shows the scene chroma key safe while it's on. The Output page also lists any competition or club colour close to the key colour (a keyer would cut holes in it); logos can't be checked, so look at those. For a graphic used on its own at its link and keyed on a colour, add `?chromaKey=true` to its link.

- **Switcher (a Blackmagic ATEM):** the **Switcher** card under Video output controls an ATEM over the network: the Mac and the switcher on the same network, ideally wired. **Set up**, then **Look for switchers**, finds every ATEM on the network the way ATEM Software Control does (the first time, macOS asks to let Studio use the local network: say Allow); pick one and Studio connects and remembers it by name, so it finds it again if the venue's router gives it another address. **Or type its address** (as set in ATEM Setup; an ATEM ships as 192.168.10.240) works too. The card shows the switcher's inputs by the names set in ATEM Software Control, red on program and green on preview: click one to put it on preview, then **Cut** or **Auto** (the switcher's own transition) to take it. **Fade to black**, and each upstream and downstream key on and off (red while on air; a downstream key's **Auto** takes it at its own rate), are there too. Studio reads what the switcher has and offers only that:
  - **ATEM Mini and Mini Pro:** 4 inputs, one upstream key, one downstream key. Fill and key into the downstream key takes two inputs and leaves two cameras; a chroma key needs the upstream key (one input, three cameras left).
  - **ATEM Mini Extreme:** 8 inputs, four upstream keys, two downstream keys.

  A link or button for something the switcher doesn't have (a second downstream key on a Mini Pro, say) is refused with what it does have.
  Into the switcher as: each output's setup (under Video output) says which of the switcher's keys its graphics come into and which inputs its fill and key are plugged into (a chroma output needs just one). The usual key is downstream key 1 for Output 1 and upstream keys 2 to 4 for Outputs 2 to 4; a chroma output goes on an upstream key, as only those key on a colour. Pick another key, or ***Not keyed by the switcher***. Studio then checks that key's settings in the switcher, read only, and lists what doesn't suit the output: its fill and key sources, its type (Chroma for a chroma output, Luma for fill and key) and **Pre Multiplied Key** (on), and asks for a Blackmagic device's **Straight** to be off. It also says when the output can't be keyed as it is: a chroma output on a downstream key, NDI with transparency or fill only (an ATEM needs a key signal or a colour), two outputs on the same key, or a key the switcher hasn't got. **Set up this key for me** changes just those settings on the switcher (and unticks Straight, which restarts the Blackmagic output for a moment), after asking; it leaves the key's clip, gain, mask, position and whether it's on air alone, and the log says what it changed. For a chroma key, pick the key colour in ATEM Software Control with its sample picker. Nothing is

sent to the switcher until it's pressed.
Cameras per output: an output on an upstream key can say which cameras its graphics **go with** (ticked under Into the switcher as; the outputs' own fill and key inputs aren't offered). Studio's cuts, autos and stingers, from the Switcher card, a Stream Deck key or a sequence or scene, then take that key on with those cameras and off with the others in the same transition, so the camera and its graphics change on the same frame (Studio picks the keys in the switcher's next transition and puts the BKGD and KEY buttons back as they were afterwards). A straight-to-program press brings the key straight after. With **Keys follow the camera** on (the Switcher card, on unless turned off), a cut made on the switcher itself or in ATEM Software Control brings the keys a frame or two later. It only acts on a change of camera, so a key turned on or off by hand stays as it is, and it sends nothing on connecting. An output with no cameras ticked, on a downstream key (over every camera), or whose key check shows a problem is left alone. The output's card says ***On air with Bench*** or ***Waiting for Bench***, and the Switcher card underlines each key in its output's colour. The switcher's own buttons, ATEM Software Control and other controllers keep working alongside Studio, and the card follows what they do. Studio sends nothing to the switcher by itself: not when it connects, and not when it reconnects. If the switcher stops answering (a pulled cable, a power cut), or is off when Studio connects, the card turns red within about five seconds and Studio keeps trying; anything pressed meanwhile is refused, not saved up for later, and the log says so. Studio connects when it goes on air and lets go when it goes off air. Stream Deck: `/deck/switcher/cut`, `auto` and `ftb`; `/deck/switcher/preview/<input>` and `program/<input>` (its number, or its name such as `Camera%202`); `/deck/switcher/usk/<n>/on`, `off` or `toggle`; `/deck/switcher/dsk/<n>/on`, `off`, `toggle` or `auto`; and `/deck/switcher/macro/<n>` (a macro saved on the switcher, numbered from 1 as in ATEM Software Control). The Stream Deck plugin's **Switcher** key presses any of these and shows the tally: a camera's key lights red while it's on program and green on preview, a key or Fade to black red while on air, and every key goes dark with a yellow dot while the switcher isn't answering.
Stinger: with the Stinger page as a source in the scene, the card's **Stinger** button (and `/deck/switcher/stinger`, `/deck/switcher/stinger/<input>`) plays it on the output on a downstream key (Output 1, unless another output that's switched on is Into the switcher as a downstream key, as only a downstream key covers the whole picture) and cuts to the preview input underneath it. The stinger tells Studio when it starts and when it covers the frame, from its own settings, and the cut lands in the middle of the cover (0.72 s in with the usual half second hold), so the cut comes about a second and a half after the press, once the page has loaded. That output's picture reaches the switcher a few frames late (more over NDI than HDMI): set its **line-up** in frames (each output keeps its own, so moving the stinger to another output keeps both) (−25 to 25, at the switcher's own frame rate) to move the cut, and press **Test the line-up** before you're live: it cuts just 2 frames after the bands cover the frame, so if the old picture flashes before them, add frames (run it two or three times and go by what you see each time: one run can be a frame out either way). Only that output's own video output starts the timing, not a preview or another output, so a stinger needs that output on. With no stinger in the scene, a stinger is an auto; if the page doesn't start within 3 s, it's a plain cut, and the log says which. A second stinger while one plays is left out (a press, a Stream Deck key, or a sequence or scene's own Stinger to), so a

double tap never makes two cuts. **Stinger to** a camera that's already on program plays the stinger with no cut (for a change of graphics under it). In a sequence or scene, actions at the same moment as a stinger wait for its cut, so a key change lands under the bands too.
Switcher actions in sequences and scenes: once a switcher is set up, a sequence's sources or scene step has a 🎥 button, and each scene has 🎥 **Switcher** under Edit scenes. Each adds up to eight actions: **Cut to**, **Auto to** or **Stinger to** an input (or what's on preview), **Put on preview**, an **Upstream key** or **Downstream key** on, off or toggled (a downstream key can go **at its rate**), **Fade to black** or back, and **Run macro**, each as the step starts or the scene is picked, or up to 60 s in (in tenths of a second). Actions at the same moment go in order, so a preview then a cut works. A step's actions still to come are called off when the step ends or the sequence stops; a scene's when another scene is picked. A scene step does the scene's actions too. They happen only when the scene or sequence plays on Output 1 (its picture is the one going into the switcher); on another output it plays without touching the switcher. The editor offers the inputs and keys the connected switcher has; one it hasn't got (upstream key 2 on a Mini Pro, say) is shown in red and skipped, with a line in the log. If the switcher isn't answering when an action's moment comes, it's skipped and logged, never done late. Scenes and sequences with switcher actions show 🎥 by their name.

The scene only changes when you change it, so it adds no PlayHQ requests beyond those of the pages in it.

Scoreboard control in detail

Every graphic uses the game night, game and court set on the Scoreboard control. That page is laid out for the operator: the game on the board in one bar with **Follow PlayHQ** or **Manual** and the next game the board will switch to, then the two scores and the clock as the biggest things on the page, with their keys printed on the buttons (**A**/**Z** and **K**/**M**, **Shift** for a super shot, **Space** for the clock); the graphics, team lists, finals and the rest are tabs beneath, with a preview of the board beside them, and the game's own set-up (night, court, automatic selection, how often PlayHQ is read) is the **Game** section at the bottom, which **Change game…** opens and which folds to its title. A setting added to a page's own link (like `?division=2` or `?round=6`) overrides it for that page only.

- **Game night:** the games list has every division's games on one date, whatever round each division is up to, so one division on Round 10 and another on Finals Round 1 the same night are listed together, each game with its own round. **Automatic** picks the first game night in the next 5 days, counting today, so a Tuesday night is picked from the Friday before (the number of days is on the competition setup page). Otherwise it's the last night played. Each night in the list names its rounds. Pick a night to use it, then set it back to **Automatic**. Pages that show one round (like the fixture) use each division's round on that night. On a day with several rounds, it moves on to each round as its first game comes up.

- **Always a game on:** when Studio starts, or the scoreboard is cleared, it puts a game on straight away with **Follow PlayHQ** on: the game for now on the court, otherwise this game night's first game on that court, otherwise its first game. The board is only left empty if PlayHQ can't be reached or the competition isn't found (check the PlayHQ link and competition on the competition setup page).
- **Game:** tick **Pick the game automatically for court** and set the court (it starts on the court set on the competition setup page, e.g. `Indoor 01` or just `1`). It shows the first game of the day on that court, then switches to each next game when its electronic scoring starts (10 minutes before by default, set on the setup page), with the quarter back to Q1 and a full clock. A final (a round with "Final" in its name) stays on until PlayHQ marks it finished, in case of extra time. With a game night picked by hand, it follows that night's games on the court by their time: the game due now, otherwise the next one to come, otherwise (a night already played) its last game.
- **Production Planner:** the Production Planner plans the court Studio streams each game night and who is on the mic for each game. When it has a court for tonight (the game night picked by hand, otherwise today), that court is set here once when the night arrives, which also ticks automatic selection, and again if the plan's court changes. The **Commentators** graphic and **Commentator stats** show whoever is planned for the game on the scoreboard, and a night that hasn't been planned takes the plan of the night before it. It says ***from the Production Planner*** beside the court. A court changed or a game picked by hand afterwards stays, and the Planner flags it.
- **Overrides:** picking a game from the list turns automatic selection off until you tick it again or change the court. Pressing a score button switches scores to manual, and any clock button switches the clock to manual.
- **Follow PlayHQ:** the score and quarter come from PlayHQ's live scoring every 5 seconds, and **Follow the PlayHQ scorer's clock** runs the clock from the scorer's clock too. Switching scores between **Follow PlayHQ** and **Manual** (on Scoreboard control, the Stream Deck, by pressing a score button or by picking a game) turns the clock option on or off to match, and Scoreboard control says so. Changing the clock after that still wins: untick the box or press a clock button to run the clock by hand while scores follow PlayHQ.
- **Pages that use the picked game's division:** ladder and finals (without `?division=`), ladder bar, top shooters, title, match up, The Clash, winner celebration, team lists, shooting stats and full time. The fixture, results, ticker and up next use the round.
- **Stream Deck:** `/deck/auto/on`, `/deck/auto/off`, `/deck/auto/toggle`, `/deck/court/2`, `/deck/clock/playhq/on` and `/deck/ticker/faster` / `slower`. The Devices tab has the rest under Links, with Copy buttons, and the plugin. A score key switches scoring to Manual (Follow PlayHQ stops for the score and the clock), and a clock key stops the clock following the PlayHQ scorer.

Socials in detail

The Socials tab makes pictures of a game night for Instagram and other social media, from the fixture, results and ladders Studio already has. Pick the **game night** (by date,

so every division playing that night is in it), then **Upcoming** (before the night) or **Results** (after it; a night with results opens on Results) and the sizes: **Square** 1080 x 1080, **Portrait** 1080 x 1350 and **Story** 1080 x 1920. Upcoming pictures:

- **Game night fixture:** every game of the night on every court, by time, with the logos, division and court.
- **Match up:** one per game, the two logos big with the date, time, court and venue.
- **On the stream:** the court Studio streams that night, each game's division and who is calling it, from the Production Planner (plan the night first).
- **Commentary team:** a picture for each game someone calls, with that game's commentators (up to three) side by side and its time and teams once under the headshots, so the social team can post them one by one or group them; each commentator has their headshot, to share before the livestream. Add each commentator's headshot on the Production Planner (click the circle beside their name); their initials show until then, and a guest gets a Special guest pill.
- **Polaroid:** a tilted white print for a match photo. Drop a photo on it, drag the preview to frame it and set the zoom; with no photo its window is see-through, so your social team can put a photo under it in Canva or a phone app.

Results pictures (games show once PlayHQ has their final score):

- **Results:** every finished game's final score, division by division, the winner in the highlight colour.
- **Full time:** one per game, the two logos and the score big, with who won and by how much.
- **Ladder:** one per division that played, as PlayHQ has it now ("after Round 5" when it is that division's latest night with results), with the finals line.
- **Top shooters:** one per division, its top six shooters that night, from each finished game's line ups (goals, plus super goals and points where there are super shots).
- **Tournament top shooters** (**Season top shooters** when the fixture runs 15 days or more, or as picked under Game night in Competitions setup): one per division, its top eight shooters over every finished game so far, labelled after this night's latest round, or after an earlier round that night (**As at**, for a carnival day with several rounds). With **Highlight movers** ticked (kept for the competition), each shows the places they climbed (green ▲) or fell (red ▼) since the round before, even on the same day (the day before for games with no round), and NEW for anyone new to the top eight. It reads the same kept line ups as Top shooters.
- **Polaroid:** one per game, with the score under the photo ("Blizzardz win 31–26") and the opponent, division, round and date under that.
- **Finals bound:** after a division's last home and away round (when it has a finals format in Competitions setup), the teams in its finals in ladder order, each with what their place earned: Double chance, Week off, Elimination final.
- **Finals draw:** with Finals bound, and again on each finals night (under Upcoming too, to post before the games): the division's finals week by week, who plays who ("Winner minor semi final" until a team is known), with the scores and winners filled in as the

games finish, from PlayHQ or entered by hand on Scoreboard control. The grand final is outlined.

- **Champions** (or **Premiers**, as picked under Finals in Competitions setup): once the grand final (or a best of three) has a winner, the winners' logo lit in their colour, their name, and the grand final's score.

On a finals night, the Match up, Full time and Results pictures name each final (Major semi final, Grand final) instead of the round.

Each picture has a **Sponsor strip** tick (your naming, elite and major sponsors along the bottom), remembered for next week, and Under **Options**, **FirstPass Studio mark** (bottom right) and **PIVT logo** (bottom left) turn those on or off for every picture. **Template** picks the look of every picture: **Default** (the stream's own), Dynamic, Junior, Retro, Professional, Thunder, Gradient, High contrast, Neon, Poster, Splash, Game Day, Hero or Final (these two stand a picked athlete in front of the title; cut-out PNGs look best), the netball sets Court, Bib and Goal, or the sporting sets Bottle, Cones, Sneakers, Stadium and Locker room, each with its own shapes and lettering. A star beside a template makes it a favourite, and **Favourites only** lists just those (and the one in use), remembered for the competition. Any template works with any colour set. Every picture's title stays on one line beside the competition logo, shrinking a little at a size where it would otherwise wrap. A short ladder or a couple of games grow to fill the picture, and a Story keeps its words and logos out of the top and bottom, where Instagram puts its own bar and reply box. A long team name shrinks to stay on one line in its row. Athletes' photos come from the Athletes tab, and only for athletes ticked **Approved for photo use**: Top shooters shows each shooter's photo, and on a Match up, Full time or Polaroid card you can pick an athlete by name, narrowed by division and team. **Colours** picks a colour set for every picture: **Competition colours** (the stream's own look), or a set made from the competition's main, highlight and accent colours (Accent first, Highlight ground, Accent ground, Night, Day, Two colours), or your own with **New colour set**. A set whose titles would be hard to read says ***low contrast***. The choice and your own sets are kept per competition. **Select all** and **Select none** tick or untick every picture at once. **Save pictures** saves the ticked ones as PNGs in **Pictures/FirstPass Studio/<competition> <date> Upcoming** (or **Results**) and opens the folder; with **As a zip** ticked it also saves them as one .zip beside the folder, and Share sends that zip; **Share…** saves them too and opens the Mac's Share menu (AirDrop to your phone, Messages, Mail). Saving needs the Mac app. Each picture is also a page of its own, `/social.html?kind=fixtures&size=square` (kinds `fixtures`, `matchup`, `onstream`, `commentators`, `polaroid`, `results`, `fulltime`, `ladder` with `&division=`, `topshooters` with `&division=`, `tournament` with `&division=` (and `&after=HH:MM`, `&movers=0|1`); `&date=`, `&game=`, `&sponsor=0|1`, `&mark=0|1`, `&pivt=0|1`, `&colours=`, `&template=`, `&home=`, `&away=`, `&athlete=`, `&title=`). Photos dropped on a polaroid stay on this Mac (the newest 40), never in a backup or iCloud.

Recordings in detail

For each PlayHQ game on the scoreboard while on air, Studio keeps a recording of what PlayHQ said and what went to air, timed: PlayHQ's answers (live score, clock, quarter, line ups, fixtures, ladders), each scoreboard press and Stream Deck press, and every change to the scene, the board and the answers the graphics read. There's no video or sound in it, only what the graphics need, a megabyte or so a game. Each recording follows a PlayHQ game: it starts when the game is due on the scoreboard (as many minutes before its scheduled start as **automatic game selection** on the competition setup page says, 10 by default) and stops a set time after PlayHQ says it's finished (**Keep recording for … minutes after full time**, 10 by default, 0 to 60), when the scoreboard moves on to another game, or when Studio goes off air. Being on air with no PlayHQ game due records nothing, and a game PlayHQ already says is finished isn't recorded. Each recording starts with the last of everything Studio already knew (the fixtures and ladders PlayHQ gave, the scene, the board), so it stands on its own. The Recordings tab lists each game's recording (named after when it started and the teams), its length and size, with **Download** (the .jsonl.gz file, to keep or send in as test data) and **Remove**, and says when the next one starts. **Record each PlayHQ game on air** is on unless you turn it off; turning it off stops the recording being made. The Output page's status board (Show) has the same switch as **Record** and **Stop**, so a recording can be stopped or started without leaving it. A recording older than 60 days is removed when Studio starts, and so is an old one with no game in it. Recordings stay on this Mac (in **Recordings/** of the data folder), never in a backup or iCloud, and hold nothing secret: no keys, devices, cookies or private answers. A recording also replays a game's graphics at 4K (below).

4K replay: Replay beside a game puts that game's graphics back on every output exactly as they went to air: the scene, the scoreboard and what each graphic read, from the recording. Turn on **4K** on the Output page and the outputs send it at 2160p, sharp, to key over the 4K camera recording of the game in a 4K switcher or recorder; Studio doesn't play the video. It starts paused at the game. To sync, pause the video on a moment you can see (the first centre pass), pick the same moment in the list of what went on air (***Scoreboard on***, ***Q1***…) and **Jump there**, then press **Play** on both together; **Nudge** moves the graphics a frame (40 ms) or a second either way. While it runs the live scene and scoreboard are kept as they were and can't be changed (the outputs can), nothing counts for the sponsors and the replay isn't recorded, and a red strip on every Studio page says so, with Stop. Stop puts the live graphics back.

Following a HyperDeck: with the 4K video on a HyperDeck, set **Follow** to ***A HyperDeck's timecode*** and type its IP address (on its setup page). The replay then goes where the HyperDeck's timecode says, as a time of day on the game's night: play, stop and scrub the HyperDeck and the graphics go with it, so they never drift, and Studio's own Play is the HyperDeck's. Studio only reads its timecode over the network (its control protocol, port 9993) and never plays, stops or records anything on it. If the HyperDeck's clock and this Mac's differed on the night, sync once: stop the HyperDeck on a moment you can see, pick the same moment in the list and press **Jump there**. **Nudge** then moves the graphics against the video. The sync is kept for each recording.

The recording needs time-of-day timecode (the HyperDeck's timecode set to Time of day, or timecode from the switcher).

Following the SDI input: any player can drive the replay through the UltraStudio instead. Send its SDI into the UltraStudio's SDI input (a HyperDeck's monitor or SDI output is enough; the 4K picture can go to the switcher as usual) and set **Follow** to ***Timecode into the UltraStudio (SDI)***. Studio reads the timecode that comes with each frame (only the timecode, not the picture) and follows it the same way: playing, stopped and scrubbing, with **Jump there** and **Nudge** to sync. HDMI carries no timecode, so the HDMI input can't be used. If the UltraStudio can't read its input while it sends the graphics, Studio says so beside Follow. Only in the Mac app.

Stream Deck: the plugin's **Replay** key plays or pauses, jumps to the next or previous moment, nudges or stops the 4K replay or practice game, whichever is running, and lights while one runs; or the links `/deck/replay/toggle`, `next`, `prev`, `nudge/-40`, `stop` and `/deck/practice/toggle`, `next`, `prev`, `stop` (the Devices tab lists them all under Links).

Practice games: Practise beside a game on the Recordings tab plays it back as if it were live. Studio asks PlayHQ nothing; the recording answers instead, with the game's date and times moved to now, so the game starts a few minutes from now: as many as **A practice game starts in** says (5 to begin with, 0 to 60). Everything runs as on a real night: the scoreboard follows the score, clock and quarter, sequences and game triggers fire, and the outputs can be on. **Jump to** Start, each quarter or Full time, **Pause** (the clock stands still) and **Stop** are at the top of the Recordings tab, and a red strip on every Studio page says a practice game is running, with Stop. Nothing it does counts for the sponsors' reports, it isn't recorded as a game, and Stop puts the scoreboard back as it was. Use it to rehearse a night, try a new sequence or show Studio to someone.

Demo games: Demo beside a recorded game plays it as one of the competition in use's games, for showing Studio to a league in their own colours. Pick a **division** and one of its finished **games**: Studio takes that game's two teams (with their logos and colours) and athletes, and the division's other teams, and everything else (the score, clock, quarters, moments and timing) comes from the recording. Athletes are matched by position, so the recorded GS's goals are their GS's; anyone with no match is left unnamed and their goals go to the team. A game with no line up (or a short one) takes the team's season squad, by the position each played most. While the demo runs, its **Athletes** table shows who each recorded athlete is shown as, and any of them can be changed. **Save demo** keeps it by name (teams, athletes, Swap) under **Saved demos**, to start again in one click; saved demos stay on this Mac. **Swap home and away** puts the teams the other way round, so either can be the winner. The competition's setup comes from PlayHQ, so a demo needs the internet. Studio never plays video: play the recorded game wherever you like and lay Studio's output over it. A red strip says a demo game is running, as for practice.